Before the Public Service Commission

The Brooklyn Union Gas Company d/b/a National Grid NY

and

KeySpan Gas East Corporation d/b/a/ National Grid

Rebuttal Testimony

of

Joshua C. Nowak

Date: September 22, 2023

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I. INTRODUCTION

Q. Please state your name, affiliation, and business address.

A. My name is Joshua C. Nowak. I am employed as Vice President by Concentric Energy Advisors, Inc. ("Concentric") at 293 Boston Post Road West, Suite 500, Marlborough, Massachusetts 01752.

Q. Are you the same Joshua C. Nowak who previously submitted Direct Testimony in these proceedings?

A. Yes. The terms defined in my Direct Testimony have the same meanings here unless otherwise indicated.

II. PURPOSE AND OVERVIEW OF REBUTTAL TESTIMONY

Q. What is the purpose of your rebuttal testimony?

A. The purpose of my rebuttal testimony in these proceedings is to respond to the prepared testimony of Department of Public Service Staff ("Staff") Witness Mr. Kwaku Duah and the testimony of Public Utility Law Project of New York, Inc. ("PULP") witness Mr. William D. Yates regarding the appropriate return on equity ("ROE") for The Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY") and KeySpan Gas East Corporation d/b/a National Grid ("KEDLI"), collectively referred to as "the Companies," wholly-owned subsidiaries of National Grid USA ("National Grid"). My analysis is supported by the data presented in Exhibits__ (JCN-1R) through (JCN-9R).

Q. What are your key conclusions regarding Staff Witness Duah's recommendations.

A. My key conclusions are as follows:

(1) Mr. Duah's overall recommendation does not meet the comparable return standard established in *Hope* and *Bluefield*[1] and considering recent market conditions and other benchmarks, such as recently authorized ROEs and interest rates.

(2) Mr. Duah places primary reliance on his Discounted Cash Flow ("DCF") analysis, which contains flaws and inconsistencies that produces a result (8.52 percent) that is below any ROEs authorized for natural gas utilities since at least 1980.[2]

(3) The cost of equity for regulated utility companies is being affected by several key factors in the current and prospective capital markets, including the interest rate environment and central bank monetary policy, as well as current inflationary pressure. Despite the effect such conditions have on the DCF and Capital Asset Pricing Model ("CAPM") approaches, Mr. Duah's approach mechanically applies the same DCF and CAPM formulas established nearly 30 years ago.

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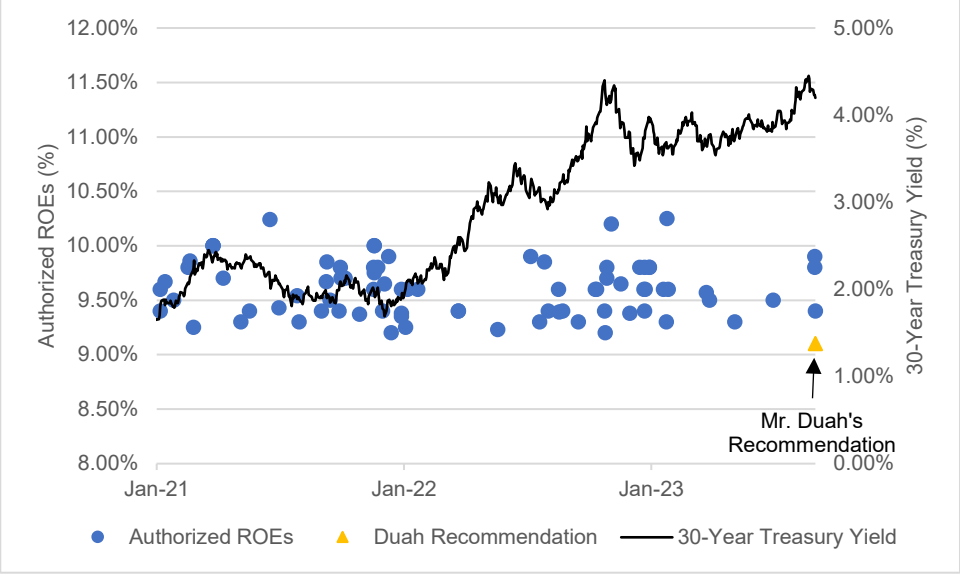
[1] *Bluefield Waterworks and Improvement Company v. Public Service Commission of West Virginia*, 262 U.S. 679 (1923) ("*Bluefield*"), and *Federal Power Commission v. Hope Natural Gas Company*, 320 U.S. 591 (1944) ("*Hope*").

[2] Source: S&P Capital IQ Pro, Regulatory Research Associates.

(4) While Mr. Duah's ROE recommendation is unreasonably low and well below the average ROEs authorized for other natural gas utilities nationwide, he fails to demonstrate that the Companies' risk profiles are lower than the average natural gas utility to support a departure from the returns available to other utilities.

As shown in Figure 1 below, ROEs authorized by other regulatory agencies for natural gas utilities since 2021 have averaged 9.61 percent and the lowest authorized ROE was 9.20 percent. Over this period, interest rates have increased substantially, demonstrating a sustained increase in the costs of capital. Despite this evidence, Mr. Duah recommends an ROE *below* any authorization by another regulatory agency over the past two years.

Figure 1: Authorized ROEs for Natural Gas Utilities, January 2021 through August 2023[3]



[3] Sources: S&P Capital IQ Pro, Regulatory Research Associates and Bloomberg Professional. Authorized ROEs for jurisdictions other than New York, excludes limited issue riders.

Q. Please summarize your concerns with PULP Witness Yates's testimony.

A. PULP Witness Yates compares the Companies' authorized ROEs to ROEs derived from Staff's testimony in other cases over the duration of prior rate plans. On that basis, Mr. Yates alleges that KEDNY and KEDLI earned above the cost of equity over the course of their respective rate plans from 2008 through 2022.[4] Mr. Yates proposes "an ROE update process for each year of the proposed rate plan," based on the GFP's recommended ROE calculation.[5] However, Mr. Yates's analysis of purported excessive revenues is flawed and fails to take into account several factors that would affect the measurement of the cost of equity, and the full spectrum of risk to which equity investors are exposed. His proposed ROE update process would undermine the settlement process and expose incremental risks to equity investors that would cause an increase in required returns.

Q. Please summarize the updated results of the ROE estimation models that you considered in your analyses.

A. As described in my Direct Testimony, I considered the results of the multi-stage form of the DCF model and two versions of the CAPM. The results of my updated analyses are summarized in Figure 2 (below). In my updated analysis, I maintained all of the assumptions I applied in my Direct Testimony, but updated for current market data, as of August 31, 2023. I

[4] Testimony of William D. Yates, at 36-38.
[5] *Id.*, at 42-43.

also excluded Hawaiian Electric Industries, Inc. from my proxy group due to its recent credit ratings downgrade to B-/B1, which is below investment grade.[6] Based on this updated analysis, the Companies' requested ROE of 9.80 percent is at the low end of this range of reasonableness and is a conservative estimate of the investor-required ROE.

Figure 2: Summary of Analytical Results[7]

	Low	Mean	High
DCF	9.69%	9.97%	10.34%
Mean CAPM	11.84%	11.86%	11.87%
67%/33% DCF/CAPM	10.41%	10.60%	10.85%
50%/50% DCF/CAPM	10.77%	10.92%	11.11%

In addition, reasonable modifications to Staff Witness Duah's proposed assumptions result in significant changes in the resulting ROE. Figure 3 summarizes the incremental effects of changes to Mr. Duah's analyses.

[6] Source: S&P Capital IQ Pro.
[7] Sources: Exhibit__ (JCN-1R) and Exhibit__ (JCN-3R).

Figure 3: Summary of Adjusted Results[8]

	Adjustment	Mean DCF Result	Mean Staff CAPM	ROE Result	Incremental Change (bps)
	Staff Estimate	8.52%	10.18%	9.10%	
1	Evenly Weight the DCF/CAPM Analyses	8.52%	10.18%	9.30%	+20 bps
2	Use the XIRR function in DCF	8.67%	10.18%	9.40%	+10 bps
3	Apply EPS Growth Rate for Near-Term Growth in the DCF Model	8.85%	10.18%	9.50%	+10 bps
4	Apply GDP Growth Rate for Long-Term Growth in the DCF Model	9.55%	10.18%	9.80%	+30 bps
5	Apply Market Risk Premium Estimate based on S&P 500 market return	9.55%	12.12%	10.80%	+100 bps
Total Change				10.80%	+170 bps

[8] Source: Exhibit__ (JCN-7R).

III. ECONOMIC AND CAPITAL MARKET CONDITIONS

Q. How have the economic and financial markets conditions changed over the past months since you submitted your Direct Testimony?

A. As I discussed in my Direct Testimony, inflation and elevated interest rates are the predominant factors in the current economic environment.[9] While inflation has declined since I filed my Direct Testimony in April 2023, it remains above the Federal Reserve's 2.0 percent target. In fact, the most recent monthly report on the Consumer Price Index reported an increase from the prior month with a 3.7 percent increase over the prior year.[10]

Q. Are conditions expected to stabilize in the near-term?

A. No, although difficult to predict in the long-term, capital market conditions remain highly uncertain. The economy and capital markets remain in a tenuous state with ongoing global supply constraints and expectations of weaker consumer demand in the face of persistent inflation and higher interest rates. Interest rates may not fully reflect the effects of inflation because no one knows exactly how high interest rates must go before inflation returns to more acceptable levels. Chairman of the Federal Reserve, Jerome Powell, recently reiterated the Fed's objective to bring inflation down to its 2.0 percent target, and is "prepared to raise rates further if appropriate, and intend to hold policy at a restrictive level until we are

[9] Direct Testimony of Joshua C. Nowak Direct, at 16.
[10] US Bureau of Labor Statistics, Economic News Release, Consumer Price Index Summary, September 13, 2023.

confident that inflation is moving sustainably down toward our objective."[11] Looking forward, the inflation risks in the market are twofold: (1) either inflation becomes embedded in the economy if the Federal Reserve does not move aggressively enough in tightening monetary policy, or (2) the Federal Reserve responds to inflationary pressure by raising short-term interest rates to a level that causes a slowdown in economic growth or a recession.

Q. Have you factored these circumstances into your cost of equity estimates for the Companies?

A. Yes. I have relied on the most recent market data and estimates available to me in my updated analysis. Long-term interest rates have increased substantially since the historical lows of 2020 and are expected to continue to increase as the Federal Reserve focuses on inflation. These circumstances also reinforce the importance of considering the results of multiple models, and the importance of the CAPM as interest rates are a direct input to the model.

Q. Has Staff Witness Duah considered the effect of economic and capital market conditions in establishing his ROE recommendation?

A. While Staff Witness Duah reviewed several factors related to capital market conditions, the fact that he did not make any adjustment to his ROE

[11] Board of Governors of the Federal Reserve System, Remarks by Jerome H. Powell, "Inflation: Progress and the Path Ahead," August 25, 2023.

approach or recommendation implies that the DCF and CAPM models are equally relevant, with the same market inputs, regardless of the capital market conditions. Specifically, Mr. Duah is using the same DCF and CAPM models, with the same respective weights that Staff has used for at least 15 years. As I described in my Direct Testimony, cases decided 15 years ago were amidst an extended period of low inflation, and without the degree of significant turmoil we have witnessed over the most recent 15 years. Since 2008, we have experienced significant, disruptive events that have substantially influenced capital markets, and caused significant monetary policy intervention. Beginning with the Global Financial Crisis, and more recently the worldwide pandemic associated with COVID-19 and the ensuing inflation crisis, capital markets have been exceptionally volatile. With that in mind, it is important to recognize that all models are subject to limitations and assumptions, and no single model is accurate under all market conditions. The presumption that the same two models, with the same inputs, weighted the same amount would be equally accurate under all conditions is highly unlikely to be accurate.

IV. RESPONSE TO STAFF WITNESS DUAH

Q. Please summarize Staff Witness Duah's ROE recommendation for the Companies.

A. Mr. Duah recommends an authorized ROE of 9.10 percent based on two-thirds weight applied to his average DCF model result of 8.52 percent and

one-third weight applied to the average of his two CAPM results of 10.18 percent.[12]

Q. What are the primary areas of disagreement between Mr. Duah and you as they relate to the authorized ROE for the Companies?

A. The primary area of disagreement between my analysis and recommendation and that of Mr. Duah is that Staff has dogmatically developed the assumptions used in their ROE formula without considering the effect of market conditions on the models, or the current financial profile of the Companies. As a result, Staff simply runs the models and recommends the results that the model generates. As will be discussed in my testimony, the results of the Staff's model and ultimate recommendations are not consistent with recently authorized returns in any other jurisdiction.

Mr. Duah's criticism of my methods for estimating the cost of equity stems from the principle of maintaining consistency with prior Commission practice. Mr. Duah frequently cites Commission precedent as the basis for critical assumptions within his ROE estimation methodology. The basis for Mr. Duah's objections to my methodology is essentially that the Commission has not historically employed the same data and methods that I have relied upon. However, Mr. Duah does not provide any substantiated

[12] Prepared Testimony of Kwaku Duah, at 100.

evidence to demonstrate that my assumptions and methods are not aligned with those used by investors or that they lack reasonability. While Mr. Duah points to numerous instances where it has adhered to "precedent" in developing assumptions, he does not acknowledge that there have been several instances where the Commission has adjusted its approach in response to changing market conditions. This fundamental disagreement between my ROE estimation approach and Mr. Duah's methodology arises from these differences in perspective.

The specific areas of disagreement related to the calculation of the ROE estimation models are as follows: (1) the screening criteria used to develop the proxy group; (2) the relative weighting of the DCF and CAPM results; (3) the application of the DCF model and the reasonableness of the results produced by the DCF model under current market conditions; (4) the application of the CAPM; and (5) the specific regulatory and business risks that have a direct bearing on the Companies' cost of equity, including the Companies' exposure to the CLCPA as compared to other utilities. The following subsections address each of these areas of disagreement.

A. PROXY GROUP COMPOSITION

Q. Please summarize Staff Witness Duah's proxy group selection.

A. Mr. Duah began with the universe of publicly traded utility holding companies that Value Line categorized as "electric utilities" and "natural gas utilities." Based on those 47 companies, Mr. Duah then required that

each company: (1) must currently maintain an investment-grade credit rating from Moody's and S&P; (2) must have regulated utility revenues of at least 70 percent of its total revenues; (3) must currently pay dividends; and (4) must not be engaged in any major mergers, acquisitions, or other transformational activities.

Q. What concerns does Mr. Duah raise with the selection criteria you applied to select your proxy group?

A. Mr. Duah alleges two concerns with my proxy group selection criteria. First, Mr. Duah argues that I should have screened based on the percentage of revenue derived from regulated utility operations, rather than screening based on the percentage of operating income derived from regulated utility operations. In addition, Mr. Duah states that I should not have limited my proxy group to transmission and distribution only utilities by excluding companies with owned generation comprising more than 60 percent of MWh sales to ultimate customers.

Q. Why is net operating income, rather than revenue, the appropriate screening criterion to utilize for selecting a proxy group for the Companies?

A. Net operating income better reflects the contribution of each business segment to a company's earnings. As described by Mr. Duah, "common equity investors only earn a return after the payment of all other

expenses."[13] In the case of regulated utilities that collect the cost of purchased gas, fuel, or power on a pass-through basis through rates, regulated revenue can, for example, fluctuate materially with changes in weather, alone, without impacting the corporation's earnings or financial position. In contrast, net operating income, which excludes pass-through commodity costs, more closely represents the effect of each business segment on the corporation's overall earnings, and therefore its risk profile.

Q. Why is it appropriate to exclude companies with substantial generation assets?

A. The risks a vertically integrated electric utility faces are quite different from the risks faced by the Companies. Investors analyzing a vertically integrated electric utility's stock consider the risks associated with the company's fleet of electricity generating resources, the impact of weather on the company's generation, and the company's ability to add renewable generation capacity. None of those are risks faced by the Companies, which operate as natural gas distribution utilities.

[13] Prepared Testimony of Kwaku Duah, at 14.

Q. What is your conclusion with regard to the differences in proxy group composition between your analysis and Mr. Duah's analysis?

A. In spite of this disagreement, my conclusion is that differences in our respective proxy groups do not account materially for the more substantial differences in our respective analyses or ROE recommendations.

B. WEIGHTING OF THE DCF AND CAPM METHODOLOGIES

Q. Please describe the reasons the Commission initiated the GFP.

A. The GFP was initiated because the Commission recognized that the DCF method was producing returns far below the returns produced by other methodologies.[14] The RD proposed that a two-third/one-third weighting be assigned to the results of the DCF and CAPM analyses, respectively, with less weight given to CAPM because it had only been used to that point as a check on the DCF model.[15] While Mr. Duah relies on the GFP to support his decision to place two-thirds weight on the DCF results, Mr. Duah does not acknowledge that the RD in the GFP left open the possibility that the weightings and methodologies could be adjusted if necessary to ensure that the results promote regulatory credibility.

[14] Case 91-M-0509, Proceeding on Motion of the Commission to Consider Financial and Regulatory Policies for New York State Utilities, Recommended Decision, (issued July 19, 1994) ("Generic Finance RD"), at 2.

[15] *Id*., at 27.

Q. Does Mr. Duah cite to the Commission's rationale in his weighting of the DCF and CAPM results?

A. Yes, Mr. Duah cites orders issues in Cases 07-G-0141, 06-E-1433, and 06-E-1547.[16] In these cases, the Commission authorized ROEs calculated by placing two-thirds weight on the DCF and one-third weight on the CAPM.

Q. What is the relevant context to evaluate the orders cited by Mr. Duah.

A. The most recent Commission decision cited by Mr. Duah is from December 2007. As I described in my Direct Testimony, from the mid-1980s until 2008, an extended period of low inflation was characterized by tighter monetary policy as compared with the prior period. The decisions cited by Mr. Duah are from this period. Subsequently, the Global Financial Crisis created a severe recession that was met with comparatively looser monetary policy (*i.e.*, lower target interest rates) and significant fiscal policy stimulus as well as the Federal Reserve's extended involvement in the financial markets that began following those events. More recently, in response to the economic effects of COVID-19, extensive monetary and fiscal policy was enacted to mitigate the effects of the pandemic. The subsequent inflationary concerns caused the Federal Reserve to pull back on its COVID-related monetary policies and apply tighter monetary policy with higher interest rates. It is important to consider that the cases cited by Mr. Duah, and the market data used to estimate the ROE in those proceedings were from a

[16] Prepared Testimony of Kwaku Duah, at 133-134

relatively stable period of low inflation and modest monetary policy changes. Subsequent to those decisions, capital markets have been significantly affected by anomalous conditions and sustained periods of Federal Reserve involvement in stabilizing markets.

With that context, it is significant to note that the Commission's conclusion more than 15 years ago was based on the circumstances of that time and did not suggest that the Commission was precluded from making changes in the future. The Commission's conclusion was that is was "not *now* inclined to deviate from our long-held view that the CAPM should not be entitled to more than one-third of the weight" [17] leaving the possibility that it may be appropriate to consider such a change in the future.

Q. Has the Commission recognized a willingness to change inputs in its calculations?

A. Yes, in fact in Case 06-E-1433, cited by Mr. Duah, the Commission notes that it changed its calculation of the market return used in the estimation of the market risk premium in the CAPM. The Commission recognized that the use of historical returns published by Ibbotson were stale and less reliable and therefore began relying on projected returns as published by Merrill Lynch. In this same case, the Commission recognized that six-

[17] Case 06-E-1433, *Orange and Rockland Utilities*, "Order Setting Permanent Rates, Reconciling Overpayments During Temporary Rate Period, and Establishing Disposition of Property Tax Refunds" (October 18, 2007) at 15 [emphasis added].

month average stock prices could be "stale."[18] Currently, the Commission's methodology relies on three-month average stock prices. These types of changes demonstrate that the Commission is willing to consider modifications to the ROE estimation methodology to include more current and relevant information as market conditions change.

Q. Why is it that current capital market conditions support reconsideration of the respective weights placed on the DCF and CAPM approaches?

A. There has been an increasing level of complexity and sophistication in investors' analytical approaches over time. Regulators have recognized such complexities and how they may be impacting DCF model results. Specifically, regulators have recognized the need to consider multiple analytical approaches to develop a reasonable cost of equity. Recognizing the important role that dividend yields play in the DCF model, the FERC determined that capital market conditions have caused the DCF model to understate equity costs for regulated utilities. In Opinion No. 531, the FERC noted:

> There is 'model risk' associated with the excessive reliance or mechanical application of a model when the surrounding conditions are outside of the normal range. 'Model risk' is the risk that a theoretical model that is used to value real world transactions fails to predict or represent the real phenomenon that is being modeled.[19]

[18] Case 06-E-1433, *Orange and Rockland Utilities*, "Order Setting Permanent Rates, Reconciling Overpayments During Temporary Rate Period, and Establishing Disposition of Property Tax Refunds" (October 18, 2007) at 11.

[19] FERC Docket No. EL11-66-001, Opinion No. 531 (June 19, 2014), fn 286.

Given the role that the Federal Reserve has in influencing capital market conditions, and its effect on interest rates, it is reasonable to place more weight on models that directly account for such inputs. As discussed in Section III, inflation and elevated interest rates are the predominant factors in the current economic environment and interest rates are a direct input to the CAPM. As such, the model is more adaptable than the DCF in accounting for changes in monetary policy and its effect on the cost of equity.

Q. How have other regulators responded to the dynamic capital market conditions in determining the cost of equity?

A. As I discussed in my Direct Testimony, several regulators, including the FERC, have recognized the challenges associated with relying primarily on the DCF to establish the authorized ROE for regulated utilities in recent years. The FERC abandoned its sole reliance on the DCF model to rely on equal weighting of multiple models, including the DCF and CAPM.[20] Other jurisdictions, including the Pennsylvania Public Utility Commission, State of Illinois Commerce Commission, and Missouri Public Service Commission have also recognized deemphasizing the DCF in recent cases.[21] Therefore, it is reasonable to place more weight on the CAPM than the one-third weight it has been given in prior cases. As shown in Exhibit___(JCN-

[20] I note that while the most recent FERC precedent incorporates equal weighting of the DCF, CAPM, and Risk Premium approaches, the inclusion of the Risk Premium model remains subject to a remand order by the DC Circuit Court of Appeals.

[21] Direct Testimony of Joshua C. Nowak Direct, at 53-54.

7R), evenly weighting the DCF and CAPM results would increase the weighted average results from 9.10 percent to 9.30 percent.

C. Application of the DCF Model

Q. Please summarize Mr. Duah's application of the DCF model.

A. Mr. Duah uses a two-stage DCF model that forecasts dividends from 2024 to 2027 using Value Line's estimates of projected dividends in that period, and a "sustainable growth rate" from 2028 forward. Using the DCF methodology, Mr. Duah calculates an average ROE for his proxy group of 8.52 percent. Mr. Duah applies a two-thirds weight to his DCF model results in deriving his overall ROE recommendation.

Q. How does Mr. Duah's DCF model differ from your approach?

A. The most significant difference between my application of the DCF model and Mr. Duah's approach is the growth rates we use in our respective analyses. I have used a consensus of analysts' EPS growth rates for the proxy group companies as the near-term growth rate, and an estimate of growth in the overall economy for the long-term growth rate. This mitigates the uncertainty associated with forecasting individual companies' growth rates over very long-time horizons. By contrast, Mr. Duah uses dividend growth projections from a single source (*i.e.*, Value Line) for their near-term growth rate, and a "sustainable growth rate" for their long-term growth rate. Mr. Duah states that it is "highly unlikely that investors would rely

exclusively on the earnings per share growth rate forecasts of Wall Street analysts in determining short-term dividend projections."[22]

Q. Do you agree with Mr. Duah's concern with earnings growth rates?

A. No, I do not. Mr. Duah prefers dividend growth rates rather than earnings growth rates as the near-term growth rate in their DCF analysis. This ignores the fact, as discussed in my Direct Testimony, that dividend growth is fundamentally driven by earnings growth.[23] While the model is, indeed, called the "Discounted Cash Flow" model, the cash flows it refers to are those an investor can expect to receive during the time they own the stock. Those cash flows are quarterly dividend payments plus any capital appreciation that occurs between when the stock is purchased and when it is sold. Dividend payments and capital appreciation are both a function of earnings per share ("EPS"), which is ultimately what determines the ROE to the investor. As discussed in my Direct Testimony, dividends are based on management decisions related to cash management and other factors, and therefore dividend growth rates are less likely to accurately reflect investors' growth expectations than earnings growth rates.[24]

[22] Prepared Testimony of Kwaku Duah, at 114.
[23] Direct Testimony of Joshua C. Nowak, at 35-38.
[24] *Id.*, at 35-36.

I have relied on earnings growth because earnings are the fundamental determinant of a company's ability to pay dividends. As noted by Brigham and Houston:

> Growth in dividends occurs primarily as a result of growth in earnings per share (EPS). Earnings growth, in turn, results from a number of factors, including (1) inflation, (2) the amount of earnings the company retains and invests, and (3) the rate of return the company earns on its equity (ROE). [25]

Investment analysts report predominant reliance on EPS growth projections. In a survey completed by 297 members of the Association for Investment Management and Research, the majority of respondents ranked earnings as the most important variable in valuing a security (more important than cash flow, dividends, or book value).[26]

Academic research also supports the use of EPS growth estimates. A 2002 study in the *Journal of Accounting Research*, examined "the valuation performance of a comprehensive list of value drivers" and found that "forward earnings explain stock prices remarkably well" and were generally superior to other value drivers analyzed.[27] A 2012 study from the journal *Contemporary Accounting Research* found that the sell-side analysts with

[25] Eugene F. Brigham and Joel F. Houston, *Fundamentals of Financial Management*, at 317 (Concise Fourth Edition, Thomson South-Western, 2004).

[26] Block, Stanley B., "A Study of Financial Analysts: Practice and Theory", Financial Analysts Journal (July/August 1999).

[27] Liu, Jing, et al., "Equity Valuation Using Multiples," Journal of Accounting Research, Vol. 40, No. 1, March 2002.

the most accurate stock price targets were those whom the researchers found to have more accurate earnings forecasts.[28]

Q. Mr. Duah claims that Value Line growth rates are produced by a group of analysts, and therefore represent a consensus view. Do you agree?

A. No, I do not. Mr. Duah provides a letter from Value Line describing the development of its growth projections, as Exhibit ___ (KXD-27). Value Line states that each company it covers is assigned to a lead analyst who is responsible for building the coverage model for that company. Specifically, the letter states, "[e]ach stock in The Value Line Investment Survey is assigned to a specific analyst."[29] The letter goes on to report the quality control procedures applied to the analyst's report. Nowhere in the letter, titled "Quality Control Procedures," does Value Line describe a process whereby multiple independent evaluations are performed and then averaged together to form a consensus view. That is the important element missing from Mr. Duah's analysis: Mr. Duah relies on a source for growth rate estimates that reflects the opinion of a single analyst.

While I agree that Value Line is a trusted source for investment professionals, it is not the only tool that investors rely on to make decisions. There are additional data sources readily available that compile the

[28] Gleason, C.A., et al., "Valuation Model Use and the Price Target Performance of Sell-Side Equity Analysts," *Contemporary Accounting Research*.

[29] Prepared Testimony of Kwaku Duah, Exhibit __ (KXD-27).

consensus viewpoints of multiple brokerage analysts, and it is reasonable to expect that investors also consider that information. In fact, there have been studies performed comparing Value Line and I/B/E/S analyst earnings forecasts in terms of accuracy, rationality and as proxies for market expectations. In 2001, a study concluded that "I/B/E/S forecasts were superior, as explained by the combination of I/B/E/S's timing advantage and the mitigation of idiosyncratic error through consensus building."[30] Furthermore, the I/B/E/S long-term forecasts were less biased and more accurate.[31]

In addition, Mr. Duah's position on the appropriate growth rate is contradictory. Mr. Duah criticizes the use of earnings growth rates because "[i]nvestors will not just look at expected earnings but also factor in all relevant information when estimating their required return."[32] If investors consider all relevant information, it is inconsistent with that principle to limit the DCF analysis to rely exclusively on a single source of estimated growth rates, as Mr. Duah has done. Based on Mr. Duah's conclusion, it is reasonable to consider the various sources of consensus growth estimates that I have relied on in addition to the Value Line earnings growth rates.

[30] Ramnath, Sundaresh, Rock, Steven, Shane, Philip, "Value Line and I/B/E/S Earnings Forecasts," November 8, 2001, at 1.
[31] *Ibid*.
[32] Prepared Testimony of Kwaku Duah, at 115.

Q. Is there additional evidence that investors give more weight to earnings growth estimates, rather than dividend growth estimates?

A. Yes, there is. As Dr. Roger Morin observed in "New Regulatory Finance," earnings growth forecasts are far more widely available than dividend growth forecasts, which indicates that the investment community places more weight on earnings growth rates than dividend growth rates. Specifically, Dr. Morin stated:

> Casual inspection of the Zacks Investment Research, First Call Thompson, and Multex Web sites reveals that earnings per share forecasts dominate the information provided. There are few, if any, dividend growth forecasts. Only Value Line provides comprehensive long-term dividend growth forecasts. The wide availability of earnings forecasts is not surprising. There is an abundance of evidence attesting to the importance of earnings in assessing investors' expectations. The sheer volume of earnings forecasts available from the investment community relative to the scarcity of dividend forecasts attests to their importance. The fact that these investment information providers focus on growth in earnings rather than growth in dividends indicates that the investment community regards earnings growth as a superior indicator of future long-term growth. Surveys of analytical techniques used by analysts reveal the dominance of earnings and conclude that earnings are considered far more important than dividends. Finally, Value Line's principle investment rating assigned to individual stocks, Timeliness Rank, is based primarily on earnings, accounting for 65% of the ranking.[33]

As Dr. Morin indicates, the fact that there is a lack of academic research analyzing dividend growth forecasts, while there is an abundance of

[33] Roger A Morin, "New Regulatory Finance," Public Utilities Reports, Inc. 2006, at 302-303.

research regarding earnings growth forecasts, is yet another indication that earnings growth is the more meaningful measure.

Q. Do you agree with Mr. Duah that reliance on Value Line growth rates lends a degree of predictability to the New York rate-setting process for the investment community?

A. No, I do not. As I noted, the Value Line growth rates are the opinion of one analyst regarding growth in dividends when the investment community is more focused on earnings growth. Contrary to Mr. Duah's assertion,[34] the investment community reviews the returns established through the New York rate-setting process based on the reasonableness and comparability of the returns to other investment alternatives.

Q. Please summarize your concerns with Mr. Duah's sustainable growth rate.

A. I have two primary concerns with Mr. Duah's sustainable growth rate. First, while Mr. Duah asserts that his calculation of the sustainable growth rate represents a measure of long-term growth for the period 2028 and beyond, the inputs to his calculation are, for the most part, shorter-term estimates for the period 2028 and earlier. For example, the "b * r" component of Mr. Duah's sustainable growth rate is derived from Value Line forecasts that only extend through 2028. Therefore, Mr. Duah's long-term growth rate

[34] Prepared Testimony of Kwaku Duah, at 117.

estimate reflects, at best, one analyst's forecast of only the very early years of the second stage of Mr. Duah's DCF model, which theoretically extends into perpetuity. In contrast, my estimate of long-term GDP growth reflects projections over 30 years or more and considers overall measures of economic growth.[35] Second, Mr. Duah's sustainable growth rate relies on Value Line's estimate of each proxy company's ROE, as the "r" in the "b * r" component of Mr. Duah's growth rate in the expected ROE. This introduces an element of circularity into Mr. Duah's calculation that he fails to reconcile with his results. That is, based on Exhibit __ (KXD-22), the mean and median ROEs assumed in Mr. Duah's DCF calculation are 10.85 percent and 10.16 percent, respectively, yet Mr. Duah's analysis suggests the proxy group average ROE is 8.52 percent. By that measure, the range of ROEs established in my analysis for my proxy group companies (*i.e.*, 9.80 percent to 10.70 percent) is quite conservative relative to the proxy group ROEs assumed in Mr. Duah's sustainable growth estimate.

Q. Is there academic research that recognizes the shortcomings of the sustainable growth rate approach?

A. Yes. Dr. Roger Morin noted his concern with the use of the sustainable growth rates in the DCF. Specifically, Dr. Morin stated:

[35] Direct Testimony of Joshua C. Nowak, at 40.

> In summary, there are three proxies for the expected growth component of the DCF model: historical growth rates, analysts' forecasts, and the sustainable growth method. Criteria in choosing among the three proxies should include ease of use, ease of understanding, theoretical and mathematical correctness, and empirical validation. The latter two are crucial. The method should be logically valid and consistent, and should possess an adequate track record in predicting and explaining security value. The retention growth method is the weakest of the three proxies on both conceptual and empirical grounds. The research in this area has shown that the first two growth proxies do a better job of explaining variations in market valuations (M/B and P/E ratios) and are more highly correlated to measures of value than is the retention growth proxy.[36]

Considering the theoretical and empirical concerns noted by both regulators and academics, it is reasonable to conclude, as I have, that the sustainable growth rate should not be relied on as the estimate of the long-term growth rate in the multi-stage DCF model.

Q. Why is it reasonable to rely on historical averages of real GDP growth to approximate future economic activity?

A. As described in my Direct Testimony, the use of historical real GDP growth is consistent with the approach relied upon by Morningstar, a leading provider of investment information which previously published data on historical stock and bond returns from Ibbotson and Associates.[37]

[36] Roger A Morin, "New Regulatory Finance," Public Utilities Reports, Inc. 2006, at 307.
[37] Direct Testimony of Joshua C. Nowak, at 41-42.

Q. Do the GDP growth projections that Mr. Duah references account for the effect of the business cycle?

A. No, they do not. Specifically, Mr. Duah references a real GDP growth projection from the 2023 Annual Energy Outlook published by EIA from 2022 through 2050.[38] EIA, however, assume expectations of smooth economic growth and do not anticipate business cycles or large economic shocks.[39] Thus, EIA's projection of real GDP growth will most likely understate the long-term growth in real GDP since the methodology does not consider business cycles and relies on current trends in economic data. This is a particularly meaningful shortcoming considering the recent economic downturn associated with the current health pandemic. Therefore, EIA's projections are not a relevant check on the reasonableness of the GDP growth rate reflected in my multi-stage DCF analysis.

Q. Has Mr. Duah relied on historical data in any of his analyses?

A. Yes. For example, in his CAPM analysis, Mr. Duah relies exclusively on Value Line Beta coefficients that use a five-year historical estimate of the relative risk of proxy companies and the overall market.

[38] Prepared Testimony of Kwaku Duah, at 87.

[39] U.S. Energy Information Administration, Annual Energy Outlook 2023, Table 20. Macroeconomic Indicators, March 16, 2023.

Q. Do you have any additional concerns with the projected GDP growth rate cited by Mr. Duah?

A. Yes. The GDP growth projections cited by Mr. Duah cover relatively short time periods. For example, Mr. Duah cites to the EIA's base case real GDP growth projection of 1.90 percent for the period 2022 through 2050 as evidence that the projected real GDP growth rate that I have relied upon is overstated.[40] Mr. Duah's multi-stage DCF analysis, however, extends 200 years, with the second stage beginning in year 6 of the analysis period. This means that Mr. Duah is attempting to assess the reasonableness of a long-term growth rate using a projection that covers approximately 15.00 percent of the period it is being applied to (*i.e.*, Mr. Duah is assessing a 195-year projection using a 30-year projection). Mr. Duah's exercise is even less instructive since as noted above EIA does not anticipate or incorporate the effect of business cycles in the projection.

Q. Why have you chosen to rely on a three-stage DCF, as opposed to a two-stage DCF?

A. My reliance on a three-stage DCF model reflects my view that growth rates will not suddenly shift from the near-term growth rate to the long-term growth rate, but rather will transition gradually. While Mr. Duah argues that I have not provided any evidence regarding the accuracy of the three-stage model as compared to the two-stage model, Mr. Duah has not

[40] Prepared Testimony of Kwaku Duah, at 87.

provided any evidence on this question either. In fact, the evidence that Mr. Duah has provided indicates that investors use the three-stage model. Specifically, Mr. Duah develops his estimate of the market return for purposes of the CAPM analysis based on reports published by Merrill Lynch. Merrill Lynch develops its projected market return based on a three-stage DCF model.[41]

Q. Are there any problems with the calculations in Mr. Duah's DCF model?

A. Yes, in calculating the ROE, Mr. Duah relies on the "IRR" Microsoft Excel function, and that dividends are assumed to be received at the end of the year.[42] However, the proxy companies issue quarterly dividends and the "XIRR" function would be more appropriate to take this into account. Both the IRR and XIRR functions are Microsoft Excel functions that are used to calculate the internal rate of return for a series of cash flows. The IRR function makes the simplifying assumption that cash flows are available at a set time in each period. In contrast, the XIRR function allows additional inputs as to the timing of cash flows. Specifically, the XIRR function allows the analysis to reflect the assumption that dividends are paid, on average, halfway through the year (*i.e.*, two dividends are paid in the first half of the year, and two dividends are paid in the second half of the year). The XIRR

[41] Exhibit__ (JCN-9R), containing Staff Response to Interrogatory NG-27.
[42] Prepared Testimony of Kwaku Duah, at 124.

function allows the analysis to reflect a more nuanced set of assumptions regarding the timing of cash flows.

Q. Please provide an example of why the XIRR function is appropriate, even for regular cash flows.

A. Suppose that there are two dividend paying utilities, identical in every respect but one: utility one pays a lump-sum dividend once per year; and utility two pays daily dividends that are equal, on an annualized basis, to the lump-sum dividend paid by utility one. According to Mr. Duah's analysis, the result of a discounted cash flow analysis of both utilities would be identical. However, it is a well-recognized financial principle that investors place value on the timing in which cash flows are received. While both utilities in this hypothetical pay "regular" dividends, the timing of cash flows matters. The XIRR function allows the analysis to better reflect the timing of cash flows. In contrast, the oversimplified and imprecise IRR function assumes that all dividends are paid at the end of the year. Every utility in my proxy groups, and in Mr. Duah's proxy group, pays dividends on a quarterly basis. Thus, the IRR function used by Mr. Duah, which assumes that investors have to wait longer to receive dividend payments from the utility stock, understates the cost of equity. As shown in Exhibit___(JCN-5R), applying the more precise XIRR function would increase Mr. Duah's DCF result from 8.52 percent to 8.67 percent.

D. Application of the CAPM Approach

Q. Please summarize Mr. Duah's application of the CAPM.

A. Mr. Duah developed his estimate of the risk-free rate of 3.82 percent by using a recent average of 10-year and 30-year Treasury bond yields. For the Beta estimate, Mr. Duah used the average Value Line Beta for his proxy group of 0.89. To estimate the market risk premium, Mr. Duah subtracted the risk-free rate estimate from an average of the forecast returns for the S&P 500 from Merrill Lynch's May, June and July 2023 Quantitative Profiles. Mr. Duah also developed a zero-Beta CAPM using similar inputs as his traditional CAPM analysis, but using a market risk premium based on weighting the Beta times the market risk premium by 0.75 and the market risk premium itself by 0.25. Mr. Duah relied on the average of his two CAPM analyses and weighted that result by one-third (as compared to two-thirds weight for the DCF) in the formulation of his overall ROE recommendation.

Q. What are the primary differences between your application of the CAPM and Mr. Duah's application of this model?

A. Because the estimation of the ROE is a forward-looking concept, and the ROE that is authorized in these cases will be in effect for some period in the future, my analysis appropriately considers both the recent historical risk-free rate and the projected risk-free rate. The analysis presented in my Direct Testimony also relies on Bloomberg estimates of Beta that more accurately reflect market conditions. Finally, I estimate the market risk

premium based on the difference between the return on large company stocks, as measured by the S&P 500, and the yield on 30-year Treasury bonds.

Q. Why is the use of 30-year Treasury bond yields more appropriate than relying on 10-year and 30-year Treasury bond yields?

A. Mr. Duah's rationale for relying on 10-year and 30-year Treasury bond yields in his CAPM analysis is that it is consistent with the approach the Commission has relied on in prior cases, and the use of these securities reflects the expectations of investors who have "intermediate as well as long-term investment horizons."[43] I disagree with Mr. Duah's rationale. As noted by Morningstar, "the horizon is a function of the investment, *not the investor*." The use of the 30-year Treasury bond yield as the risk-free rate is consistent with the investment horizon for natural gas utility assets. Taking Mr. Duah's rationale to the limits, an investor who plans to hold a position in a utility equity share for only six months would use a certificate of deposit rate to evaluate the potential return. That is clearly not the case for any rational investor considering return requirements.

[43] Prepared Testimony of Kwaku Duah, 94.

Q. Have other regulatory agencies commented on the appropriate securities to use for the risk-free rate in the CAPM?

A. Yes. In Opinion No. 531, the FERC relies on the yield on the 30-year Treasury bond.

> As noted above, the 30-year U.S. Treasury bond yields are a generally accepted proxy for the risk-free rate in a CAPM analysis, and are also considered superior to short- and intermediate-term bonds for this purpose.[44]

FERC made no modification to this approach in its recently proposed ROE methodology.[45]

Q. Why are projected treasury bond yields informative in the CAPM analysis?

A. Just as investors have available to them actual and projected growth rates for individual companies, they also have available current and projected yields on Treasury bonds. It is reasonable to expect that investors can and do consider this information, as they do other market projections. Furthermore, the use of projected Treasury bond yields is consistent with the use of a forward-looking market risk premium. Therefore, it is reasonable, and consistent with Mr. Duah's position regarding the calculation of the market risk premium, to rely on expectations of market conditions to the extent that those expected conditions differ from recent

[44] FERC Order 531-B at para 114. Citing also to Roger A. Morin, New Regulatory Finance 151-152 (Public Utility Reports, Inc. 2006) ('the yield on very long-term government bonds, namely, the yield on 30-year Treasury bonds, is the best measure of the risk-free rate for use in the CAPM and Risk Premium methods.").

[45] FERC Order 569-B.

history. From a practical perspective, the ROE that is approved by the Commission will be the return to the Companies that will be reflected in rates over the rate period, until such time as another rate proceeding is filed. It is important to reflect investors' expectations of returns over that time period. Therefore, I conclude that it is appropriate to consider consensus estimates of investors' expectation of yields on Treasury bonds in the CAPM analysis.

Q. Does Mr. Duah misconstrue your testimony with respect to the use of 30-year Treasury bonds in the CAPM?

A. Yes. Mr. Duah suggests that in my Direct Testimony I conclude that "all utility equity investors have an investment horizon of 30 years."[46] Mr. Duah then states that this conclusion is unsubstantiated and erroneous. Mr. Duah provides no citation for where this conclusion appears in my Direct Testimony.

Q. What is your testimony with respect to the use of the yield on 30-year Treasury bonds as the risk-free rate?

A. It is my testimony that it is important to select the term (or maturity) that best matches the life of the underlying investment.[47] I also cite to Morningstar, which notes that the Treasury security relied on should match the time horizon of what is being valued, and that the time horizon is a

[46] Prepared Testimony of Kwaku Duah, at 129.
[47] Direct Testimony of Joshua C. Nowak, at 48.

function of the investment, not the investor. My testimony does not suggest that equity investors have an investment horizon of 30 years, nor does my testimony suggest that this would be a relevant factor in determining the appropriate security for the risk-free rate even if investors' time horizons were that long.

Q. Please summarize Mr. Duah's concern with your use of Bloomberg's Beta coefficients in the CAPM analysis.

A. Mr. Duah contends that the Commission has consistently preferred Value Line's Beta coefficients because they are more accurate, and they reflect the exercise of expert judgment. To support this position, Mr. Duah cites a study analyzing the relative accuracy of Value Line's Beta coefficients compared to other sources.[48]

Q. Why is it reasonable to rely on Bloomberg's Beta coefficients?

A. As a practical matter, the Bloomberg Beta coefficients that I have relied on are comparable to those reported by Value Line. Furthermore, it is generally reasonable to consider several measures of market conditions in estimating the ROE. The Bloomberg Beta coefficient is widely used, and because it is based on a ten-year period as compared to Value Line's five-year period, it mitigates the effect of exogenous events on utility Beta coefficients. As discussed in my Direct Testimony, utility companies have

[48] Prepared Testimony of Kwaku Duah, at 127.

been significantly affected by multiple exogenous events, namely the Global Financial Crisis, the Tax Cuts and Jobs Act of 2017, and the COVID-19 pandemic.[49] The academic study cited by Mr. Duah was published before such exogenous events occurred.

Q. What research did Mr. Duah cite regarding Value Line's Beta coefficients?

A. Mr. Duah cites a 2011 finance text by H. Kent Baker and Gerald S. Martin titled "Capital Structure and Corporate Financing Decisions: Theory, Evidence, and Practice" to suggest that the Value Line estimate of Beta coefficients is superior.[50]

Q. Does that source demonstrate that Mr. Duah's Beta coefficients are superior to those reflected in your CAPM analysis?

A. No, it does not. As an initial matter, I have included Value Line's Beta coefficients in my CAPM analysis. As such, we agree that Value Line Beta coefficients provide relevant information to investors. Further, the conclusion of the article that Mr. Duah relies on is consistent with the process I have used in my application of Beta coefficients in the CAPM analysis. Baker and Martin criticize estimates that "involve no input from analysts" and go on to conclude "Value Line's adjustments also appear informed by more than the type of simple smoothing done by Bloomberg,

[49] *See, e.g.,* Direct Testimony of Joshua C. Nowak at 18.
[50] Prepared Testimony of Kwaku Duah, at 126.

which uses the following formula: adjusted beta = historic beta (0.67) +0.33" [51]. The smoothing performed by Value Line, however, is similar to Bloomberg's (Value Line uses an adjusted beta = historic beta (0.67) +0.35). Further, in my application of Bloomberg Beta coefficients, I adjusted the measurement period to mitigate the effects of the exogenous events as described above. This is consistent with Baker and Martin's conclusion on Beta coefficients: "The objective is to capture investors' view of future risks." Therefore, I disagree with Mr. Duah's conclusion that this source demonstrates that Value Line Beta coefficients are superior to my application of Bloomberg coefficients that were specified with parameters based on an analysis of the market conditions and my informed judgment.

Q. Please summarize Mr. Duah's concern with your estimation of the Market Risk Premium.

A. My market risk premium was developed by applying the Constant Growth DCF model to the companies in the S&P 500. Mr. Duah argues that this approach is not appropriate because the companies in the S&P 500 cannot sustain the projected three-to-five-year growth rates in perpetuity.[52]

Q. Do you agree with that criticism?

A. No, I do not. The calculation of the MRP is based on the return on the broader market, as measured by S&P 500, less the return on a risk-free

[51] *Ibid*.
[52] *Id.*, at 123.

instrument. The S&P 500 is an index that includes the largest 500 companies by market capitalization. Over time, the specific companies that are included in the S&P 500 Index will vary, but investor expectations of growth and return overall for the index as a whole may not, based on the selection process involved in the index. Therefore, it is reasonable to assume that the average growth of the index could be sustainable in the long run.

Q. Is there additional support for the reasonableness of the market return you have used to calculate the forward-looking market risk premium?

A. Yes, other alternative sources provide reputable forecasts of market returns that are significantly higher than the estimated market return relied on by Mr. Duah. For example, S&P regularly publishes earnings growth projections for the S&P 500. As of September 7, 2023, S&P projected earnings growth of 12.92 percent for the S&P 500.[53] When combined with the twelve-month average dividend yield of 1.56 percent for the S&P 500 reported by S&P, that results in a projected return on the S&P 500 of 14.58 percent. Additionally, as of September 12, 2023, State Street projected long-term EPS growth of 13.83 percent for an exchange-traded fund tracking the performance of the S&P 500 index.[54] When combined with the reported dividend yield of 1.57 percent, that results in a projected market return of 15.51 percent. By contrast, the market return that Mr. Duah relied

[53] S&P Dow Jones Indices, S&P 500 Earnings and Estimate Report, September 7, 2023.
[54] https://www.ssga.com/us/en/individual/etfs/funds/spdr-sp-500-etf-trust-spy.

on is 10.85 percent, which is more than 350-basis points below the market return estimates provided by these reputable sources.

Q. What are your conclusions regarding the CAPM analysis?

A. In principle, I agree with Mr. Duah that the development of the inputs and assumptions for the CAPM should be done on a forward-looking basis, and that *ex-post* analysis can be flawed in some circumstances. Furthermore, both our CAPM analyses contain some historical data where we believe it to be the best estimate in current market conditions. In particular, I note that recent market conditions should not be considered a proxy for future ROE results. As such, I have developed the assumptions used in the CAPM to be forward-looking to the period that the Companies' rates will be in effect. In the instance of Beta, I relied on Bloomberg Beta coefficients in addition to Value Line's Beta coefficients to account for the effects of short-term market dislocations on utility Beta coefficients.

Based on the specification of the methodology, and the support for this methodology that has been offered by other regulatory agencies, in addition to its long, three decades use by this Commission, I believe that it is appropriate to give this form of the CAPM an equal weighting in the estimation of the ROE.

E. Regulatory Framework and Business Risks

Q. Please summarize Mr. Duah's position regarding the Companies' business risks.

A. Mr. Duah does not perform any comparative analysis of the KEDNY and KEDLI to his proxy companies. Rather, Mr. Duah disagrees with my conclusion that he characterizes as "the Commission's efforts in implementing risk-reducing measures have resulted in no significant difference in business risk between New York and other jurisdictions within his proxy group."[55] In support of his argument, he points to a report from Regulatory Research Associates ("RRA") profile which acknowledges the various regulatory mechanisms available to New York utilities.[56] Mr. Duah also claims that the service quality performance and gas safety benchmarks are industry-wide standards, but fails to provide any evidence that other utilities, or his proxy companies are subject to the same potential negative revenue adjustments.[57] In addition, Mr. Duah dismisses the risks associated with the CLCPA because "the Companies are not likely to be exposed to any CLCPA-related risk during the Rate Year," again without any evidence to support this statement[58]

[55] Prepared Testimony of Kwaku Duah, at 136.
[56] *Id.*, at 139-140.
[57] *Id.*, at Duah at 141.
[58] *Id.*, at 143.

Q. Do you agree with Mr. Duah's conclusion regarding the profile of the Commission by RRA?

A. Mr. Duah provides only New York's RRA's profile, which acknowledges the Commission's constructive regulatory mechanisms, which include multi-year settlements, revenue decoupling mechanisms, and fully forecasted test years, among other provisions. But Mr. Duah has not performed any comparative analysis to demonstrate that these mechanisms are unique to New York, or otherwise reduce the Companies risk profile as compared to the industry or the proxy companies. In fact, RRA agrees that New York's risk profile offers no comparative advantage compared to other jurisdictions. As Mr. Duah explains, RRA assigned a ranking of "Average/2" to New York, which "is consistent with the majority of regulatory jurisdictions across the United States." [59] To be clear, as shown in Exhibit___(KXD-32), Average/2 is the median ranking given by RRA, and there are 18 jurisdictions with a higher ranking, 19 jurisdictions with a lower ranking, and 15 jurisdictions with the same ranking. This puts New York squarely in middle of the range supporting the contention that, even with the risk reducing measures implemented in New York, those measures result is no significant difference in regulatory risk between New York and other jurisdictions. This is consistent with my findings presented in my Direct Testimony. As shown in Exhibit ___ (JCN-6), KEDNY and KEDLI have comparable regulatory protection to the proxy group companies.

[59] *Id.*, at 138-139.

Q. What is your response to Mr. Duah's claim that the service quality performance and gas safety benchmarks are industry-wide standards?

A. While I agree that the service quality performance metrics and gas safety benchmarks are metrics to ensure reliable, safe, and satisfactory services for customers and such objectives are common across the industry, Mr. Duah has provided no evidence that any other utilities are subject to the same penalties embedded directly into a rate plan in a way that is comparable to the Companies' exposure. That is, while other utilities may be held to performance standards, there is no evidence that there is a direct, downside-only penalty in other jurisdictions. It is the punitive mechanism by which the standards are enforced that poses a unique and incremental risk to KEDNY and KEDLI.

Q. Is there any evidence that the risk from such mechanisms has increased from the Companies' last case?

A. Yes, I am advised that there is. For example, Staff's Pipeline Safety Panel ("SPSP") testified that KEDNY and KEDLI can currently incur up to 150 basis points of negative revenue adjustments ("NRA"), and up to 22 basis points of positive revenue adjustments ("PRA"). For the NRAs, that panel recommends that the Commission maintain the current 150 basis points exposure but would reduce the basis points for PRAs from 22 to 14. Similarly, the Staff Customer Service Panel ("SCSP") testified that under current metrics "if performance on any customer service performance indicator metric falls below the specified thresholds", the Companies will

"incur a negative revenue adjustment, or NRA consisting of a total of $11.7 million and $9.9 million in potential NRAs… for KEDNY and KEDLI, respectively, divided among each of their performance measures." SCSP at 54. The Panel, however, proposes to increase some of the customer service standards and dramatically increase the combined total NRA values to $30.9 million and $19.5 million for KEDNY and KEDLI, respectively, sums that Staff concedes "represent a significant dollar value increase…" SCSP at 75-76. Clearly, this is a significant increase in risk that is not factored into Staff's ROE calculation. Similarly, I am informed that the ability of the Companies to earn incentive awards through Earnings Adjustment Mechanisms, or EAMs, for things such as promoting energy efficiency has been truncated in Staff's case. Again, this is a negative development that diminishes the supportiveness of the regulatory framework and Mr. Duah does not factor this into his analysis.

Q. Do you agree with Mr. Duah's assessment that the CLCPA risks are not relevant because the Companies are not likely to be exposed to the risk during the Rate Year?

A. No, I do not. As a preliminary matter, Mr. Duah has provided no explanation as to why the Companies would be insulated from such risks during the rate year. The CLCPA includes aggressive targets to achieve a net zero carbon economy that will require significant action as soon as practicable in order to achieve the policy objectives. In addition, it is entirely possible that policy developments will occur during the rate year,

and such policies or regulatory decisions would affect the Companies' operations and cash flows, which would have an immediate effect on the intrinsic value of the Companies. Nonetheless, the timing is not the relevant issue for equity investors. Equity investments in utility companies represent long-duration investments. As such, even if there is no direct effect of the CLCPA during the Rate Year, the long-term expectations for the CLCPA will affect the cost of equity for investors. That is, consistent with Mr. Duah's DCF model, investors base their return requirements on expected cash flows over a long period of time - 200 years in the case of Mr. Duah's DCF model. To limit risk assessment to those factors that only have an immediate effect on the Rate Year is inconsistent with financial theory.

Q. What are your conclusions regarding the Companies' business risk relative to the proxy group?

A. As discussed in my Direct Testimony, KEDNY and KEDLI have comparable regulatory protection to the proxy group companies. Mr. Duah's acknowledgment of New York's ranking from RRA as "Average/2" confirms that assessment. In my Direct Testimony, I demonstrated that KEDNY and KEDLI have comparable regulatory protection to the proxy group companies, but the magnitude of the Companies' exposure to negative revenues adjustments places the Companies at greater risk than proxy companies on average. Further, despite the regulatory protections in place for the Companies, Mr. Duah points to RRA's note "the Commission

generally authorizes electric and gas return on equity values lower than industry averages."[60] As also demonstrated in my Direct Testimony, it is clear that the Companies' capital expenditure requirements as a percentage of net utility plant are higher than the majority of the proxy group companies. In addition, the uncertainty surrounding the CLCPA creates significant financial and market risk for utility companies as compared to utility companies outside New York. The heightened level of business risk faced by the Companies due to their capital spending plans increases the importance of setting a return for the Companies that is within the range of reasonableness.

V. RESPONSE TO PULP WITNESS YATES

Q. Does PULP Witness. Yates provide any recommendation regarding the Companies' authorized ROE?

A. Based on a comparison of the Companies' authorized ROEs to ROEs calculated by Staff in other cases, Mr. Yates contends that KEDNY and KEDLI's allowed ROE was above the cost of equity over the course of their rate plans from 2008 through 2022.[61] To remedy this illusory "problem," Mr. Yates proposes "an ROE update process for each year of the proposed rate plan," based on the GFP's recommended ROE calculation.[62]

[60] Prepared Testimony of Kwaku Duah, at 139.
[61] Testimony of William D. Yates, at 36-38.
[62] *Id.*, at 42-43.

Q. Are there any issues with Mr. Yates's analysis?

A. Yes, there are several issues with Mr. Yates's analysis that call into question the conclusions he draws from it. First, Mr. Yates appears fundamentally to misunderstand the purpose of incentive regulation and the earnings sharing mechanisms. As the Commission observed more than three decades ago:

> In particular, CPB sees the sharing mechanism - including the 100-basis-point buffer within which the company may retain all earnings - as an incentive for more efficient operations and not as a device for increasing the probability that the target will be met.
>
> * * *
>
> As for the substantive issues, CPB makes a telling point in distinguishing between a mechanism intended to provide incentives and one designed to limit variations from the targeted earnings level. The latter should be symmetric, for shortfalls might be as objectionable as overearnings. The former need not be symmetric, for it simply provides the company the opportunity to retain a portion of any excess earnings, while not relieving it of the risks of shortfalls. The sharing mechanism here is such an incentive device, and ATTCOM has shown no need to modify it.[63]

The mere fact that the Companies' might have "exceeded" an earnings threshold in any given year is an integral part of the incentive regulation scheme inherent in multi-year settlements that can drive efficiencies that ultimately inure to customers.

Second, when Mr. Yates reviews earned returns in relation to his assessment of the effectiveness of earnings sharing mechanisms, he biases his analysis

[63] AT&T Communications of New York, Inc., 28 NYPSC 619 (Apr. 22, 1988).

by excluding years in which the Companies' underearned allowed ROEs.[64] As the ATT order quoted above explains, adjusting for alleged "overearnings" requires remedies for "underearnings." It also obscures the views equity investors have of the earnings potential of the Companies and how they view the cost of equity. Mr. Yates's analysis is incomplete as it does not assess whether shareholders exceeded the returns used to set rates over the full duration of the rate plan. In effect, Mr. Yates's analysis is limited to the one side of a two-sided distribution. That is, shareholders are exposed to both the upside and downside associated with over/under earnings and Mr. Yates overlooks the degree to which shareholders were required to bear losses associated with underearning.

Third, Mr. Yates's analysis assumes that the DPS Staff analyses from other cases are an appropriate proxy for the cost of equity of KEDNY and KEDLI. However, the cost of equity depends on the risk profile of the subject company. As a practical matter, the cost of equity for combination utilities, such as Niagara Mohawk Power Corporation, Consolidated Edison Company of New York, New York State Electric and Gas, Rochester Gas & Electric, Central Hudson Gas & Electric will be different from the cost of equity for a natural gas-only utility such as KEDNY and KEDLI. Specifically, the proxy group composition may be significantly different for the underlying utilities used in each case. Staff has testified on this point in

[64] Testimony of William D. Yates, at 41.

prior cases, recognizing the importance of including natural gas companies in the proxy group for the gas distribution utility, stating that (emphasis added):

> The starting point for establishing the proxy group is confirming that all of the companies in the group are in the same specific industry classification as the target company. This is done in order to identify each company as a true peer within that proxy group. [65]

In this case, Mr. Duah similarly noted "it is important to determine the specific industry classification of the subject company to identify its true peers."[66]

Fourth, as I discuss extensively in my response to Mr. Duah, the cost of equity for regulated utility companies is being affected by several key factors in the current and prospective capital markets. Despite the effect such conditions have on the DCF and CAPM approaches, Mr. Yates effectively adopts Staff's mechanical applications of the same DCF and CAPM formulas established nearly 30 years ago. Mr. Yates's implicit assumption that the same two models, with the same inputs, weighted the same amount would be equally accurate under all conditions is highly unlikely to be accurate.

[65] Case 16-G-0257, Prepared Testimony of the Staff Finance Panel, at 45-46.
[66] Prepared Testimony of Kwaku Duah, at 65.

As a final point, the ROE that Mr. Yates applies to calculate the cost of equity is not what the Commission authorized in the majority of the cases cited. That is, Mr. Yates cites Staff's analytical result, which is often stale at the time of the decision and different from what the Commission authorized as the appropriate ROE. The obvious implication is that the Commission erred in these cases by authorizing an ROE above the cost of equity in each case. New York authorized ROEs over the last 15 years have been low compared to the utility industry overall, yet Mr. Yates's analysis would suggest authorized ROEs should have been below Commission authorizations, which is inconsistent with the *Hope* and *Bluefield* comparable return standards. Mr. Yates is effectively suggesting that KEDNY and KEDLI overearned under the presumption that more than 15 years of precedent on ROE should be relitigated.

Based on these flaws, Mr. Yates's analysis fails to establish that KEDNY and KEDLI ratepayers incurred costs above the cost of equity. Further, past trends are not guarantees of future performance and there is no validity to his assumption that the cost of equity in future rate plan years is likely to be lower (rather than higher) than the ROE determined in this proceeding.

Q. What are the ratemaking implications of Mr. Yates's proposal to include an ROE update process for each year of the proposed rate plan?

A. Mr. Yates is suggesting a solution to a problem that does not exist. His

proposal would undermine the potential for settlement agreements and multi-year rate plans that are a key feature of the New York regulatory environment and introduce both single-issue and retroactive ratemaking. Further, Mr. Yates's proposal would create incremental burdens and costs on stakeholders by introducing an additional administrative process.

Q. Please elaborate on how Mr. Yates's proposal would undermine the potential for settlement agreements for multi-year rate plans.

A. Settlement agreements represent comprehensive rate plans and include agreement on many issues, not just the cost of capital, and represents the inevitable give-and-take required to reach a compromise acceptable to all parties involved. One important "give" of a settlement agreement involves the Companies foregoing the option to file a new rate case in exchange for a reasonable assurance of rate certainty. In doing so, however, a company is exposed to changes in the underlying costs that may deviate from expected levels of inflation, national and international events, and other unanticipated changes such as pandemics. That is, by staying out of rate cases, the utility may not fully recover material amounts of capital expenditures and may be required to absorb losses due to differences between the cost of service established in the rate plan and actual levels of revenue and expense. As such, the option of seeking rate relief has value. By foregoing this option, equity investors must be compensated for the incremental risk they bear as with regard to uncertainty around cost

recovery. Simply put, by requiring updates to the cost of equity each year, a company has less certainty of associated with its revenues, and therefore has less incentive to enter into a settlement agreement.

Q. Please describe how Mr. Yates's proposal constitutes single-issue ratemaking.

A. Mr. Yates has focused on a single element of the Companies' total cost of service in his proposal to update the cost of equity on an annual basis. As described in my Direct Testimony, the Companies' most recent rate plan included a period of historically high inflation, reaching 9.10 percent in June of 2022. The Companies had limited protections for cost recovery associated with inflation during this period due to the constraints of the rate plan. Mr. Yates's proposal to expose the Companies to changes in the cost of equity during the rate plan, but without protections for other macroeconomic drivers, such as inflation, introduces single-issue ratemaking that would introduce greater risk to shareholders requiring higher returns. In addition to adjusting for a single issue, there is an element of retroactivity to his proposal. While resetting the ROE prospectively on an annual basis may escape the onus of being classical "retroactive ratemaking" it, nevertheless, is still rooted in redress of earnings of the past and abnegation of earnings allowances of the past and prior settlement agreements.

Q. What is the effect of Mr. Yates's proposal on the cost of equity for the Companies?

A. As Mr. Duah pointed out in his testimony, RRA affords the New York regulatory environment a ranking of "Average/2" based on the Commission's constructive regulatory mechanisms, which include multi-year settlements. Mr. Yates's proposal would discourage multi-year settlements, and if agreed upon, introduce incremental uncertainty of cost recovery in any agreed upon multi-year settlement. By diminishing the constructive regulatory mechanisms available to New York utilities, this would increase utilities' regulatory risk. Accordingly, with a higher degree of regulatory risk, investors would require a higher return on equity.

Q. What is your conclusion with respect to Mr. Yates's proposed ROE update process for each year of the proposed rate plan?

A. Mr. Yates's flawed analysis of alleged excessive revenues fails to take into account several factors that would affect the measurement of the cost of equity, and the full spectrum of risk to which equity investors are exposed. His proposed ROE update process would undermine the settlement process and expose equity investors to incremental risks that would cause an increase in required returns.

VI. CONCLUSION AND RECOMMENDATION

Q. What is your conclusion regarding a fair return on book equity for KEDNY and KEDLI?

A. My recommended return on equity considers the results of the DCF and CAPM models, summarized in Figure 4 below, updated to reflect market information through August 31, 2023.

Figure 4: Summary of Analytical Results[67]

	Low	Mean	High
DCF	9.69%	9.97%	10.34%
Mean CAPM	11.84%	11.86%	11.87%
67%/33% DCF/CAPM	10.41%	10.60%	10.85%
50%/50% DCF/CAPM	10.77%	10.92%	11.11%

In addition, reasonable modifications to Staff Witness Duah's proposed assumptions result in significant changes in the resulting ROE. Figure 5 summarizes the incremental effects of changes to Mr. Duah's analyses.

[67] Sources: Exhibit__ (JCN-1R) and Exhibit__ (JCN-3R).

Figure 5: Summary of Adjusted Results[68]

Adjustment	Mean DCF Result	Mean Staff CAPM	ROE Result
Staff Estimate	8.52%	10.18%	9.10%
Evenly Weight the DCF/CAPM Analyses	8.52%	10.18%	9.30%
Use the XIRR function in DCF	8.67%	10.18%	9.40%
Apply EPS Growth Rate for Near-Term Growth in the DCF Model	8.85%	10.18%	9.50%
Apply GDP Growth Rate for Long-Term Growth in the DCF Model	9.55%	10.18%	9.80%
Apply Market Risk Premium Estimate based on S&P 500 market return	9.55%	12.12%	10.80%

Based on these analyses, I continue to conclude that the Companies' requested ROE of 9.80 percent is at the low end of this range of reasonableness and is a conservative estimate of the investor-required ROE.

Q. Does this conclude your rebuttal testimony?

A. Yes, it does.

[68] Source: Exhibit__ (JCN-7R).

List of Exhibits

Exhibit__ (JCN-1R)

Exhibit__ (JCN-2R)

Exhibit__ (JCN-3R)

Exhibit__ (JCN-4R)

Exhibit__ (JCN-5R)

Exhibit__ (JCN-6R)

Exhibit__ (JCN-7R)

Exhibit__ (JCN-8R)

Exhibit__ (JCN-9R)

Exhibit__ (JCN-1R)

THREE-MONTH MULTI-STAGE DCF - PROXY GROUP ANALYSIS - MEAN GROWTH RATE

		[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]	[11]	[12]	[13]
								Second Stage Growth						
Company	Ticker	Stock Price	Annualized Dividend	Value Line EPS Growth	Yahoo! Finance EPS Growth	Zacks EPS Growth	First Stage Growth	Year 6	Year 7	Year 8	Year 9	Year 10	Third Stage Growth	ROE
ALLETE, Inc.	ALE	$57.85	$2.71	6.00%	8.10%	8.10%	7.40%	7.07%	6.74%	6.41%	6.08%	5.75%	5.42%	11.27%
American Electric Power Company, Inc.	AEP	$83.22	$3.32	6.00%	5.20%	5.60%	5.60%	5.57%	5.54%	5.51%	5.48%	5.45%	5.42%	9.87%
Atmos Energy Corporation	ATO	$117.92	$2.96	7.00%	7.50%	7.30%	7.27%	6.96%	6.65%	6.34%	6.04%	5.73%	5.42%	8.51%
Avista Corporation	AVA	$37.88	$1.84	6.50%	6.30%	6.30%	6.37%	6.21%	6.05%	5.89%	5.74%	5.58%	5.42%	11.12%
Black Hills Corporation	BKH	$59.29	$2.50	3.00%	5.40%	2.20%	3.53%	3.85%	4.16%	4.48%	4.79%	5.11%	5.42%	9.54%
CMS Energy Corporation	CMS	$59.19	$1.95	6.50%	7.80%	7.80%	7.37%	7.04%	6.72%	6.39%	6.07%	5.74%	5.42%	9.52%
Edison International	EIX	$69.60	$2.95	4.50%	4.53%	3.70%	4.24%	4.44%	4.64%	4.83%	5.03%	5.22%	5.42%	9.76%
Eversource Energy	ES	$69.39	$2.70	6.50%	6.70%	5.70%	6.30%	6.15%	6.01%	5.86%	5.71%	5.57%	5.42%	9.95%
Exelon Corporation	EXC	$40.76	$1.44	NMF	6.30%	6.30%	6.30%	6.15%	6.01%	5.86%	5.71%	5.57%	5.42%	9.53%
NiSource Inc.	NI	$27.18	$1.00	9.50%	6.70%	7.00%	7.73%	7.35%	6.96%	6.58%	6.19%	5.81%	5.42%	10.10%
Northwest Natural Gas Company	NWN	$42.19	$1.94	6.50%	2.80%	3.70%	4.33%	4.51%	4.70%	4.88%	5.06%	5.24%	5.42%	10.17%
NorthWestern Corporation	NWE	$55.70	$2.56	3.50%	3.66%	5.20%	4.12%	4.34%	4.55%	4.77%	4.99%	5.20%	5.42%	10.10%
OGE Energy Corporation	OGE	$35.51	$1.66	6.50%	negative	3.70%	5.10%	5.15%	5.21%	5.26%	5.31%	5.37%	5.42%	10.48%
ONE Gas, Inc.	OGS	$77.54	$2.60	6.50%	5.00%	5.00%	5.50%	5.49%	5.47%	5.46%	5.45%	5.43%	5.42%	9.12%
PPL Corporation	PPL	$26.44	$0.96	8.00%	17.21%	7.40%	10.87%	9.96%	9.05%	8.14%	7.24%	6.33%	5.42%	10.99%
Public Service Enterprise Group Inc.	PEG	$62.07	$2.28	4.00%	5.50%	5.50%	5.00%	5.07%	5.14%	5.21%	5.28%	5.35%	5.42%	9.35%
Spire, Inc.	SR	$62.50	$2.88	8.00%	n/a	4.20%	6.10%	5.99%	5.87%	5.76%	5.65%	5.53%	5.42%	10.74%
Xcel Energy Inc.	XEL	$61.80	$2.08	6.00%	6.30%	6.10%	6.13%	6.01%	5.90%	5.78%	5.66%	5.54%	5.42%	9.29%
Mean														9.97%
Median														9.91%

Notes
[1] Source: Bloomberg Professional, equals 3-month average as of 08/31/2023
[2] Source: Bloomberg Professional
[3] Source: Value Line
[4] Source: Yahoo! Finance
[5] Source: Zacks
[6] Equals Average ([3], [4], [5])
[7] Equals [6] + ([12] - [6]) / 6
[8] Equals [7] + ([12] - [6]) / 6
[9] Equals [8] + ([12] - [6]) / 6
[10] Equals [9] + ([12] - [6]) / 6
[11] Equals [10] + ([12] - [6]) / 6
[12] Source: Exhibit__(JCN-2)
[13] Equals internal rate of return of cash flows for Year 0 through Year 200

THREE-MONTH MULTI-STAGE DCF - PROXY GROUP ANALYSIS - LOW GROWTH RATE

		[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]	[11]	[12]	[13]
								Second Stage Growth						
Company	Ticker	Stock Price	Annualized Dividend	Value Line EPS Growth	Yahoo! Finance EPS Growth	Zacks EPS Growth	First Stage Growth	Year 6	Year 7	Year 8	Year 9	Year 10	Third Stage Growth	ROE
ALLETE, Inc.	ALE	$57.85	$2.71	6.00%	8.10%	8.10%	6.00%	5.90%	5.81%	5.71%	5.61%	5.52%	5.42%	10.80%
American Electric Power Company, Inc.	AEP	$83.22	$3.32	6.00%	5.20%	5.60%	5.20%	5.24%	5.27%	5.31%	5.35%	5.38%	5.42%	9.76%
Atmos Energy Corporation	ATO	$117.92	$2.96	7.00%	7.50%	7.30%	7.00%	6.74%	6.47%	6.21%	5.95%	5.68%	5.42%	8.45%
Avista Corporation	AVA	$37.88	$1.84	6.50%	6.30%	6.30%	6.30%	6.15%	6.01%	5.86%	5.71%	5.57%	5.42%	11.10%
Black Hills Corporation	BKH	$59.29	$2.50	3.00%	5.40%	2.20%	2.20%	2.74%	3.27%	3.81%	4.35%	4.88%	5.42%	9.19%
CMS Energy Corporation	CMS	$59.19	$1.95	6.50%	7.80%	7.80%	6.50%	6.32%	6.14%	5.96%	5.78%	5.60%	5.42%	9.30%
Edison International	EIX	$69.60	$2.95	4.50%	4.53%	3.70%	3.70%	3.99%	4.27%	4.56%	4.85%	5.13%	5.42%	9.61%
Eversource Energy	ES	$69.39	$2.70	6.50%	6.70%	5.70%	5.70%	5.65%	5.61%	5.56%	5.51%	5.47%	5.42%	9.78%
Exelon Corporation	EXC	$40.76	$1.44	NMF	6.30%	6.30%	6.30%	6.15%	6.01%	5.86%	5.71%	5.57%	5.42%	9.53%
NiSource Inc.	NI	$27.18	$1.00	9.50%	6.70%	7.00%	6.70%	6.49%	6.27%	6.06%	5.85%	5.63%	5.42%	9.81%
Northwest Natural Gas Company	NWN	$42.19	$1.94	6.50%	2.80%	3.70%	2.80%	3.24%	3.67%	4.11%	4.55%	4.98%	5.42%	9.72%
NorthWestern Corporation	NWE	$55.70	$2.56	3.50%	3.66%	5.20%	3.50%	3.82%	4.14%	4.46%	4.78%	5.10%	5.42%	9.92%
OGE Energy Corporation	OGE	$35.51	$1.66	6.50%	negative	3.70%	3.70%	3.99%	4.27%	4.56%	4.85%	5.13%	5.42%	10.05%
ONE Gas, Inc.	OGS	$77.54	$2.60	6.50%	5.00%	5.00%	5.00%	5.07%	5.14%	5.21%	5.28%	5.35%	5.42%	9.00%
PPL Corporation	PPL	$26.44	$0.96	8.00%	17.21%	7.40%	7.40%	7.07%	6.74%	6.41%	6.08%	5.75%	5.42%	9.95%
Public Service Enterprise Group Inc.	PEG	$62.07	$2.28	4.00%	5.50%	5.50%	4.00%	4.24%	4.47%	4.71%	4.95%	5.18%	5.42%	9.10%
Spire, Inc.	SR	$62.50	$2.88	8.00%	n/a	4.20%	4.20%	4.40%	4.61%	4.81%	5.01%	5.22%	5.42%	10.14%
Xcel Energy Inc.	XEL	$61.80	$2.08	6.00%	6.30%	6.10%	6.00%	5.90%	5.81%	5.71%	5.61%	5.52%	5.42%	9.26%
Mean														9.69%
Median														9.74%

Notes
[1] Source: Bloomberg Professional, equals 3-month average as of 08/31/2023
[2] Source: Bloomberg Professional
[3] Source: Value Line
[4] Source: Yahoo! Finance
[5] Source: Zacks
[6] Equals Minimum ([3], [4], [5])
[7] Equals [6] + ([12] - [6]) / 6
[8] Equals [7] + ([12] - [6]) / 6
[9] Equals [8] + ([12] - [6]) / 6
[10] Equals [9] + ([12] - [6]) / 6
[11] Equals [10] + ([12] - [6]) / 6
[12] Source: Exhibit__(JCN-2)
[13] Equals internal rate of return of cash flows for Year 0 through Year 200

THREE-MONTH MULTI-STAGE DCF - PROXY GROUP ANALYSIS - HIGH GROWTH RATE

		[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]	[11]	[12]	[13]
								Second Stage Growth						
Company	Ticker	Stock Price	Annualized Dividend	Value Line EPS Growth	Yahoo! Finance EPS Growth	Zacks EPS Growth	First Stage Growth	Year 6	Year 7	Year 8	Year 9	Year 10	Third Stage Growth	ROE
ALLETE, Inc.	ALE	$57.85	$2.71	6.00%	8.10%	8.10%	8.10%	7.65%	7.21%	6.76%	6.31%	5.87%	5.42%	11.52%
American Electric Power Company, Inc.	AEP	$83.22	$3.32	6.00%	5.20%	5.60%	6.00%	5.90%	5.81%	5.71%	5.61%	5.52%	5.42%	9.98%
Atmos Energy Corporation	ATO	$117.92	$2.96	7.00%	7.50%	7.30%	7.50%	7.15%	6.81%	6.46%	6.11%	5.77%	5.42%	8.56%
Avista Corporation	AVA	$37.88	$1.84	6.50%	6.30%	6.30%	6.50%	6.32%	6.14%	5.96%	5.78%	5.60%	5.42%	11.17%
Black Hills Corporation	BKH	$59.29	$2.50	3.00%	5.40%	2.20%	5.40%	5.40%	5.41%	5.41%	5.41%	5.42%	5.42%	10.07%
CMS Energy Corporation	CMS	$59.19	$1.95	6.50%	7.80%	7.80%	7.80%	7.40%	7.01%	6.61%	6.21%	5.82%	5.42%	9.63%
Edison International	EIX	$69.60	$2.95	4.50%	4.53%	3.70%	4.53%	4.68%	4.83%	4.97%	5.12%	5.27%	5.42%	9.84%
Eversource Energy	ES	$69.39	$2.70	6.50%	6.70%	5.70%	6.70%	6.49%	6.27%	6.06%	5.85%	5.63%	5.42%	10.07%
Exelon Corporation	EXC	$40.76	$1.44	NMF	6.30%	6.30%	6.30%	6.15%	6.01%	5.86%	5.71%	5.57%	5.42%	9.53%
NiSource Inc.	NI	$27.18	$1.00	9.50%	6.70%	7.00%	9.50%	8.82%	8.14%	7.46%	6.78%	6.10%	5.42%	10.63%
Northwest Natural Gas Company	NWN	$42.19	$1.94	6.50%	2.80%	3.70%	6.50%	6.32%	6.14%	5.96%	5.78%	5.60%	5.42%	10.86%
NorthWestern Corporation	NWE	$55.70	$2.56	3.50%	3.66%	5.20%	5.20%	5.24%	5.27%	5.31%	5.35%	5.38%	5.42%	10.44%
OGE Energy Corporation	OGE	$35.51	$1.66	6.50%	negative	3.70%	6.50%	6.32%	6.14%	5.96%	5.78%	5.60%	5.42%	10.94%
ONE Gas, Inc.	OGS	$77.54	$2.60	6.50%	5.00%	5.00%	6.50%	6.32%	6.14%	5.96%	5.78%	5.60%	5.42%	9.37%
PPL Corporation	PPL	$26.44	$0.96	8.00%	17.21%	7.40%	17.21%	15.24%	13.28%	11.31%	9.35%	7.38%	5.42%	13.28%
Public Service Enterprise Group Inc.	PEG	$62.07	$2.28	4.00%	5.50%	5.50%	5.50%	5.49%	5.47%	5.46%	5.45%	5.43%	5.42%	9.48%
Spire, Inc.	SR	$62.50	$2.88	8.00%	n/a	4.20%	8.00%	7.57%	7.14%	6.71%	6.28%	5.85%	5.42%	11.38%
Xcel Energy Inc.	XEL	$61.80	$2.08	6.00%	6.30%	6.10%	6.30%	6.15%	6.01%	5.86%	5.71%	5.57%	5.42%	9.33%
Mean														10.34%
Median														10.07%

Notes
[1] Source: Bloomberg Professional, equals 3-month average as of 08/31/2023
[2] Source: Bloomberg Professional
[3] Source: Value Line
[4] Source: Yahoo! Finance
[5] Source: Zacks
[6] Equals Maximum ([3], [4], [5])
[7] Equals [6] + ([12] - [6]) / 6
[8] Equals [7] + ([12] - [6]) / 6
[9] Equals [8] + ([12] - [6]) / 6
[10] Equals [9] + ([12] - [6]) / 6
[11] Equals [10] + ([12] - [6]) / 6
[12] Source: Exhibit__(JCN-2)
[13] Equals internal rate of return of cash flows for Year 0 through Year 200

Exhibit__ (JCN-2R)

CALCULATION OF LONG-TERM GDP GROWTH RATE

Step 1	
Real GDP ($ Billions) [1]	
1929	$ 1,110.2
2022	$ 20,015.4
Compound Annual Growth Rate	**3.16%**
Step 2	
Consumer Price Index (YoY % Change) [2]	
2030-2034	2.00%
Average	2.00%
Consumer Price Index (All-Urban) [3]	
2031	3.62
2050	5.54
Compound Annual Growth Rate	2.26%
GDP Chain-type Price Index (20`1=1.000) [3]	
2031	1.58
2050	2.43
Compound Annual Growth Rate	2.31%
Average Inflation Forecast	**2.19%**
Long-Term GDP Growth Rate	**5.42%**

Notes:
[1] Bureau of Economic Analysis, August 31, 2023
[2] Blue Chip Financial Forecasts, Vol. 42, No.62, June 1, 2023 at 14
[3] Energy Information Administration, Annual Energy Outlook 2023 at Table 20, March 2023

Exhibit__ (JCN-3R)

CAPITAL ASSET PRICING MODEL -- CURRENT RISK-FREE RATE & VL BETA
K = Rf + β (Rm − Rf)

Company	Ticker	[1] Current 3-month average of 30-year U.S. Treasury bond yield	[2] Beta (β)	[3] Market Return (Rm)	[4] Market Risk Premium (Rm − Rf)	[5] ROE (K)
ALLETE, Inc.	ALE	4.04%	0.90	12.99%	8.95%	12.10%
American Electric Power Company, Inc.	AEP	4.04%	0.75	12.99%	8.95%	10.76%
Atmos Energy Corporation	ATO	4.04%	0.85	12.99%	8.95%	11.65%
Avista Corporation	AVA	4.04%	0.90	12.99%	8.95%	12.10%
Black Hills Corporation	BKH	4.04%	1.00	12.99%	8.95%	12.99%
CMS Energy Corporation	CMS	4.04%	0.80	12.99%	8.95%	11.20%
Edison International	EIX	4.04%	1.00	12.99%	8.95%	12.99%
Eversource Energy	ES	4.04%	0.90	12.99%	8.95%	12.10%
Exelon Corporation	EXC	4.04%	NMF	12.99%	8.95%	
NiSource Inc.	NI	4.04%	0.90	12.99%	8.95%	12.10%
Northwest Natural Gas Company	NWN	4.04%	0.80	12.99%	8.95%	11.20%
NorthWestern Corporation	NWE	4.04%	0.95	12.99%	8.95%	12.55%
OGE Energy Corporation	OGE	4.04%	1.00	12.99%	8.95%	12.99%
ONE Gas, Inc.	OGS	4.04%	0.80	12.99%	8.95%	11.20%
PPL Corporation	PPL	4.04%	1.10	12.99%	8.95%	13.89%
Public Service Enterprise Group Inc.	PEG	4.04%	0.95	12.99%	8.95%	12.55%
Spire, Inc.	SR	4.04%	0.85	12.99%	8.95%	11.65%
Xcel Energy Inc.	XEL	4.04%	0.85	12.99%	8.95%	11.65%
Mean			0.90			12.10%
Median			0.90			12.10%

Notes:
[1] Source: Bloomberg Professional
[2] Source: Value Line
[3] Source: Exhibit __ (JCN-5)
[4] Equals [3] - [1]
[5] Equals [1] + [2] x [4]

CAPITAL ASSET PRICING MODEL -- NEAR-TERM PROJECTED RISK-FREE RATE & VL BETA
K = Rf + β (Rm − Rf)

Company	Ticker	[1] Near-term projected 30-year U.S. Treasury bond yield (Q4 2023 - Q4 2024)	[2] Beta (β)	[3] Market Return (Rm)	[4] Market Risk Premium (Rm − Rf)	[5] ROE (K)
ALLETE, Inc.	ALE	4.04%	0.90	12.99%	8.95%	12.10%
American Electric Power Company, Inc.	AEP	4.04%	0.75	12.99%	8.95%	10.76%
Atmos Energy Corporation	ATO	4.04%	0.85	12.99%	8.95%	11.65%
Avista Corporation	AVA	4.04%	0.90	12.99%	8.95%	12.10%
Black Hills Corporation	BKH	4.04%	1.00	12.99%	8.95%	12.99%
CMS Energy Corporation	CMS	4.04%	0.80	12.99%	8.95%	11.20%
Edison International	EIX	4.04%	1.00	12.99%	8.95%	12.99%
Eversource Energy	ES	4.04%	0.90	12.99%	8.95%	12.10%
Exelon Corporation	EXC	4.04%	NMF	12.99%	8.95%	
NiSource Inc.	NI	4.04%	0.90	12.99%	8.95%	12.10%
Northwest Natural Gas Company	NWN	4.04%	0.80	12.99%	8.95%	11.20%
NorthWestern Corporation	NWE	4.04%	0.95	12.99%	8.95%	12.55%
OGE Energy Corporation	OGE	4.04%	1.00	12.99%	8.95%	12.99%
ONE Gas, Inc.	OGS	4.04%	0.80	12.99%	8.95%	11.20%
PPL Corporation	PPL	4.04%	1.10	12.99%	8.95%	13.89%
Public Service Enterprise Group Inc.	PEG	4.04%	0.95	12.99%	8.95%	12.55%
Spire, Inc.	SR	4.04%	0.85	12.99%	8.95%	11.65%
Xcel Energy Inc.	XEL	4.04%	0.85	12.99%	8.95%	11.65%
Mean			0.90			12.10%
Median			0.90			12.10%

Notes:
[1] Source: Blue Chip Financial Forecasts, Vol. 42, No. 9, September 1, 2023 at 2
[2] Source: Value Line
[3] Source: Exhibit __ (JCN-5)
[4] Equals [3] - [1]
[5] Equals [1] + [2] x [4]

CAPITAL ASSET PRICING MODEL -- LONG-TERM PROJECTED RISK-FREE RATE & VL BETA
K = Rf + β (Rm − Rf)

Company	Ticker	[1] Projected 30-year U.S. Treasury bond yield (2025 - 2029) [3]	[2] Beta (β)	[3] Market Return (Rm)	[4] Market Risk Premium (Rm − Rf)	[5] ROE (K)
ALLETE, Inc.	ALE	3.80%	0.90	12.99%	9.19%	12.08%
American Electric Power Company, Inc.	AEP	3.80%	0.75	12.99%	9.19%	10.70%
Atmos Energy Corporation	ATO	3.80%	0.85	12.99%	9.19%	11.62%
Avista Corporation	AVA	3.80%	0.90	12.99%	9.19%	12.08%
Black Hills Corporation	BKH	3.80%	1.00	12.99%	9.19%	12.99%
CMS Energy Corporation	CMS	3.80%	0.80	12.99%	9.19%	11.16%
Edison International	EIX	3.80%	1.00	12.99%	9.19%	12.99%
Eversource Energy	ES	3.80%	0.90	12.99%	9.19%	12.08%
Exelon Corporation	EXC	3.80%	NMF	12.99%	9.19%	
NiSource Inc.	NI	3.80%	0.90	12.99%	9.19%	12.08%
Northwest Natural Gas Company	NWN	3.80%	0.80	12.99%	9.19%	11.16%
NorthWestern Corporation	NWE	3.80%	0.95	12.99%	9.19%	12.53%
OGE Energy Corporation	OGE	3.80%	1.00	12.99%	9.19%	12.99%
ONE Gas, Inc.	OGS	3.80%	0.80	12.99%	9.19%	11.16%
PPL Corporation	PPL	3.80%	1.10	12.99%	9.19%	13.91%
Public Service Enterprise Group Inc.	PEG	3.80%	0.95	12.99%	9.19%	12.53%
Spire, Inc.	SR	3.80%	0.85	12.99%	9.19%	11.62%
Xcel Energy Inc.	XEL	3.80%	0.85	12.99%	9.19%	11.62%
Mean			0.90			12.08%
Median			0.90			12.08%

Notes:
[1] Source: Blue Chip Financial Forecasts, Vol. 42, No. 6, June 1, 2023 at 14
[2] Source: Value Line
[3] Source: Exhibit __ (JCN-5)
[4] Equals [3] - [1]
[5] Equals [1] + [2] x [4]

CAPITAL ASSET PRICING MODEL -- CURRENT RISK-FREE RATE & BLOOMBERG BETA
$K = Rf + \beta (Rm - Rf)$

Company	Ticker	[1] Current 3-month average of 30-year U.S. Treasury bond yield	[2] Beta (β)	[3] Market Return (Rm)	[4] Market Risk Premium (Rm − Rf)	[5] ROE (K)
ALLETE, Inc.	ALE	4.04%	0.82	12.99%	8.95%	11.43%
American Electric Power Company, Inc.	AEP	4.04%	0.76	12.99%	8.95%	10.83%
Atmos Energy Corporation	ATO	4.04%	0.75	12.99%	8.95%	10.75%
Avista Corporation	AVA	4.04%	0.75	12.99%	8.95%	10.75%
Black Hills Corporation	BKH	4.04%	0.90	12.99%	8.95%	12.11%
CMS Energy Corporation	CMS	4.04%	0.75	12.99%	8.95%	10.76%
Edison International	EIX	4.04%	0.85	12.99%	8.95%	11.63%
Eversource Energy	ES	4.04%	0.81	12.99%	8.95%	11.27%
Exelon Corporation	EXC	4.04%	0.86	12.99%	8.95%	11.77%
NiSource Inc.	NI	4.04%	0.82	12.99%	8.95%	11.34%
Northwest Natural Gas Company	NWN	4.04%	0.70	12.99%	8.95%	10.32%
NorthWestern Corporation	NWE	4.04%	0.86	12.99%	8.95%	11.79%
OGE Energy Corporation	OGE	4.04%	0.92	12.99%	8.95%	12.25%
ONE Gas, Inc.	OGS	4.04%	0.78	12.99%	8.95%	11.06%
PPL Corporation	PPL	4.04%	0.95	12.99%	8.95%	12.53%
Public Service Enterprise Group Inc.	PEG	4.04%	0.85	12.99%	8.95%	11.67%
Spire, Inc.	SR	4.04%	0.77	12.99%	8.95%	10.94%
Xcel Energy Inc.	XEL	4.04%	0.74	12.99%	8.95%	10.69%
Mean			0.81			11.33%
Median			0.81			11.31%

Notes:
[1] Source: Bloomberg Professional
[2] Source: Bloomberg Professional
[3] Source: Exhibit __ (JCN-5)
[4] Equals [3] - [1]
[5] Equals [1] + [2] x [4]

CAPITAL ASSET PRICING MODEL -- NEAR-TERM PROJECTED RISK-FREE RATE & BLOOMBERG BETA
K = Rf + β (Rm − Rf)

Company	Ticker	[1] Near-term projected 30-year U.S. Treasury bond yield (Q4 2023 - Q4 2024)	[2] Beta (β)	[3] Market Return (Rm)	[4] Market Risk Premium (Rm − Rf)	[5] ROE (K)
ALLETE, Inc.	ALE	4.04%	0.82	12.99%	8.95%	11.43%
American Electric Power Company, Inc.	AEP	4.04%	0.76	12.99%	8.95%	10.83%
Atmos Energy Corporation	ATO	4.04%	0.75	12.99%	8.95%	10.75%
Avista Corporation	AVA	4.04%	0.75	12.99%	8.95%	10.75%
Black Hills Corporation	BKH	4.04%	0.90	12.99%	8.95%	12.11%
CMS Energy Corporation	CMS	4.04%	0.75	12.99%	8.95%	10.76%
Edison International	EIX	4.04%	0.85	12.99%	8.95%	11.63%
Eversource Energy	ES	4.04%	0.81	12.99%	8.95%	11.27%
Exelon Corporation	EXC	4.04%	0.86	12.99%	8.95%	11.77%
NiSource Inc.	NI	4.04%	0.82	12.99%	8.95%	11.34%
Northwest Natural Gas Company	NWN	4.04%	0.70	12.99%	8.95%	10.32%
NorthWestern Corporation	NWE	4.04%	0.86	12.99%	8.95%	11.79%
OGE Energy Corporation	OGE	4.04%	0.92	12.99%	8.95%	12.25%
ONE Gas, Inc.	OGS	4.04%	0.78	12.99%	8.95%	11.06%
PPL Corporation	PPL	4.04%	0.95	12.99%	8.95%	12.53%
Public Service Enterprise Group Inc.	PEG	4.04%	0.85	12.99%	8.95%	11.67%
Spire, Inc.	SR	4.04%	0.77	12.99%	8.95%	10.94%
Xcel Energy Inc.	XEL	4.04%	0.74	12.99%	8.95%	10.69%
Mean			0.81			11.33%
Median			0.81			11.31%

Notes:
[1] Source: Blue Chip Financial Forecasts, Vol. 42, No. 9, September 1, 2023 at 2
[2] Source: Bloomberg Professional
[3] Source: Exhibit __ (JCN-5)
[4] Equals [3] - [1]
[5] Equals [1] + [2] x [4]

CAPITAL ASSET PRICING MODEL -- LONG-TERM PROJECTED RISK-FREE RATE & BLOOMBERG BETA
K = Rf + β (Rm − Rf)

Company	Ticker	[1] Projected 30-year U.S. Treasury bond yield (2025 - 2029) [3]	[2] Beta (β)	[3] Market Return (Rm)	[4] Market Risk Premium (Rm − Rf)	[5] ROE (K)
ALLETE, Inc.	ALE	3.80%	0.82	12.99%	9.19%	11.38%
American Electric Power Company, Inc.	AEP	3.80%	0.76	12.99%	9.19%	10.77%
Atmos Energy Corporation	ATO	3.80%	0.75	12.99%	9.19%	10.69%
Avista Corporation	AVA	3.80%	0.75	12.99%	9.19%	10.69%
Black Hills Corporation	BKH	3.80%	0.90	12.99%	9.19%	12.09%
CMS Energy Corporation	CMS	3.80%	0.75	12.99%	9.19%	10.70%
Edison International	EIX	3.80%	0.85	12.99%	9.19%	11.60%
Eversource Energy	ES	3.80%	0.81	12.99%	9.19%	11.23%
Exelon Corporation	EXC	3.80%	0.86	12.99%	9.19%	11.74%
NiSource Inc.	NI	3.80%	0.82	12.99%	9.19%	11.30%
Northwest Natural Gas Company	NWN	3.80%	0.70	12.99%	9.19%	10.25%
NorthWestern Corporation	NWE	3.80%	0.86	12.99%	9.19%	11.75%
OGE Energy Corporation	OGE	3.80%	0.92	12.99%	9.19%	12.23%
ONE Gas, Inc.	OGS	3.80%	0.78	12.99%	9.19%	11.00%
PPL Corporation	PPL	3.80%	0.95	12.99%	9.19%	12.52%
Public Service Enterprise Group Inc.	PEG	3.80%	0.85	12.99%	9.19%	11.63%
Spire, Inc.	SR	3.80%	0.77	12.99%	9.19%	10.88%
Xcel Energy Inc.	XEL	3.80%	0.74	12.99%	9.19%	10.63%
Mean			0.81			11.28%
Median			0.81			11.26%

Notes:
[1] Source: Blue Chip Financial Forecasts, Vol. 42, No. 6, June 1, 2023 at 14
[2] Source: Bloomberg Professional
[3] Source: Exhibit __ (JCN-5)
[4] Equals [3] - [1]
[5] Equals [1] + [2] x [4]

ZERO BETA CAPITAL ASSET PRICING MODEL -- CURRENT RISK-FREE RATE & VL BETA
K = Rf + 0.25 (Rm − Rf) + 0.75 (β (Rm − Rf))

Company	Ticker	[1] Current 3-month average of 30-year U.S. Treasury bond yield	[2] Beta (β)	[3] Market Return (Rm)	[4] Market Risk Premium (Rm − Rf)	[5] ROE (K)
Atmos Energy Corporation	ATO	4.04%	0.85	12.99%	8.95%	11.99%
ALLETE, Inc.	ALE	4.04%	0.90	12.99%	8.95%	12.32%
American Electric Power Company, Inc.	AEP	4.04%	0.75	12.99%	8.95%	11.32%
Avista Corporation	AVA	4.04%	0.90	12.99%	8.95%	12.32%
Black Hills Corporation	BKH	4.04%	1.00	12.99%	8.95%	12.99%
CMS Energy Corporation	CMS	4.04%	0.80	12.99%	8.95%	11.65%
Edison International	EIX	4.04%	1.00	12.99%	8.95%	12.99%
Eversource Energy	ES	4.04%	0.90	12.99%	8.95%	12.32%
Exelon Corporation	EXC	4.04%	NMF	12.99%	8.95%	
NiSource Inc.	NI	4.04%	0.90	12.99%	8.95%	12.32%
Northwest Natural Gas Company	NWN	4.04%	0.80	12.99%	8.95%	11.65%
NorthWestern Corporation	NWE	4.04%	0.95	12.99%	8.95%	12.66%
OGE Energy Corporation	OGE	4.04%	1.00	12.99%	8.95%	12.99%
ONE Gas, Inc.	OGS	4.04%	0.80	12.99%	8.95%	11.65%
PPL Corporation	PPL	4.04%	1.10	12.99%	8.95%	13.67%
Public Service Enterprise Group Inc.	PEG	4.04%	0.95	12.99%	8.95%	12.66%
Spire, Inc.	SR	4.04%	0.85	12.99%	8.95%	11.99%
Xcel Energy Inc.	XEL	4.04%	0.85	12.99%	8.95%	11.99%
Mean			0.90			12.32%
Median			0.90			12.32%

Notes:
[1] Source: Bloomberg Professional
[2] Source: Value Line
[3] Source: Exhibit __ (JCN-5)
[4] Equals [3] - [1]
[5] Equals [1] + [2] x [4]

ZERO BETA CAPITAL ASSET PRICING MODEL -- NEAR-TERM PROJECTED RISK-FREE RATE & VL BETA
K = Rf + 0.25 (Rm − Rf) + 0.75 (β (Rm − Rf))

Company	Ticker	[1] Near-term projected 30-year U.S. Treasury bond yield (Q3 2020 - Q3 2021)	[2] Beta (β)	[3] Market Return (Rm)	[4] Market Risk Premium (Rm − Rf)	[5] ROE (K)
Atmos Energy Corporation	ATO	4.04%	0.85	12.99%	8.95%	11.99%
ALLETE, Inc.	ALE	4.04%	0.90	12.99%	8.95%	12.32%
American Electric Power Company, Inc.	AEP	4.04%	0.75	12.99%	8.95%	11.32%
Avista Corporation	AVA	4.04%	0.90	12.99%	8.95%	12.32%
Black Hills Corporation	BKH	4.04%	1.00	12.99%	8.95%	12.99%
CMS Energy Corporation	CMS	4.04%	0.80	12.99%	8.95%	11.65%
Edison International	EIX	4.04%	1.00	12.99%	8.95%	12.99%
Eversource Energy	ES	4.04%	0.90	12.99%	8.95%	12.32%
Exelon Corporation	EXC	4.04%	NMF	12.99%	8.95%	
NiSource Inc.	NI	4.04%	0.90	12.99%	8.95%	12.32%
Northwest Natural Gas Company	NWN	4.04%	0.80	12.99%	8.95%	11.65%
NorthWestern Corporation	NWE	4.04%	0.95	12.99%	8.95%	12.66%
OGE Energy Corporation	OGE	4.04%	1.00	12.99%	8.95%	12.99%
ONE Gas, Inc.	OGS	4.04%	0.80	12.99%	8.95%	11.65%
PPL Corporation	PPL	4.04%	1.10	12.99%	8.95%	13.67%
Public Service Enterprise Group Inc.	PEG	4.04%	0.95	12.99%	8.95%	12.66%
Spire, Inc.	SR	4.04%	0.85	12.99%	8.95%	11.99%
Xcel Energy Inc.	XEL	4.04%	0.85	12.99%	8.95%	11.99%
Mean			0.90			12.32%
Median			0.90			12.32%

Notes:
[1] Source: Blue Chip Financial Forecasts, Vol. 42, No. 9, September 1, 2023 at 2
[2] Source: Value Line
[3] Source: Exhibit __ (JCN-5)
[4] Equals [3] - [1]
[5] Equals [1] + [2] x [4]

ZERO BETA CAPITAL ASSET PRICING MODEL -- LONG-TERM PROJECTED RISK-FREE RATE & VL BETA
K = Rf + 0.25 (Rm − Rf) + 0.75 (β (Rm − Rf))

Company	Ticker	[1] Projected 30-year U.S. Treasury bond yield (2022 - 2026)	[2] Beta (β)	[3] Market Return (Rm)	[4] Market Risk Premium (Rm − Rf)	[5] ROE (K)
Atmos Energy Corporation	ATO	3.80%	0.85	12.99%	9.19%	11.96%
ALLETE, Inc.	ALE	3.80%	0.90	12.99%	9.19%	12.30%
American Electric Power Company, Inc.	AEP	3.80%	0.75	12.99%	9.19%	11.27%
Avista Corporation	AVA	3.80%	0.90	12.99%	9.19%	12.30%
Black Hills Corporation	BKH	3.80%	1.00	12.99%	9.19%	12.99%
CMS Energy Corporation	CMS	3.80%	0.80	12.99%	9.19%	11.62%
Edison International	EIX	3.80%	1.00	12.99%	9.19%	12.99%
Eversource Energy	ES	3.80%	0.90	12.99%	9.19%	12.30%
Exelon Corporation	EXC	3.80%	NMF	12.99%	9.19%	
NiSource Inc.	NI	3.80%	0.90	12.99%	9.19%	12.30%
Northwest Natural Gas Company	NWN	3.80%	0.80	12.99%	9.19%	11.62%
NorthWestern Corporation	NWE	3.80%	0.95	12.99%	9.19%	12.65%
OGE Energy Corporation	OGE	3.80%	1.00	12.99%	9.19%	12.99%
ONE Gas, Inc.	OGS	3.80%	0.80	12.99%	9.19%	11.62%
PPL Corporation	PPL	3.80%	1.10	12.99%	9.19%	13.68%
Public Service Enterprise Group Inc.	PEG	3.80%	0.95	12.99%	9.19%	12.65%
Spire, Inc.	SR	3.80%	0.85	12.99%	9.19%	11.96%
Xcel Energy Inc.	XEL	3.80%	0.85	12.99%	9.19%	11.96%
Mean			0.90			12.30%
Median			0.90			12.30%

Notes:
[1] Source: Blue Chip Financial Forecasts, Vol. 42, No. 6, June 1, 2023 at 14
[2] Source: Value Line
[3] Source: Exhibit __ (JCN-5)
[4] Equals [3] - [1]
[5] Equals [1] + [2] x [4]

ZERO BETA CAPITAL ASSET PRICING MODEL -- CURRENT RISK-FREE RATE & BLOOMBERG BETA
K = Rf + 0.25 (Rm − Rf) + 0.75 (β (Rm − Rf))

Company	Ticker	[1] Current 3-month average of 30-year U.S. Treasury bond yield	[2] Beta (β)	[3] Market Return (Rm)	[4] Market Risk Premium (Rm − Rf)	[5] ROE (K)
Atmos Energy Corporation	ATO	4.04%	0.75	12.99%	8.95%	11.31%
ALLETE, Inc.	ALE	4.04%	0.82	12.99%	8.95%	11.82%
American Electric Power Company, Inc.	AEP	4.04%	0.76	12.99%	8.95%	11.37%
Avista Corporation	AVA	4.04%	0.75	12.99%	8.95%	11.31%
Black Hills Corporation	BKH	4.04%	0.90	12.99%	8.95%	12.33%
CMS Energy Corporation	CMS	4.04%	0.75	12.99%	8.95%	11.32%
Edison International	EIX	4.04%	0.85	12.99%	8.95%	11.97%
Eversource Energy	ES	4.04%	0.81	12.99%	8.95%	11.70%
Exelon Corporation	EXC	4.04%	0.86	12.99%	8.95%	12.08%
NiSource Inc.	NI	4.04%	0.82	12.99%	8.95%	11.76%
Northwest Natural Gas Company	NWN	4.04%	0.70	12.99%	8.95%	10.99%
NorthWestern Corporation	NWE	4.04%	0.86	12.99%	8.95%	12.09%
OGE Energy Corporation	OGE	4.04%	0.92	12.99%	8.95%	12.44%
ONE Gas, Inc.	OGS	4.04%	0.78	12.99%	8.95%	11.54%
PPL Corporation	PPL	4.04%	0.95	12.99%	8.95%	12.65%
Public Service Enterprise Group Inc.	PEG	4.04%	0.85	12.99%	8.95%	12.00%
Spire, Inc.	SR	4.04%	0.77	12.99%	8.95%	11.45%
Xcel Energy Inc.	XEL	4.04%	0.74	12.99%	8.95%	11.27%
Mean			0.81			11.74%
Median			0.81			11.73%

Notes:
[1] Source: Bloomberg Professional
[2] Source: Bloomberg Professional
[3] Source: Exhibit __ (JCN-5)
[4] Equals [3] - [1]
[5] Equals [1] + [2] x [4]

ZERO BETA CAPITAL ASSET PRICING MODEL -- NEAR-TERM PROJECTED RISK-FREE RATE & BLOOMBERG BETA
K = Rf + 0.25 (Rm − Rf) + 0.75 (β (Rm − Rf))

Company	Ticker	[1] Near-term projected 30-year U.S. Treasury bond yield (Q3 2020 - Q3 2021)	[2] Beta (β)	[3] Market Return (Rm)	[4] Market Risk Premium (Rm − Rf)	[5] ROE (K)
Atmos Energy Corporation	ATO	4.04%	0.75	12.99%	8.95%	11.31%
ALLETE, Inc.	ALE	4.04%	0.82	12.99%	8.95%	11.82%
American Electric Power Company, Inc.	AEP	4.04%	0.76	12.99%	8.95%	11.37%
Avista Corporation	AVA	4.04%	0.75	12.99%	8.95%	11.31%
Black Hills Corporation	BKH	4.04%	0.90	12.99%	8.95%	12.33%
CMS Energy Corporation	CMS	4.04%	0.75	12.99%	8.95%	11.32%
Edison International	EIX	4.04%	0.85	12.99%	8.95%	11.97%
Eversource Energy	ES	4.04%	0.81	12.99%	8.95%	11.70%
Exelon Corporation	EXC	4.04%	0.86	12.99%	8.95%	12.08%
NiSource Inc.	NI	4.04%	0.82	12.99%	8.95%	11.75%
Northwest Natural Gas Company	NWN	4.04%	0.70	12.99%	8.95%	10.99%
NorthWestern Corporation	NWE	4.04%	0.86	12.99%	8.95%	12.09%
OGE Energy Corporation	OGE	4.04%	0.92	12.99%	8.95%	12.43%
ONE Gas, Inc.	OGS	4.04%	0.78	12.99%	8.95%	11.54%
PPL Corporation	PPL	4.04%	0.95	12.99%	8.95%	12.65%
Public Service Enterprise Group Inc.	PEG	4.04%	0.85	12.99%	8.95%	12.00%
Spire, Inc.	SR	4.04%	0.77	12.99%	8.95%	11.45%
Xcel Energy Inc.	XEL	4.04%	0.74	12.99%	8.95%	11.27%
Mean			0.81			11.74%
Median			0.81			11.73%

Notes:
[1] Source: Blue Chip Financial Forecasts, Vol. 42, No. 9, September 1, 2023 at 2
[2] Source: Bloomberg Professional
[3] Source: Exhibit __ (JCN-5)
[4] Equals [3] - [1]
[5] Equals [1] + [2] x [4]

ZERO BETA CAPITAL ASSET PRICING MODEL -- LONG-TERM PROJECTED RISK-FREE RATE & BLOOMBERG BETA
K = Rf + 0.25 (Rm − Rf) + 0.75 (β (Rm − Rf))

Company	Ticker	[1] Projected 30-year U.S. Treasury bond yield (2022 - 2026)	[2] Beta (β)	[3] Market Return (Rm)	[4] Market Risk Premium (Rm − Rf)	[5] ROE (K)
Atmos Energy Corporation	ATO	3.80%	0.75	12.99%	9.19%	11.27%
ALLETE, Inc.	ALE	3.80%	0.82	12.99%	9.19%	11.79%
American Electric Power Company, Inc.	AEP	3.80%	0.76	12.99%	9.19%	11.33%
Avista Corporation	AVA	3.80%	0.75	12.99%	9.19%	11.27%
Black Hills Corporation	BKH	3.80%	0.90	12.99%	9.19%	12.32%
CMS Energy Corporation	CMS	3.80%	0.75	12.99%	9.19%	11.27%
Edison International	EIX	3.80%	0.85	12.99%	9.19%	11.95%
Eversource Energy	ES	3.80%	0.81	12.99%	9.19%	11.67%
Exelon Corporation	EXC	3.80%	0.86	12.99%	9.19%	12.05%
NiSource Inc.	NI	3.80%	0.82	12.99%	9.19%	11.72%
Northwest Natural Gas Company	NWN	3.80%	0.70	12.99%	9.19%	10.93%
NorthWestern Corporation	NWE	3.80%	0.86	12.99%	9.19%	12.06%
OGE Energy Corporation	OGE	3.80%	0.92	12.99%	9.19%	12.42%
ONE Gas, Inc.	OGS	3.80%	0.78	12.99%	9.19%	11.50%
PPL Corporation	PPL	3.80%	0.95	12.99%	9.19%	12.64%
Public Service Enterprise Group Inc.	PEG	3.80%	0.85	12.99%	9.19%	11.97%
Spire, Inc.	SR	3.80%	0.77	12.99%	9.19%	11.41%
Xcel Energy Inc.	XEL	3.80%	0.74	12.99%	9.19%	11.22%
Mean			0.81			11.71%
Median			0.81			11.70%

Notes:
[1] Source: Blue Chip Financial Forecasts, Vol. 42, No. 6, June 1, 2023 at 14
[2] Source: Bloomberg Professional
[3] Source: Exhibit __ (JCN-5)
[4] Equals [3] - [1]
[5] Equals [1] + [2] x [4]

Exhibit__ (JCN-4R)

MARKET RISK PREMIUM - DERIVED FROM ANALYSTS' LONG-TERM GROWTH ESTIMATES

[1] Estimated Weighted Average Dividend Yield	1.51%
[2] Estimated Weighted Average Long-Term Growth Rate	13.53%
[3] S&P 500 Estimated Required Market Return	15.14%

STANDARD AND POOR'S 500 INDEX

Name	Ticker	[4] Weight in Index	[5] Current Dividend Yield	[6] Cap-Weighted Dividend Yield	[7] Bloomberg Long-Term Growth Est.	[8] Value Line Long-Term Growth Est.	[9] Cap-Weighted Long-Term Growth Est.
LyondellBasell Industries NV	LYB	0.08%	5.06%	0.00%	10.50%	2.00%	0.005%
American Express Co	AXP	0.30%	1.52%	0.00%	11.89%	8.50%	0.031%
Verizon Communications Inc	VZ	0.38%	7.46%	0.03%		2.00%	0.008%
Broadcom Inc	AVGO	0.98%	1.99%	0.02%	12.40%	30.00%	0.208%
Boeing Co/The	BA	n/a	n/a	n/a			n/a
Caterpillar Inc	CAT	0.37%	1.85%	0.01%	15.00%	13.50%	0.053%
JPMorgan Chase & Co	JPM	1.09%	2.73%	0.03%	-0.50%	8.50%	0.044%
Chevron Corp	CVX	0.77%	3.75%	0.03%	14.77%	21.50%	0.140%
Coca-Cola Co/The	KO	0.67%	3.08%	0.02%	7.19%	7.50%	0.049%
AbbVie Inc	ABBV	0.67%	4.03%	0.03%	6.50%	2.00%	0.028%
Walt Disney Co/The	DIS	0.39%	n/a	n/a	22.27%	65.00%	0.172%
FleetCor Technologies Inc	FLT	0.05%	n/a	n/a	12.30%	13.50%	0.007%
Extra Space Storage Inc	EXR	0.07%	1.90%	0.00%	2.46%	6.50%	0.003%
Exxon Mobil Corp	XOM	1.14%	3.27%	0.04%	13.89%	7.00%	0.120%
Phillips 66	PSX	0.13%	3.68%	0.00%	13.29%	15.50%	0.019%
General Electric Co	GE	0.32%	0.28%	0.00%	7.00%	26.00%	0.053%
HP Inc	HPQ	0.08%	3.53%	0.00%	-5.48%	12.50%	0.003%
Home Depot Inc/The	HD	0.85%	2.53%	0.02%	3.44%	6.50%	0.042%
Monolithic Power Systems Inc	MPWR	0.06%	0.77%	0.00%		15.00%	0.010%
International Business Machines Corp	IBM	0.34%	4.52%	0.02%	3.35%	3.00%	0.011%
Johnson & Johnson	JNJ	1.00%	2.94%	0.03%	4.00%	5.00%	0.045%
McDonald's Corp	MCD	0.53%	2.16%	0.01%	10.40%	10.50%	0.055%
Merck & Co Inc	MRK	0.71%	2.68%	0.02%	49.31%	8.50%	0.205%
3M Co	MMM	0.15%	5.62%	0.01%	10.00%	4.50%	0.011%
American Water Works Co Inc	AWK	0.07%	2.04%	0.00%	8.00%	3.00%	0.004%
Bank of America Corp	BAC	0.59%	3.35%	0.02%	-5.00%	0.00%	-0.015%
Pfizer Inc	PFE	0.51%	4.64%	0.02%	-3.70%	2.00%	-0.004%
Procter & Gamble Co/The	PG	0.94%	2.44%	0.02%	6.38%	5.50%	0.056%
AT&T Inc	T	0.27%	7.51%	0.02%	2.44%	1.50%	0.005%
Travelers Cos Inc/The	TRV	0.09%	2.48%	0.00%	14.92%	7.50%	0.011%
RTX Corp	RTX	0.32%	2.74%	0.01%	8.88%	15.00%	0.038%
Analog Devices Inc	ADI	0.23%	1.89%	0.00%	6.50%	11.50%	0.021%
Walmart Inc	WMT	1.13%	1.40%	0.02%	8.00%	6.50%	0.082%
Cisco Systems Inc	CSCO	0.60%	2.72%	0.02%	7.50%	8.50%	0.048%
Intel Corp	INTC	0.38%	1.42%	0.01%	5.65%		0.021%
General Motors Co	GM	0.12%	1.07%	0.00%	0.36%	8.50%	0.005%
Microsoft Corp	MSFT	6.26%	0.83%	0.05%	16.62%	12.50%	0.912%
Dollar General Corp	DG	0.08%	1.70%	0.00%	-0.10%	5.50%	0.002%
Cigna Group/The	CI	0.21%	1.78%	0.00%	9.80%	10.00%	0.021%
Kinder Morgan Inc	KMI	0.10%	6.56%	0.01%	2.00%	17.50%	0.010%
Citigroup Inc	C	0.20%	5.13%	0.01%	-8.06%	3.50%	-0.005%
American International Group Inc	AIG	0.11%	2.46%	0.00%	10.00%	4.00%	0.007%
Altria Group Inc	MO	0.20%	8.86%	0.02%	6.00%	6.00%	0.012%
HCA Healthcare Inc	HCA	0.19%	0.87%	0.00%	7.58%	12.50%	0.019%
International Paper Co	IP	0.03%	5.30%	0.00%	-2.00%	4.50%	0.000%
Hewlett Packard Enterprise Co	HPE	0.06%	2.83%	0.00%	3.34%	7.50%	0.003%
Abbott Laboratories	ABT	0.46%	1.98%	0.01%	2.18%	4.50%	0.015%
Aflac Inc	AFL	0.11%	2.25%	0.00%	5.98%	8.00%	0.008%
Air Products and Chemicals Inc	APD	0.17%	2.37%	0.00%	10.27%	10.50%	0.018%
Royal Caribbean Cruises Ltd	RCL	0.07%	n/a	n/a	124.32%		0.081%
Hess Corp	HES	0.12%	1.13%	0.00%	-23.46%	23.50%	0.000%
Archer-Daniels-Midland Co	ADM	0.11%	2.27%	0.00%	-6.10%	7.50%	0.001%
Automatic Data Processing Inc	ADP	0.27%	1.96%	0.01%	16.00%	11.00%	0.036%
Verisk Analytics Inc	VRSK	0.09%	0.56%	0.00%	11.58%	8.00%	0.009%
AutoZone Inc	AZO	0.12%	n/a	n/a	13.48%	13.00%	0.016%
Avery Dennison Corp	AVY	0.04%	1.72%	0.00%	7.00%	9.50%	0.003%
Enphase Energy Inc	ENPH	0.04%	n/a	n/a	23.17%	27.50%	0.011%
MSCI Inc	MSCI	0.11%	1.02%	0.00%	15.26%	12.50%	0.015%
Ball Corp	BALL	0.04%	1.47%	0.00%	10.30%	13.00%	0.005%
Axon Enterprise Inc	AXON	0.04%	n/a	n/a		24.00%	0.010%
Ceridian HCM Holding Inc	CDAY	n/a	n/a	n/a			n/a
Carrier Global Corp	CARR	0.12%	1.29%	0.00%	10.65%	13.00%	0.015%
Bank of New York Mellon Corp/The	BK	0.09%	3.74%	0.00%	10.00%	7.00%	0.008%
Otis Worldwide Corp	OTIS	0.09%	1.59%	0.00%		11.00%	0.010%
Baxter International Inc	BAX	0.05%	2.86%	0.00%	0.66%	6.00%	0.002%
Becton Dickinson & Co	BDX	0.21%	1.30%	0.00%	9.36%	5.00%	0.015%
Berkshire Hathaway Inc	BRK/B	1.21%	n/a	n/a		6.00%	0.073%

MARKET RISK PREMIUM - DERIVED FROM ANALYSTS' LONG-TERM GROWTH ESTIMATES

[1] Estimated Weighted Average Dividend Yield	1.51%
[2] Estimated Weighted Average Long-Term Growth Rate	13.53%
[3] S&P 500 Estimated Required Market Return	15.14%

STANDARD AND POOR'S 500 INDEX

Name	Ticker	[4] Weight in Index	[5] Current Dividend Yield	[6] Cap-Weighted Dividend Yield	[7] Bloomberg Long-Term Growth Est.	[8] Value Line Long-Term Growth Est.	[9] Cap-Weighted Long-Term Growth Est.
Best Buy Co Inc	BBY	0.04%	4.81%	0.00%	3.21%	3.00%	0.001%
Boston Scientific Corp	BSX	0.20%	n/a	n/a	12.10%	13.00%	0.025%
Bristol-Myers Squibb Co	BMY	0.33%	3.70%	0.01%	3.10%		0.010%
Brown-Forman Corp	BF/B	0.05%	1.24%	0.00%	7.04%	12.50%	0.005%
Coterra Energy Inc	CTRA	0.05%	2.84%	0.00%	23.02%		0.013%
Campbell Soup Co	CPB	0.03%	3.55%	0.00%	3.06%	5.00%	0.001%
Hilton Worldwide Holdings Inc	HLT	0.10%	0.40%	0.00%	17.14%		0.017%
Carnival Corp	CCL	n/a	n/a	n/a			n/a
Qorvo Inc	QRVO	0.03%	n/a	n/a	2.83%	14.50%	0.002%
UDR Inc	UDR	0.03%	4.21%	0.00%	7.46%	15.50%	0.004%
Clorox Co/The	CLX	0.05%	3.07%	0.00%	17.90%	7.00%	0.006%
Paycom Software Inc	PAYC	0.05%	0.51%	0.00%		19.50%	0.009%
CMS Energy Corp	CMS	0.04%	3.47%	0.00%		6.50%	0.003%
Newell Brands Inc	NWL	n/a	2.65%	n/a			n/a
Colgate-Palmolive Co	CL	0.16%	2.61%	0.00%	7.85%	8.50%	0.013%
EPAM Systems Inc	EPAM	0.04%	n/a	n/a	4.70%	20.50%	0.005%
Comerica Inc	CMA	0.02%	5.90%	0.00%	-6.12%	4.00%	0.000%
Conagra Brands Inc	CAG	0.04%	4.69%	0.00%	1.31%	4.50%	0.001%
Consolidated Edison Inc	ED	0.08%	3.64%	0.00%	4.00%	6.00%	0.004%
Corning Inc	GLW	0.07%	3.41%	0.00%	6.58%	17.50%	0.009%
Cummins Inc	CMI	0.08%	2.92%	0.00%		10.00%	0.008%
Caesars Entertainment Inc	CZR	n/a	n/a	n/a			n/a
Danaher Corp	DHR	0.50%	0.41%	0.00%	1.00%	11.00%	0.030%
Target Corp	TGT	0.15%	3.48%	0.01%	2.51%	12.00%	0.011%
Deere & Co	DE	0.30%	1.31%	0.00%	18.05%	13.50%	0.048%
Dominion Energy Inc	D	0.10%	5.50%	0.01%	0.30%	2.50%	0.001%
Dover Corp	DOV	0.05%	1.38%	0.00%	13.00%	6.50%	0.005%
Alliant Energy Corp	LNT	0.03%	3.61%	0.00%	6.48%	6.50%	0.002%
Steel Dynamics Inc	STLD	0.05%	1.59%	0.00%	-16.45%	2.00%	-0.003%
Duke Energy Corp	DUK	0.18%	4.62%	0.01%	6.10%	5.00%	0.010%
Regency Centers Corp	REG	0.03%	4.18%	0.00%	5.02%	10.50%	0.002%
Eaton Corp PLC	ETN	0.24%	1.49%	0.00%	15.00%	12.00%	0.032%
Ecolab Inc	ECL	0.13%	1.15%	0.00%	16.00%	10.00%	0.018%
Revvity Inc	RVTY	0.04%	0.24%	0.00%	46.45%	-1.50%	0.008%
Emerson Electric Co	EMR	0.14%	2.12%	0.00%	11.80%	6.50%	0.013%
EOG Resources Inc	EOG	0.19%	2.57%	0.00%	11.33%	15.00%	0.025%
Aon PLC	AON	0.17%	0.74%	0.00%	9.17%	9.50%	0.016%
Entergy Corp	ETR	0.05%	4.49%	0.00%	6.33%	0.50%	0.002%
Equifax Inc	EFX	0.07%	0.75%	0.00%	11.40%	12.00%	0.008%
EQT Corp	EQT	0.05%	1.39%	0.00%	22.19%		0.010%
IQVIA Holdings Inc	IQV	0.10%	n/a	n/a	13.16%	14.50%	0.014%
Gartner Inc	IT	0.07%	n/a	n/a	7.22%	10.50%	0.006%
FedEx Corp	FDX	0.17%	1.93%	0.00%	13.00%	7.00%	0.017%
FMC Corp	FMC	0.03%	2.69%	0.00%	8.00%	10.50%	0.003%
Brown & Brown Inc	BRO	0.05%	0.62%	0.00%	9.00%	6.50%	0.004%
Ford Motor Co	F	0.12%	4.95%	0.01%	10.96%	45.50%	0.035%
NextEra Energy Inc	NEE	0.35%	2.80%	0.01%	8.75%	9.50%	0.032%
Franklin Resources Inc	BEN	0.03%	4.49%	0.00%	-6.13%	2.00%	-0.001%
Garmin Ltd	GRMN	0.05%	2.75%	0.00%	5.60%	5.00%	0.003%
Freeport-McMoRan Inc	FCX	0.15%	1.50%	0.00%		19.00%	0.028%
Dexcom Inc	DXCM	0.10%	n/a	n/a	30.96%		0.031%
General Dynamics Corp	GD	0.16%	2.33%	0.00%	10.90%	9.50%	0.016%
General Mills Inc	GIS	0.10%	3.49%	0.00%	8.00%	4.50%	0.006%
Genuine Parts Co	GPC	0.06%	2.47%	0.00%	8.95%	9.00%	0.005%
Atmos Energy Corp	ATO	0.04%	2.55%	0.00%	7.50%	7.00%	0.003%
WW Grainger Inc	GWW	0.09%	1.04%	0.00%		11.00%	0.010%
Halliburton Co	HAL	0.09%	1.66%	0.00%	23.40%	30.00%	0.024%
L3Harris Technologies Inc	LHX	0.09%	2.56%	0.00%	2.50%	19.50%	0.010%
Healthpeak Properties Inc	PEAK	0.03%	5.83%	0.00%	4.72%	14.50%	0.003%
Insulet Corp	PODD	0.03%	n/a	n/a	36.33%		0.013%
Catalent Inc	CTLT	0.02%	n/a	n/a	12.00%	21.00%	0.004%
Fortive Corp	FTV	0.07%	0.36%	0.00%	7.93%	16.00%	0.009%
Hershey Co/The	HSY	0.08%	2.22%	0.00%	9.50%	9.50%	0.008%
Synchrony Financial	SYF	0.03%	3.10%	0.00%	64.00%	47.00%	0.019%
Hormel Foods Corp	HRL	0.05%	2.85%	0.00%	2.50%	7.50%	0.003%
Arthur J Gallagher & Co	AJG	0.13%	0.95%	0.00%	12.19%	22.00%	0.022%
Mondelez International Inc	MDLZ	0.25%	2.39%	0.01%	8.04%	10.00%	0.022%

MARKET RISK PREMIUM - DERIVED FROM ANALYSTS' LONG-TERM GROWTH ESTIMATES

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STANDARD AND POOR'S 500 INDEX

Name	Ticker	[4] Weight in Index	[5] Current Dividend Yield	[6] Cap-Weighted Dividend Yield	[7] Bloomberg Long-Term Growth Est.	[8] Value Line Long-Term Growth Est.	[9] Cap-Weighted Long-Term Growth Est.
CenterPoint Energy Inc	CNP	0.05%	2.72%	0.00%		6.50%	0.003%
Humana Inc	HUM	0.15%	0.77%	0.00%	12.32%	12.50%	0.018%
Willis Towers Watson PLC	WTW	0.06%	1.63%	0.00%	10.82%	9.50%	0.006%
Illinois Tool Works Inc	ITW	0.19%	2.26%	0.00%	3.94%	11.00%	0.014%
CDW Corp/DE	CDW	0.07%	1.12%	0.00%	13.10%	7.00%	0.007%
Trane Technologies PLC	TT	0.12%	1.46%	0.00%	11.68%	13.00%	0.015%
Interpublic Group of Cos Inc/The	IPG	0.03%	3.80%	0.00%	6.99%	8.50%	0.002%
International Flavors & Fragrances Inc	IFF	0.05%	4.60%	0.00%	-1.16%	8.00%	0.002%
Generac Holdings Inc	GNRC	0.02%	n/a	n/a	4.50%	19.00%	0.002%
NXP Semiconductors NV	NXPI	0.14%	1.97%	0.00%	20.50%	8.50%	0.020%
Kellogg Co	K	0.05%	3.93%	0.00%	4.51%	3.00%	0.002%
Broadridge Financial Solutions Inc	BR	0.06%	1.72%	0.00%		8.50%	0.005%
Kimberly-Clark Corp	KMB	0.11%	3.66%	0.00%	9.71%	7.00%	0.009%
Kimco Realty Corp	KIM	0.03%	4.86%	0.00%	4.65%	11.00%	0.002%
Oracle Corp	ORCL	0.84%	1.33%	0.01%	15.00%	10.00%	0.105%
Kroger Co/The	KR	0.09%	2.50%	0.00%	4.76%	6.00%	0.005%
Lennar Corp	LEN	0.08%	1.26%	0.00%	-3.15%	2.50%	0.000%
Eli Lilly & Co	LLY	1.35%	0.82%	0.01%	23.35%	15.50%	0.263%
Bath & Body Works Inc	BBWI	0.02%	2.17%	0.00%	11.38%	26.50%	0.004%
Charter Communications Inc	CHTR	0.17%	n/a	n/a	15.90%	15.50%	0.026%
Lincoln National Corp	LNC	0.01%	7.01%	0.00%		28.00%	0.003%
Loews Corp	L	0.04%	0.40%	0.00%		25.50%	0.009%
Lowe's Cos Inc	LOW	0.34%	1.91%	0.01%	20.64%	8.00%	0.049%
IDEX Corp	IEX	0.04%	1.13%	0.00%	10.00%	8.00%	0.004%
Marsh & McLennan Cos Inc	MMC	0.25%	1.46%	0.00%	11.25%	9.00%	0.025%
Masco Corp	MAS	0.03%	1.93%	0.00%	6.74%	6.00%	0.002%
S&P Global Inc	SPGI	0.32%	0.92%	0.00%	13.72%	7.50%	0.034%
Medtronic PLC	MDT	0.28%	3.39%	0.01%	3.17%	7.50%	0.015%
Viatris Inc	VTRS	0.03%	4.47%	0.00%	-2.18%		-0.001%
CVS Health Corp	CVS	0.22%	3.71%	0.01%	7.13%	8.50%	0.017%
DuPont de Nemours Inc	DD	0.09%	1.87%	0.00%	12.85%	10.00%	0.010%
Micron Technology Inc	MU	0.20%	0.66%	0.00%	-15.93%	9.50%	-0.006%
Motorola Solutions Inc	MSI	0.12%	1.24%	0.00%		11.00%	0.013%
Cboe Global Markets Inc	CBOE	0.04%	1.47%	0.00%		12.50%	0.005%
Laboratory Corp of America Holdings	LH	0.05%	1.38%	0.00%	-4.73%	1.00%	-0.001%
Newmont Corp	NEM	0.08%	4.06%	0.00%	11.86%	8.00%	0.008%
NIKE Inc	NKE	0.32%	1.34%	0.00%	15.34%	18.00%	0.053%
NiSource Inc	NI	0.03%	3.74%	0.00%	7.50%	9.50%	0.002%
Norfolk Southern Corp	NSC	0.12%	2.63%	0.00%	4.34%	8.50%	0.008%
Principal Financial Group Inc	PFG	0.05%	3.35%	0.00%	7.38%	5.50%	0.003%
Eversource Energy	ES	0.06%	4.23%	0.00%	4.99%	6.50%	0.003%
Northrop Grumman Corp	NOC	0.17%	1.73%	0.00%	4.06%	9.50%	0.011%
Wells Fargo & Co	WFC	0.39%	3.39%	0.01%	13.41%	12.00%	0.049%
Nucor Corp	NUE	0.11%	1.19%	0.00%	-10.89%	2.00%	-0.005%
Occidental Petroleum Corp	OXY	0.14%	1.15%	0.00%	-13.74%	17.00%	0.002%
Omnicom Group Inc	OMC	0.04%	3.46%	0.00%	6.31%	7.00%	0.003%
ONEOK Inc	OKE	0.08%	5.86%	0.00%	7.08%	12.00%	0.007%
Raymond James Financial Inc	RJF	0.06%	1.61%	0.00%		15.00%	0.008%
PG&E Corp	PCG	0.09%	n/a	n/a	8.50%	7.50%	0.007%
Parker-Hannifin Corp	PH	0.14%	1.42%	0.00%	14.56%	14.50%	0.020%
Rollins Inc	ROL	0.05%	1.31%	0.00%	13.72%	10.50%	0.006%
PPL Corp	PPL	0.05%	3.85%	0.00%	7.21%	8.00%	0.004%
ConocoPhillips	COP	0.37%	0.50%	0.00%	-0.50%	9.00%	0.016%
PulteGroup Inc	PHM	0.05%	0.78%	0.00%	-3.91%	2.50%	0.000%
Pinnacle West Capital Corp	PNW	0.02%	4.48%	0.00%	6.46%	2.50%	0.001%
PNC Financial Services Group Inc/The	PNC	0.12%	5.14%	0.01%	-0.12%	7.50%	0.005%
PPG Industries Inc	PPG	0.09%	1.83%	0.00%	11.30%	3.00%	0.006%
Progressive Corp/The	PGR	0.20%	0.30%	0.00%	38.38%	12.00%	0.051%
Public Service Enterprise Group Inc	PEG	0.08%	3.73%	0.00%	6.73%	4.00%	0.004%
Robert Half Inc	RHI	0.02%	2.60%	0.00%	0.78%	9.50%	0.001%
Edison International	EIX	0.07%	4.28%	0.00%	5.35%	4.50%	0.003%
Schlumberger NV	SLB	0.22%	1.70%	0.00%	27.56%	26.00%	0.058%
Charles Schwab Corp/The	SCHW	0.27%	1.69%	0.00%	5.31%	9.00%	0.019%
Sherwin-Williams Co/The	SHW	0.18%	0.89%	0.00%	8.49%	7.00%	0.014%
West Pharmaceutical Services Inc	WST	0.08%	0.19%	0.00%	18.65%	17.00%	0.014%
J M Smucker Co/The	SJM	0.04%	2.93%	0.00%	6.09%	6.00%	0.002%
Snap-on Inc	SNA	0.04%	2.41%	0.00%	4.87%	6.00%	0.002%

MARKET RISK PREMIUM - DERIVED FROM ANALYSTS' LONG-TERM GROWTH ESTIMATES

[1] Estimated Weighted Average Dividend Yield	1.51%
[2] Estimated Weighted Average Long-Term Growth Rate	13.53%
[3] S&P 500 Estimated Required Market Return	15.14%

STANDARD AND POOR'S 500 INDEX

Name	Ticker	[4] Weight in Index	[5] Current Dividend Yield	[6] Cap-Weighted Dividend Yield	[7] Bloomberg Long-Term Growth Est.	[8] Value Line Long-Term Growth Est.	[9] Cap-Weighted Long-Term Growth Est.
AMETEK Inc	AME	0.09%	0.63%	0.00%	9.74%	10.00%	0.009%
Southern Co/The	SO	0.19%	4.13%	0.01%	4.50%	6.50%	0.010%
Truist Financial Corp	TFC	0.10%	6.81%	0.01%	4.13%	6.00%	0.005%
Southwest Airlines Co	LUV	0.05%	2.28%	0.00%	29.08%		0.014%
W R Berkley Corp	WRB	0.04%	0.71%	0.00%	12.50%	15.00%	0.006%
Stanley Black & Decker Inc	SWK	0.04%	3.43%	0.00%		1.00%	0.000%
Public Storage	PSA	0.12%	4.34%	0.01%	3.73%	7.50%	0.007%
Arista Networks Inc	ANET	0.16%	n/a	n/a	19.35%	13.00%	0.025%
Sysco Corp	SYY	0.09%	2.87%	0.00%		18.50%	0.017%
Corteva Inc	CTVA	0.09%	1.27%	0.00%	17.92%	15.50%	0.015%
Texas Instruments Inc	TXN	0.39%	2.95%	0.01%	7.80%	3.50%	0.022%
Textron Inc	TXT	0.04%	0.10%	0.00%	11.73%	16.00%	0.005%
Thermo Fisher Scientific Inc	TMO	0.55%	0.25%	0.00%		9.50%	0.053%
TJX Cos Inc/The	TJX	0.27%	1.44%	0.00%	10.00%	17.00%	0.037%
Globe Life Inc	GL	0.03%	0.81%	0.00%		8.50%	0.002%
Johnson Controls International plc	JCI	0.10%	2.51%	0.00%	15.62%	11.50%	0.014%
Ulta Beauty Inc	ULTA	0.05%	n/a	n/a	6.54%	13.50%	0.005%
Union Pacific Corp	UNP	0.35%	2.36%	0.01%	6.50%	6.50%	0.022%
Keysight Technologies Inc	KEYS	0.06%	n/a	n/a	2.52%	13.00%	0.005%
UnitedHealth Group Inc	UNH	1.13%	1.58%	0.02%	11.90%	12.00%	0.136%
Marathon Oil Corp	MRO	0.04%	1.52%	0.00%	-10.70%	22.50%	0.002%
Bio-Rad Laboratories Inc	BIO	0.02%	n/a	n/a	6.00%	11.50%	0.002%
Ventas Inc	VTR	0.05%	4.12%	0.00%	8.12%	23.50%	0.007%
VF Corp	VFC	0.02%	6.07%	0.00%	11.54%	9.00%	0.002%
Vulcan Materials Co	VMC	0.07%	0.79%	0.00%	21.48%	9.00%	0.011%
Weyerhaeuser Co	WY	0.06%	2.32%	0.00%		5.00%	0.003%
Whirlpool Corp	WHR	0.02%	5.00%	0.00%	-1.35%	-1.50%	0.000%
Williams Cos Inc/The	WMB	0.11%	5.18%	0.01%	3.50%	10.50%	0.008%
Constellation Energy Corp	CEG	0.09%	1.08%	0.00%	23.30%		0.020%
WEC Energy Group Inc	WEC	0.07%	3.71%	0.00%	6.26%	6.00%	0.004%
Adobe Inc	ADBE	0.66%	n/a	n/a	16.90%	11.00%	0.091%
AES Corp/The	AES	0.03%	3.70%	0.00%	9.12%	14.00%	0.004%
Amgen Inc	AMGN	0.35%	3.32%	0.01%	4.00%	6.00%	0.018%
Apple Inc	AAPL	7.55%	0.51%	0.04%	11.00%	10.50%	0.812%
Autodesk Inc	ADSK	0.12%	n/a	n/a	13.86%	10.00%	0.015%
Cintas Corp	CTAS	0.13%	1.07%	0.00%	9.74%	14.00%	0.016%
Comcast Corp	CMCSA	0.49%	2.48%	0.01%	8.68%	8.50%	0.043%
Molson Coors Beverage Co	TAP	0.03%	2.58%	0.00%	7.07%	35.00%	0.007%
KLA Corp	KLAC	0.18%	1.04%	0.00%	9.27%	13.50%	0.020%
Marriott International Inc/MD	MAR	0.16%	1.02%	0.00%	17.05%	17.50%	0.027%
Fiserv Inc	FI	0.19%	n/a	n/a	14.63%	9.50%	0.023%
McCormick & Co Inc/MD	MKC	0.05%	1.90%	0.00%	7.01%	4.50%	0.003%
PACCAR Inc	PCAR	0.11%	1.31%	0.00%	12.00%	5.00%	0.009%
Costco Wholesale Corp	COST	0.63%	0.74%	0.00%	12.46%	10.50%	0.072%
Stryker Corp	SYK	0.28%	1.06%	0.00%	7.07%	7.00%	0.019%
Tyson Foods Inc	TSN	0.04%	3.60%	0.00%	-22.91%	6.00%	-0.003%
Lamb Weston Holdings Inc	LW	0.04%	1.15%	0.00%	12.14%	15.50%	0.005%
Applied Materials Inc	AMAT	0.33%	0.84%	0.00%	3.73%	5.00%	0.014%
American Airlines Group Inc	AAL	0.02%	n/a	n/a	80.75%		0.020%
Cardinal Health Inc	CAH	0.06%	2.29%	0.00%		6.50%	0.004%
Cincinnati Financial Corp	CINF	0.04%	2.84%	0.00%	17.66%	10.50%	0.006%
Paramount Global	PARA	0.02%	1.33%	0.00%	5.71%	1.50%	0.001%
DR Horton Inc	DHI	0.10%	0.84%	0.00%	-8.43%	1.00%	-0.004%
Electronic Arts Inc	EA	0.08%	0.63%	0.00%	5.64%	16.00%	0.009%
Fair Isaac Corp	FICO	0.06%	n/a	n/a		16.00%	0.009%
Expeditors International of Washington Inc	EXPD	0.04%	1.18%	0.00%		10.00%	0.004%
Fastenal Co	FAST	0.08%	2.43%	0.00%		6.50%	0.005%
M&T Bank Corp	MTB	0.05%	4.16%	0.00%	11.10%	6.50%	0.005%
Xcel Energy Inc	XEL	0.08%	3.64%	0.00%	6.14%	6.00%	0.005%
Fifth Third Bancorp	FITB	0.05%	4.97%	0.00%	25.00%	4.50%	0.007%
Gilead Sciences Inc	GILD	0.25%	3.92%	0.01%	0.73%	12.00%	0.016%
Hasbro Inc	HAS	0.03%	3.89%	0.00%	8.64%	8.50%	0.002%
Huntington Bancshares Inc/OH	HBAN	0.04%	5.59%	0.00%	-5.65%	10.50%	0.001%
Welltower Inc	WELL	0.11%	2.94%	0.00%	10.72%	12.00%	0.013%
Biogen Inc	BIIB	0.10%	n/a	n/a	0.06%	-10.50%	-0.005%
Northern Trust Corp	NTRS	0.04%	3.94%	0.00%	13.00%	5.50%	0.004%
Packaging Corp of America	PKG	0.03%	3.35%	0.00%	3.00%	9.00%	0.002%

MARKET RISK PREMIUM - DERIVED FROM ANALYSTS' LONG-TERM GROWTH ESTIMATES

[1] Estimated Weighted Average Dividend Yield	1.51%
[2] Estimated Weighted Average Long-Term Growth Rate	13.53%
[3] S&P 500 Estimated Required Market Return	15.14%

STANDARD AND POOR'S 500 INDEX

Name	Ticker	[4] Weight in Index	[5] Current Dividend Yield	[6] Cap-Weighted Dividend Yield	[7] Bloomberg Long-Term Growth Est.	[8] Value Line Long-Term Growth Est.	[9] Cap-Weighted Long-Term Growth Est.
Paychex Inc	PAYX	0.11%	2.91%	0.00%	7.00%	9.50%	0.009%
QUALCOMM Inc	QCOM	0.33%	2.79%	0.01%		5.50%	0.018%
Ross Stores Inc	ROST	0.11%	1.10%	0.00%	10.00%	14.00%	0.013%
IDEXX Laboratories Inc	IDXX	0.11%	n/a	n/a	17.57%	10.50%	0.015%
Starbucks Corp	SBUX	0.29%	2.18%	0.01%	19.71%	16.00%	0.051%
KeyCorp	KEY	0.03%	7.24%	0.00%	7.53%	7.50%	0.002%
Fox Corp	FOXA	0.02%	1.57%	0.00%	12.00%	8.50%	0.002%
Fox Corp	FOX	0.02%	1.70%	0.00%	12.00%		0.002%
State Street Corp	STT	0.06%	4.02%	0.00%	1.31%	9.00%	0.003%
Norwegian Cruise Line Holdings Ltd	NCLH	n/a	n/a	n/a			n/a
US Bancorp	USB	0.15%	5.26%	0.01%	8.00%	4.00%	0.009%
A O Smith Corp	AOS	0.02%	1.66%	0.00%		9.50%	0.002%
Gen Digital Inc	GEN	0.03%	2.47%	0.00%		10.50%	0.003%
T Rowe Price Group Inc	TROW	0.06%	4.35%	0.00%	-3.36%	2.00%	0.000%
Waste Management Inc	WM	0.16%	1.79%	0.00%	9.80%	6.50%	0.013%
Constellation Brands Inc	STZ	0.12%	1.37%	0.00%	9.73%	5.50%	0.009%
DENTSPLY SIRONA Inc	XRAY	0.02%	1.51%	0.00%	9.78%	12.00%	0.002%
Zions Bancorp NA	ZION	0.01%	4.62%	0.00%	-3.00%	6.50%	0.000%
Alaska Air Group Inc	ALK	0.01%	n/a	n/a	23.98%		0.003%
Invesco Ltd	IVZ	0.02%	5.03%	0.00%	4.26%	6.50%	0.001%
Intuit Inc	INTU	0.39%	0.66%	0.00%	18.84%	14.50%	0.065%
Morgan Stanley	MS	0.36%	3.99%	0.01%	3.76%	7.50%	0.020%
Microchip Technology Inc	MCHP	0.11%	2.00%	0.00%	12.06%	10.00%	0.013%
Chubb Ltd	CB	0.21%	1.71%	0.00%	14.50%	15.00%	0.031%
Hologic Inc	HOLX	0.05%	n/a	n/a	-14.09%	25.00%	0.003%
Citizens Financial Group Inc	CFG	0.03%	5.97%	0.00%	-6.14%	7.50%	0.000%
O'Reilly Automotive Inc	ORLY	0.15%	n/a	n/a	12.13%	12.00%	0.018%
Allstate Corp/The	ALL	0.07%	3.30%	0.00%	-4.00%	10.50%	0.002%
Equity Residential	EQR	0.06%	4.09%	0.00%	5.68%	-5.00%	0.000%
BorgWarner Inc	BWA	0.02%	1.08%	0.00%	5.31%	9.50%	0.002%
Keurig Dr Pepper Inc	KDP	0.12%	2.38%	0.00%	6.35%	12.50%	0.011%
Organon & Co	OGN	0.01%	5.10%	0.00%	7.34%		0.001%
Host Hotels & Resorts Inc	HST	0.03%	3.80%	0.00%		51.00%	0.015%
Incyte Corp	INCY	0.04%	n/a	n/a	65.18%	32.00%	0.018%
Simon Property Group Inc	SPG	0.10%	6.70%	0.01%	2.04%	3.50%	0.003%
Eastman Chemical Co	EMN	0.03%	3.72%	0.00%	5.93%	6.00%	0.002%
AvalonBay Communities Inc	AVB	0.07%	3.59%	0.00%	10.28%	6.00%	0.005%
Prudential Financial Inc	PRU	0.09%	5.28%	0.00%	10.60%	3.00%	0.006%
United Parcel Service Inc	UPS	0.32%	3.83%	0.01%	-3.89%	7.50%	0.006%
Walgreens Boots Alliance Inc	WBA	0.06%	7.59%	0.00%	-6.57%	1.00%	-0.002%
STERIS PLC	STE	0.06%	0.91%	0.00%		10.00%	0.006%
McKesson Corp	MCK	0.14%	0.60%	0.00%	10.03%	9.00%	0.014%
Lockheed Martin Corp	LMT	0.29%	2.68%	0.01%	6.98%	7.00%	0.020%
Cencora Inc	COR	0.09%	1.10%	0.00%	9.44%		0.009%
Capital One Financial Corp	COF	0.10%	2.34%	0.00%	-2.93%	4.00%	0.001%
Waters Corp	WAT	0.04%	n/a	n/a	5.79%	10.00%	0.003%
Nordson Corp	NDSN	0.04%	1.11%	0.00%		9.00%	0.003%
Dollar Tree Inc	DLTR	0.07%	n/a	n/a	7.37%	9.00%	0.006%
Darden Restaurants Inc	DRI	0.05%	3.37%	0.00%	10.79%	15.00%	0.006%
Evergy Inc	EVRG	0.03%	4.46%	0.00%	4.74%	7.50%	0.002%
Match Group Inc	MTCH	0.03%	n/a	n/a	62.00%	16.50%	0.013%
Domino's Pizza Inc	DPZ	0.03%	1.25%	0.00%	13.94%	12.00%	0.005%
NVR Inc	NVR	0.05%	n/a	n/a	-3.60%	1.50%	-0.001%
NetApp Inc	NTAP	0.04%	2.61%	0.00%	7.40%	8.50%	0.003%
DXC Technology Co	DXC	0.01%	n/a	n/a	6.84%	9.00%	0.001%
Old Dominion Freight Line Inc	ODFL	0.12%	0.37%	0.00%	4.45%	9.00%	0.008%
DaVita Inc	DVA	0.02%	n/a	n/a	15.78%	7.00%	0.003%
Hartford Financial Services Group Inc/The	HIG	0.06%	2.37%	0.00%	7.00%	8.00%	0.004%
Iron Mountain Inc	IRM	0.05%	4.09%	0.00%	4.00%	4.00%	0.002%
Estee Lauder Cos Inc/The	EL	0.10%	1.64%	0.00%	8.40%	12.00%	0.010%
Cadence Design Systems Inc	CDNS	0.17%	n/a	n/a	19.00%	12.00%	0.026%
Tyler Technologies Inc	TYL	0.04%	n/a	n/a		10.50%	0.005%
Universal Health Services Inc	UHS	0.02%	0.59%	0.00%	11.82%	6.00%	0.002%
Skyworks Solutions Inc	SWKS	0.04%	2.50%	0.00%	4.99%	4.00%	0.002%
Quest Diagnostics Inc	DGX	0.04%	2.16%	0.00%	-0.67%	4.00%	0.001%
Activision Blizzard Inc	ATVI	0.19%	1.08%	0.00%	7.00%	13.50%	0.019%
Rockwell Automation Inc	ROK	0.09%	1.51%	0.00%	15.59%	9.50%	0.012%

MARKET RISK PREMIUM - DERIVED FROM ANALYSTS' LONG-TERM GROWTH ESTIMATES

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STANDARD AND POOR'S 500 INDEX

Name	Ticker	[4] Weight in Index	[5] Current Dividend Yield	[6] Cap-Weighted Dividend Yield	[7] Bloomberg Long-Term Growth Est.	[8] Value Line Long-Term Growth Est.	[9] Cap-Weighted Long-Term Growth Est.
Kraft Heinz Co/The	KHC	0.10%	4.84%	0.01%	3.92%	6.00%	0.005%
American Tower Corp	AMT	0.22%	3.46%	0.01%	13.29%	5.00%	0.020%
Regeneron Pharmaceuticals Inc	REGN	0.23%	n/a	n/a	1.00%	1.50%	0.003%
Amazon.com Inc	AMZN	3.66%	n/a	n/a	51.21%	19.50%	1.294%
Jack Henry & Associates Inc	JKHY	0.03%	1.33%	0.00%	7.41%	7.00%	0.002%
Ralph Lauren Corp	RL	0.01%	2.57%	0.00%	10.73%	12.50%	0.001%
Boston Properties Inc	BXP	0.03%	5.87%	0.00%	3.79%	-1.00%	0.000%
Amphenol Corp	APH	0.14%	0.95%	0.00%	5.46%	12.50%	0.012%
Howmet Aerospace Inc	HWM	0.05%	0.32%	0.00%	19.27%	12.00%	0.008%
Pioneer Natural Resources Co	PXD	0.14%	3.09%	0.00%	-0.73%	8.50%	0.006%
Valero Energy Corp	VLO	0.12%	3.14%	0.00%	-7.69%	1.50%	-0.004%
Synopsys Inc	SNPS	0.18%	n/a	n/a	16.27%	15.00%	0.028%
Etsy Inc	ETSY	0.02%	n/a	n/a	8.15%	10.00%	0.002%
CH Robinson Worldwide Inc	CHRW	0.03%	2.70%	0.00%	5.00%	6.00%	0.001%
Accenture PLC	ACN	0.53%	1.38%	0.01%	10.00%	12.50%	0.059%
TransDigm Group Inc	TDG	0.13%	n/a	n/a	26.65%	26.00%	0.034%
Yum! Brands Inc	YUM	0.09%	1.87%	0.00%	11.45%	11.50%	0.011%
Prologis Inc	PLD	0.30%	2.80%	0.01%	8.95%	2.50%	0.017%
FirstEnergy Corp	FE	0.05%	4.32%	0.00%	-6.66%	4.00%	-0.001%
VeriSign Inc	VRSN	0.06%	n/a	n/a	12.30%	13.00%	0.007%
Quanta Services Inc	PWR	0.08%	0.15%	0.00%	8.00%	15.00%	0.009%
Henry Schein Inc	HSIC	0.03%	n/a	n/a	5.16%	9.00%	0.002%
Ameren Corp	AEE	0.05%	3.18%	0.00%	6.93%	6.50%	0.004%
ANSYS Inc	ANSS	0.07%	n/a	n/a	11.14%	8.50%	0.007%
FactSet Research Systems Inc	FDS	0.04%	0.90%	0.00%	11.97%	10.50%	0.005%
NVIDIA Corp	NVDA	3.13%	0.03%	0.00%	56.84%	33.50%	1.416%
Sealed Air Corp	SEE	0.01%	2.16%	0.00%	0.93%	9.00%	0.001%
Cognizant Technology Solutions Corp	CTSH	0.09%	1.62%	0.00%	12.00%	8.00%	0.009%
Intuitive Surgical Inc	ISRG	0.28%	n/a	n/a	16.14%	12.50%	0.040%
Take-Two Interactive Software Inc	TTWO	0.06%	n/a	n/a	53.59%		0.033%
Republic Services Inc	RSG	0.12%	1.48%	0.00%	9.26%	12.50%	0.013%
eBay Inc	EBAY	0.06%	2.23%	0.00%	6.50%	9.50%	0.005%
Goldman Sachs Group Inc/The	GS	0.28%	3.36%	0.01%	9.00%	5.00%	0.019%
SBA Communications Corp	SBAC	0.06%	1.51%	0.00%		23.50%	0.015%
Sempra	SRE	0.11%	3.39%	0.00%	3.45%	7.00%	0.006%
Moody's Corp	MCO	0.16%	0.91%	0.00%	13.87%	16.00%	0.024%
ON Semiconductor Corp	ON	0.11%	n/a	n/a	8.50%	18.50%	0.015%
Booking Holdings Inc	BKNG	0.28%	n/a	n/a	20.00%	22.00%	0.060%
F5 Inc	FFIV	0.02%	n/a	n/a	10.19%	10.00%	0.003%
Akamai Technologies Inc	AKAM	0.04%	n/a	n/a	10.00%	5.00%	0.003%
Charles River Laboratories International Inc	CRL	0.03%	n/a	n/a	14.00%	8.00%	0.003%
MarketAxess Holdings Inc	MKTX	0.02%	1.20%	0.00%		10.50%	0.002%
Devon Energy Corp	DVN	0.08%	3.84%	0.00%	-4.00%	10.50%	0.003%
Bio-Techne Corp	TECH	0.03%	0.41%	0.00%		13.00%	0.004%
Alphabet Inc	GOOGL	2.08%	n/a	n/a	18.01%		0.374%
Teleflex Inc	TFX	0.03%	0.64%	0.00%	7.03%	10.00%	0.002%
Bunge Ltd	BG	0.04%	2.32%	0.00%	-5.14%	1.50%	-0.001%
Allegion plc	ALLE	0.03%	1.58%	0.00%	5.43%	10.50%	0.002%
Netflix Inc	NFLX	0.49%	n/a	n/a	32.28%	13.00%	0.112%
Agilent Technologies Inc	A	0.09%	0.74%	0.00%	11.00%	13.50%	0.011%
Warner Bros Discovery Inc	WBD	n/a	n/a	n/a			n/a
Elevance Health Inc	ELV	0.27%	1.34%	0.00%	12.13%	12.50%	0.033%
Trimble Inc	TRMB	0.03%	n/a	n/a		5.50%	0.002%
CME Group Inc	CME	0.19%	2.17%	0.00%	6.14%	7.50%	0.013%
Juniper Networks Inc	JNPR	0.02%	3.02%	0.00%	7.89%	11.00%	0.002%
BlackRock Inc	BLK	0.27%	2.85%	0.01%	9.20%	7.50%	0.022%
DTE Energy Co	DTE	0.05%	3.69%	0.00%		4.50%	0.002%
Nasdaq Inc	NDAQ	0.07%	1.68%	0.00%	2.68%	6.00%	0.003%
Celanese Corp	CE	0.04%	2.22%	0.00%	3.07%	6.50%	0.002%
Philip Morris International Inc	PM	0.38%	5.29%	0.02%	7.99%	5.00%	0.025%
Salesforce Inc	CRM	0.55%	n/a	n/a	21.67%	18.00%	0.110%
Ingersoll Rand Inc	IR	0.07%	0.11%	0.00%		12.00%	0.009%
Roper Technologies Inc	ROP	0.14%	0.55%	0.00%		8.00%	0.011%
Huntington Ingalls Industries Inc	HII	0.02%	2.25%	0.00%	40.00%	10.00%	0.006%
MetLife Inc	MET	0.12%	3.28%	0.00%	13.07%	7.50%	0.013%
Tapestry Inc	TPR	0.02%	4.20%	0.00%	14.00%	12.00%	0.003%
CSX Corp	CSX	0.16%	1.46%	0.00%	3.11%	8.50%	0.009%

MARKET RISK PREMIUM - DERIVED FROM ANALYSTS' LONG-TERM GROWTH ESTIMATES

[1] Estimated Weighted Average Dividend Yield	1.51%
[2] Estimated Weighted Average Long-Term Growth Rate	13.53%
[3] S&P 500 Estimated Required Market Return	15.14%

STANDARD AND POOR'S 500 INDEX

Name	Ticker	[4] Weight in Index	[5] Current Dividend Yield	[6] Cap-Weighted Dividend Yield	[7] Bloomberg Long-Term Growth Est.	[8] Value Line Long-Term Growth Est.	[9] Cap-Weighted Long-Term Growth Est.
Edwards Lifesciences Corp	EW	0.12%	n/a	n/a	10.65%	10.50%	0.013%
Ameriprise Financial Inc	AMP	0.09%	1.60%	0.00%	17.59%	11.00%	0.013%
Zebra Technologies Corp	ZBRA	0.04%	n/a	n/a		1.50%	0.001%
Zimmer Biomet Holdings Inc	ZBH	0.06%	0.81%	0.00%	9.48%	6.50%	0.005%
CBRE Group Inc	CBRE	0.07%	n/a	n/a		8.50%	0.006%
Camden Property Trust	CPT	0.03%	3.72%	0.00%	7.34%	-4.00%	0.000%
Mastercard Inc	MA	0.99%	0.55%	0.01%	18.18%	16.00%	0.169%
CarMax Inc	KMX	0.03%	n/a	n/a	15.54%	-3.50%	0.002%
Intercontinental Exchange Inc	ICE	0.17%	1.42%	0.00%	9.87%	6.00%	0.013%
Fidelity National Information Services Inc	FIS	0.09%	3.72%	0.00%	2.68%	23.50%	0.011%
Chipotle Mexican Grill Inc	CMG	0.14%	n/a	n/a	26.95%	20.00%	0.032%
Wynn Resorts Ltd	WYNN	0.03%	0.99%	0.00%		27.00%	0.008%
Live Nation Entertainment Inc	LYV	n/a	n/a	n/a			n/a
Assurant Inc	AIZ	0.02%	2.01%	0.00%	13.68%	10.50%	0.002%
NRG Energy Inc	NRG	0.02%	4.02%	0.00%	4.03%	-2.50%	0.000%
Regions Financial Corp	RF	0.04%	5.23%	0.00%	2.08%	11.50%	0.003%
Monster Beverage Corp	MNST	0.15%	n/a	n/a	15.05%	11.00%	0.020%
Mosaic Co/The	MOS	0.03%	2.06%	0.00%	38.00%	3.50%	0.007%
Baker Hughes Co	BKR	0.09%	2.21%	0.00%	57.62%		0.054%
Expedia Group Inc	EXPE	0.04%	n/a	n/a	17.50%		0.007%
CF Industries Holdings Inc	CF	0.04%	2.08%	0.00%	44.50%	9.00%	0.010%
Leidos Holdings Inc	LDOS	0.03%	1.48%	0.00%	6.45%	7.00%	0.002%
APA Corp	APA	0.03%	2.28%	0.00%	-4.03%	21.00%	0.003%
Alphabet Inc	GOOG	2.05%	n/a	n/a	18.01%	10.50%	0.292%
First Solar Inc	FSLR	0.05%	n/a	n/a	19.80%	24.50%	0.012%
TE Connectivity Ltd	TEL	0.11%	1.78%	0.00%	3.10%	10.50%	0.007%
Cooper Cos Inc/The	COO	0.05%	0.02%	0.00%	7.00%	12.00%	0.004%
Discover Financial Services	DFS	0.06%	3.11%	0.00%	6.93%	4.00%	0.003%
Linde PLC	LIN	0.49%	1.32%	0.01%	9.20%	8.50%	0.043%
Visa Inc	V	1.01%	0.73%	0.01%	14.91%	13.50%	0.144%
Mid-America Apartment Communities Inc	MAA	0.04%	3.86%	0.00%		-12.50%	-0.005%
Xylem Inc/NY	XYL	0.06%	1.27%	0.00%		6.00%	0.004%
Marathon Petroleum Corp	MPC	0.15%	2.10%	0.00%	32.45%	14.50%	0.034%
Tractor Supply Co	TSCO	0.06%	1.89%	0.00%	10.00%	13.50%	0.007%
Advanced Micro Devices Inc	AMD	0.44%	n/a	n/a	26.26%	25.50%	0.114%
ResMed Inc	RMD	0.06%	1.20%	0.00%	9.21%	9.50%	0.006%
Mettler-Toledo International Inc	MTD	0.07%	n/a	n/a	9.75%	11.00%	0.007%
Jacobs Solutions Inc	J	0.04%	0.77%	0.00%	9.26%	11.00%	0.004%
Copart Inc	CPRT	0.11%	n/a	n/a	10.00%	7.00%	0.009%
VICI Properties Inc	VICI	0.08%	5.06%	0.00%	6.33%	8.00%	0.006%
Albemarle Corp	ALB	0.06%	0.81%	0.00%	31.93%	-4.50%	0.008%
Fortinet Inc	FTNT	0.12%	n/a	n/a	18.00%	24.00%	0.026%
Moderna Inc	MRNA	0.11%	n/a	n/a	-60.35%	-20.00%	-0.044%
Essex Property Trust Inc	ESS	0.04%	3.88%	0.00%	9.80%	-3.00%	0.001%
CoStar Group Inc	CSGP	0.09%	n/a	n/a	20.00%	14.00%	0.015%
Realty Income Corp	O	0.10%	5.47%	0.01%		5.50%	0.006%
Westrock Co	WRK	0.02%	3.36%	0.00%	-6.74%	8.00%	0.000%
Westinghouse Air Brake Technologies Corp	WAB	0.05%	0.60%	0.00%	11.33%	10.50%	0.006%
Pool Corp	POOL	0.04%	1.20%	0.00%	-4.92%	14.00%	0.002%
Western Digital Corp	WDC	0.04%	n/a	n/a	-22.46%	3.00%	-0.004%
PepsiCo Inc	PEP	0.63%	2.84%	0.02%	8.64%	5.50%	0.045%
Diamondback Energy Inc	FANG	0.07%	2.21%	0.00%	8.97%		0.006%
Palo Alto Networks Inc	PANW	0.19%	n/a	n/a	20.50%		0.039%
ServiceNow Inc	NOW	0.31%	n/a	n/a	30.00%	61.00%	0.141%
Church & Dwight Co Inc	CHD	0.06%	1.13%	0.00%	5.85%	6.00%	0.004%
Federal Realty Investment Trust	FRT	0.02%	4.45%	0.00%	6.85%	2.50%	0.001%
MGM Resorts International	MGM	0.04%	n/a	n/a		25.00%	0.010%
American Electric Power Co Inc	AEP	0.10%	4.23%	0.00%		6.00%	0.006%
SolarEdge Technologies Inc	SEDG	0.02%	n/a	n/a	26.90%	27.00%	0.006%
Invitation Homes Inc	INVH	0.05%	3.05%	0.00%	7.96%		0.004%
PTC Inc	PTC	0.04%	n/a	n/a	16.99%	15.00%	0.007%
JB Hunt Transport Services Inc	JBHT	0.05%	0.89%	0.00%	15.00%	9.00%	0.006%
Lam Research Corp	LRCX	0.24%	1.14%	0.00%	12.20%	9.50%	0.026%
Mohawk Industries Inc	MHK	0.02%	n/a	n/a	-1.83%	4.00%	0.000%
GE HealthCare Technologies Inc	GEHC	0.08%	0.17%	0.00%	12.75%		0.011%
Pentair PLC	PNR	0.03%	1.25%	0.00%	6.14%	12.00%	0.003%
Vertex Pharmaceuticals Inc	VRTX	0.23%	n/a	n/a	13.72%	12.00%	0.030%

MARKET RISK PREMIUM - DERIVED FROM ANALYSTS' LONG-TERM GROWTH ESTIMATES

[1] Estimated Weighted Average Dividend Yield	1.51%
[2] Estimated Weighted Average Long-Term Growth Rate	13.53%
[3] S&P 500 Estimated Required Market Return	15.14%

STANDARD AND POOR'S 500 INDEX

		[4]	[5]	[6]	[7]	[8]	[9]
Name	Ticker	Weight in Index	Current Dividend Yield	Cap-Weighted Dividend Yield	Bloomberg Long-Term Growth Est.	Value Line Long-Term Growth Est.	Cap-Weighted Long-Term Growth Est.
Amcor PLC	AMCR	0.04%	5.03%	0.00%	2.20%	13.00%	0.003%
Meta Platforms Inc	META	1.69%	n/a	n/a	27.44%	9.00%	0.308%
T-Mobile US Inc	TMUS	0.41%	n/a	n/a	5.00%	16.00%	0.043%
United Rentals Inc	URI	0.08%	1.24%	0.00%	20.04%	17.00%	0.015%
Honeywell International Inc	HON	0.32%	2.19%	0.01%	9.50%	11.00%	0.033%
Alexandria Real Estate Equities Inc	ARE	0.05%	4.26%	0.00%	4.05%	11.00%	0.004%
Delta Air Lines Inc	DAL	0.07%	0.93%	0.00%	37.89%		0.027%
Seagate Technology Holdings PLC	STX	0.04%	3.96%	0.00%	1.21%	7.00%	0.002%
United Airlines Holdings Inc	UAL	n/a	n/a	n/a			n/a
News Corp	NWS	0.01%	0.91%	0.00%	8.00%		0.001%
Centene Corp	CNC	0.09%	n/a	n/a	8.43%	10.00%	0.008%
Martin Marietta Materials Inc	MLM	0.07%	0.66%	0.00%	19.03%	6.00%	0.009%
Teradyne Inc	TER	0.04%	0.41%	0.00%	15.00%	13.50%	0.006%
PayPal Holdings Inc	PYPL	0.18%	n/a	n/a	15.96%	12.00%	0.025%
Tesla Inc	TSLA	2.11%	n/a	n/a	16.00%	26.00%	0.442%
Arch Capital Group Ltd	ACGL	0.07%	n/a	n/a	14.50%	21.00%	0.013%
Dow Inc	DOW	0.10%	5.13%	0.01%	2.78%	8.50%	0.006%
Everest Group Ltd	EG	0.04%	1.83%	0.00%	33.24%	10.00%	0.009%
Teledyne Technologies Inc	TDY	0.05%	n/a	n/a	6.36%	9.50%	0.004%
News Corp	NWSA	0.02%	0.93%	0.00%	8.00%		0.002%
Exelon Corp	EXC	0.10%	3.59%	0.00%	5.30%		0.005%
Global Payments Inc	GPN	0.08%	0.79%	0.00%	13.63%	13.50%	0.011%
Crown Castle Inc	CCI	0.11%	6.23%	0.01%		7.00%	0.008%
Aptiv PLC	APTV	0.07%	n/a	n/a	12.44%	30.00%	0.016%
Align Technology Inc	ALGN	0.07%	n/a	n/a	17.54%	17.00%	0.013%
Illumina Inc	ILMN	0.07%	n/a	n/a	-32.22%	6.50%	-0.009%
Kenvue Inc	KVUE	n/a	3.47%	n/a			n/a
Targa Resources Corp	TRGP	0.05%	2.32%	0.00%	15.00%		0.007%
LKQ Corp	LKQ	0.04%	2.09%	0.00%		13.00%	0.005%
Zoetis Inc	ZTS	0.23%	0.79%	0.00%	10.91%	9.00%	0.022%
Equinix Inc	EQIX	0.19%	1.75%	0.00%	15.43%	15.00%	0.029%
Digital Realty Trust Inc	DLR	0.10%	3.70%	0.00%	6.59%	-1.00%	0.003%
Molina Healthcare Inc	MOH	0.05%	n/a	n/a	11.74%	11.50%	0.005%
Las Vegas Sands Corp	LVS	n/a	1.46%	n/a			n/a

Notes:
[1] Equals Sum ([6])
[2] Equals Sum ([9])
[3] Equals ([1] x (1 + (0.5 x [2]))) + [2]
[4] Equals weight in S&P 500 based on market capitalization
[5] Source: Bloomberg Professional
[6] Equals [4] x [5]
[7] Source: Bloomberg Professional
[8] Source: Value Line
[9] Equals [4] x (([7] + [8]) / 2)

MARKET RISK PREMIUM - DERIVED FROM MERRILL LYNCH ESTIMATES

Merrill Lynch Cost of Market	Implied	Required
Jun-23	10.70%	11.10%
Jul-23	10.70%	11.10%
Aug-23	10.50%	11.00%

Merrill Lynch Expected Market Return [1]: 10.85%

Sources:
[1] *Merrill Lynch cost of market figure is average of Implied and Required Returns for the 3 months ending August 2023*

Exhibit__ (JCN-5R)

REPLICATED STAFF DCF APPROACH - AS FILED

Company	Beta	Price	EPS 2026-28	DPS 2023	DPS 2024	DPS 2026-28	BPS 2023	BPS 2024	BPS 2026-28	# of Shares 2023	# of Shares 2026-28	DPS Growth 2026-28
ALLETE, Inc.	0.90	$59.70	5.00	2.71	2.79	3.00	49.10	51.25	54.00	58.00	61.00	2.45%
Alliant Energy Corp.	0.85	$53.13	3.80	1.81	1.92	2.29	26.55	27.80	31.90	255.80	257.00	6.05%
Ameren Corp.	0.85	$84.24	5.50	2.52	2.65	3.30	40.20	42.90	55.00	267.00	285.00	7.59%
American Electric Power	0.75	$85.69	6.80	3.35	3.52	4.16	52.60	55.05	62.55	523.00	550.00	5.73%
Avista Corp.	0.90	$40.41	3.00	1.84	1.92	2.15	32.95	34.50	37.25	77.00	85.00	3.84%
Black Hills Corp.	1.00	$61.50	4.50	2.53	2.65	3.01	46.75	48.70	55.00	67.50	71.00	4.34%
Centerpoint Energy	1.10	$29.46	1.95	0.77	0.83	0.95	16.85	17.70	19.50	631.00	634.00	4.60%
CMS Energy Corp.	0.80	$59.89	3.75	1.95	2.04	2.30	25.20	25.45	26.00	292.00	300.00	4.08%
Consolidated Edison,Inc.	0.75	$94.15	6.00	3.24	3.34	3.86	58.70	60.55	66.75	345.00	345.00	4.94%
Dominion Energy, Inc	0.85	$52.46	5.00	2.67	2.75	3.18	32.95	34.85	42.15	842.00	870.00	4.96%
Duke Energy Corp.	0.85	$92.13	7.00	4.06	4.14	4.30	64.50	66.25	70.00	770.00	770.00	1.27%
Edison International	1.00	$69.69	6.45	2.95	3.05	3.65	37.00	40.00	48.50	384.00	390.00	6.17%
Entergy Corp.	0.90	$100.55	6.50	4.30	4.45	5.00	63.25	65.70	73.90	214.00	230.00	3.96%
Evergy Inc.	0.90	$59.37	4.85	2.53	2.61	3.05	42.70	44.10	47.50	230.00	230.00	5.33%
Eversource Energy	0.90	$71.89	5.60	2.70	2.86	3.48	46.40	48.65	55.50	351.50	360.00	6.76%
IDACORP, Inc.	0.80	$105.08	6.30	3.20	3.40	4.15	57.85	60.20	67.00	51.00	53.00	6.87%
NextEra Energy, Inc.	0.95	$74.47	4.40	1.87	2.06	2.74	22.20	23.50	30.00	2025.00	2050.00	9.98%
Northwestern Corporation	0.95	$57.45	4.15	2.56	2.60	2.76	47.50	48.50	52.30	62.00	62.00	2.01%
OGE Energy Corp.	1.00	$36.08	3.15	1.66	1.78	1.85	22.25	23.10	26.00	200.20	200.20	1.29%
Pinnacle West Capital Corp.	0.90	$80.53	5.70	3.48	3.54	3.75	54.10	56.75	62.00	113.50	120.00	1.94%
PPL Corp.	1.05	$27.00	2.10	0.96	1.03	1.26	19.50	20.15	22.45	737.00	738.00	6.95%
Portland General Electric Co.	0.90	$48.54	3.65	1.88	1.98	2.36	33.25	34.80	38.70	97.00	100.00	6.03%
Public Service Enterprise Group, Inc.	0.90	$62.09	4.50	2.28	2.40	2.80	28.75	30.10	35.00	500.00	500.00	5.27%
Sempra Energy	1.00	$148.09	12.35	4.76	5.00	6.10	85.55	90.20	105.65	305.00	300.00	6.85%
Southern Co. (The)	0.90	$71.38	5.15	2.78	2.86	3.10	28.00	29.90	32.25	1070.00	1070.00	2.72%
WEC Energy Group	0.80	$90.35	5.90	3.12	3.33	3.80	37.35	37.90	42.00	315.43	315.43	4.50%
Xcel Energy, Inc.	0.85	$64.61	4.25	2.08	2.22	2.66	31.60	33.15	38.25	550.00	560.00	6.21%
Atmos Energy Corp	0.85	$116.92	7.85	2.96	3.20	3.90	70.20	73.05	79.40	147.00	170.00	6.82%
Northwest Natural Holding Co.	0.80	$43.88	3.15	1.95	1.97	2.00	34.95	34.65	34.40	36.50	40.00	0.51%
ONE Gas Inc.	0.80	$78.99	5.60	2.60	2.72	3.15	52.70	50.90	64.45	55.50	57.00	5.01%
Spire Inc.	0.80	$65.36	5.50	2.88	3.00	3.45	53.40	59.75	67.10	53.00	55.00	4.77%
Median Beta	***0.9000***											
Average	**0.8887**										Median	**4.96%**
Data Source	Exhibit___(KXD-22)										Avg.	**4.83%**

REPLICATED STAFF DCF APPROACH - AS FILED

Company	Retention Rate 2027	Return on Equity 2027	B x R	Increase in Shares	MBR 2023	MBR-1 V Factor	(Accretion /dilutive) Growth S x V	Sustainable Growth	Long-Form ROE	
ALLETE, Inc.	40.00%	9.34%	3.74%	1.27%	1.22	0.22	0.27%	4.01%	8.44%	1
Alliant Energy Corp.	39.74%	12.19%	4.84%	0.12%	2.00	1.00	0.12%	4.96%	8.57%	2
Ameren Corp.	40.00%	10.41%	4.17%	1.64%	2.10	1.10	1.80%	5.97%	9.14%	3
American Electric Power	38.82%	11.10%	4.31%	1.27%	1.63	0.63	0.80%	5.11%	9.17%	4
Avista Corp.	28.33%	8.16%	2.31%	2.50%	1.23	0.23	0.57%	2.88%	7.66%	5
Black Hills Corp.	33.11%	8.35%	2.76%	1.27%	1.32	0.32	0.40%	3.17%	7.52%	6
Centerpoint Energy	51.28%	10.16%	5.21%	0.12%	1.75	0.75	0.09%	5.30%	7.99%	7
CMS Energy Corp.	38.67%	14.47%	5.60%	0.68%	2.38	1.38	0.93%	6.53%	9.65%	8
Consolidated Edison,Inc.	35.67%	9.13%	3.26%	0.00%	1.60	0.60	0.00%	3.26%	6.88%	9
Dominion Energy, Inc	36.40%	12.24%	4.45%	0.82%	1.59	0.59	0.49%	4.94%	10.07%	10
Duke Energy Corp.	38.57%	10.09%	3.89%	0.00%	1.43	0.43	0.00%	3.89%	8.05%	11
Edison International	43.41%	13.73%	5.96%	0.39%	1.88	0.88	0.34%	6.30%	10.54%	12
Entergy Corp.	23.08%	8.97%	2.07%	1.82%	1.59	0.59	1.07%	3.14%	7.58%	13
Evergy Inc.	37.11%	10.34%	3.84%	0.00%	1.39	0.39	0.00%	3.84%	8.30%	14
Eversource Energy	37.86%	10.31%	3.90%	0.60%	1.55	0.55	0.33%	4.23%	8.36%	15
IDACORP, Inc.	34.13%	9.57%	3.27%	0.97%	1.82	0.82	0.79%	4.06%	7.43%	16
NextEra Energy, Inc.	37.73%	15.26%	5.76%	0.31%	3.35	2.35	0.72%	6.48%	9.38%	17
Northwestern Corporation	33.49%	8.03%	2.69%	0.00%	1.21	0.21	0.00%	2.69%	7.09%	18
OGE Energy Corp.	41.27%	12.35%	5.10%	0.00%	1.62	0.62	0.00%	5.10%	9.52%	19
Pinnacle West Capital Corp.	34.21%	9.33%	3.19%	1.40%	1.49	0.49	0.68%	3.88%	8.01%	20
PPL Corp.	40.00%	9.52%	3.81%	0.03%	1.38	0.38	0.01%	3.82%	7.84%	21
Portland General Electric Co.	35.34%	9.60%	3.39%	0.76%	1.46	0.46	0.35%	3.74%	7.96%	22
Public Service Enterprise Group, Inc.	37.78%	13.18%	4.98%	0.00%	2.16	1.16	0.00%	4.98%	8.78%	23
Sempra Energy	50.61%	12.00%	6.07%	0.00%	1.73	0.73	0.00%	6.07%	9.41%	24
Southern Co. (The)	39.81%	16.17%	6.44%	0.00%	2.55	1.55	0.00%	6.44%	10.01%	25
WEC Energy Group	35.59%	14.29%	5.09%	0.00%	2.42	1.42	0.00%	5.09%	8.63%	26
Xcel Energy, Inc.	37.41%	11.38%	4.26%	0.45%	2.04	1.04	0.47%	4.73%	8.20%	27
Atmos Energy Corp	50.32%	10.02%	5.04%	3.70%	1.67	0.67	2.46%	7.51%	10.09%	28
Northwest Natural Holding Co.	36.51%	9.15%	3.34%	2.32%	1.26	0.26	0.59%	3.93%	8.01%	29
ONE Gas Inc.	43.75%	9.03%	3.95%	0.67%	1.50	0.50	0.33%	4.28%	7.71%	30
Spire Inc.	37.27%	8.36%	3.11%	0.93%	1.22	0.22	0.21%	3.32%	7.99%	31
Average	***38.30%***	***10.85%***	***4.19%***	***0.78%***	***172.67%***	***72.67%***	***0.45%***	***4.63%***	***8.52%***	
Median	***37.78%***	***10.16%***	***3.95%***	***0.60%***	***160.40%***	***60.40%***	***0.33%***	***4.28%***	***8.30%***	

REPLICATED STAFF DCF APPROACH - APPLY XIRR FORMULA

Company	Beta	Price	EPS 2026-28	DPS 2023	DPS 2024	DPS 2026-28	BPS 2023	BPS 2024	BPS 2026-28	# of Shares 2023	# of Shares 2026-28	DPS Growth 2026-28
ALLETE, Inc.	0.90	$59.70	5.00	2.71	2.79	3.00	49.10	51.25	54.00	58.00	61.00	2.45%
Alliant Energy Corp.	0.85	$53.13	3.80	1.81	1.92	2.29	26.55	27.80	31.90	255.80	257.00	6.05%
Ameren Corp.	0.85	$84.24	5.50	2.52	2.65	3.30	40.20	42.90	55.00	267.00	285.00	7.59%
American Electric Power	0.75	$85.69	6.80	3.35	3.52	4.16	52.60	55.05	62.55	523.00	550.00	5.73%
Avista Corp.	0.90	$40.41	3.00	1.84	1.92	2.15	32.95	34.50	37.25	77.00	85.00	3.84%
Black Hills Corp.	1.00	$61.50	4.50	2.53	2.65	3.01	46.75	48.70	55.00	67.50	71.00	4.34%
Centerpoint Energy	1.10	$29.46	1.95	0.77	0.83	0.95	16.85	17.70	19.50	631.00	634.00	4.60%
CMS Energy Corp.	0.80	$59.89	3.75	1.95	2.04	2.30	25.20	25.45	26.00	292.00	300.00	4.08%
Consolidated Edison,Inc.	0.75	$94.15	6.00	3.24	3.34	3.86	58.70	60.55	66.75	345.00	345.00	4.94%
Dominion Energy, Inc	0.85	$52.46	5.00	2.67	2.75	3.18	32.95	34.85	42.15	842.00	870.00	4.96%
Duke Energy Corp.	0.85	$92.13	7.00	4.06	4.14	4.30	64.50	66.25	70.00	770.00	770.00	1.27%
Edison International	1.00	$69.69	6.45	2.95	3.05	3.65	37.00	40.00	48.50	384.00	390.00	6.17%
Entergy Corp.	0.90	$100.55	6.50	4.30	4.45	5.00	63.25	65.70	73.90	214.00	230.00	3.96%
Evergy Inc.	0.90	$59.37	4.85	2.53	2.61	3.05	42.70	44.10	47.50	230.00	230.00	5.33%
Eversource Energy	0.90	$71.89	5.60	2.70	2.86	3.48	46.40	48.65	55.50	351.50	360.00	6.76%
IDACORP, Inc.	0.80	$105.08	6.30	3.20	3.40	4.15	57.85	60.20	67.00	51.00	53.00	6.87%
NextEra Energy, Inc.	0.95	$74.47	4.40	1.87	2.06	2.74	22.20	23.50	30.00	2025.00	2050.00	9.98%
Northwestern Corporation	0.95	$57.45	4.15	2.56	2.60	2.76	47.50	48.50	52.30	62.00	62.00	2.01%
OGE Energy Corp.	1.00	$36.08	3.15	1.66	1.78	1.85	22.25	23.10	26.00	200.20	200.20	1.29%
Pinnacle West Capital Corp.	0.90	$80.53	5.70	3.48	3.54	3.75	54.10	56.75	62.00	113.50	120.00	1.94%
PPL Corp.	1.05	$27.00	2.10	0.96	1.03	1.26	19.50	20.15	22.45	737.00	738.00	6.95%
Portland General Electric Co.	0.90	$48.54	3.65	1.88	1.98	2.36	33.25	34.80	38.70	97.00	100.00	6.03%
Public Service Enterprise Group, Inc.	0.90	$62.09	4.50	2.28	2.40	2.80	28.75	30.10	35.00	500.00	500.00	5.27%
Sempra Energy	1.00	$148.09	12.35	4.76	5.00	6.10	85.55	90.20	105.65	305.00	300.00	6.85%
Southern Co. (The)	0.90	$71.38	5.15	2.78	2.86	3.10	28.00	29.90	32.25	1070.00	1070.00	2.72%
WEC Energy Group	0.80	$90.35	5.90	3.12	3.33	3.80	37.35	37.90	42.00	315.43	315.43	4.50%
Xcel Energy, Inc.	0.85	$64.61	4.25	2.08	2.22	2.66	31.60	33.15	38.25	550.00	560.00	6.21%
Atmos Energy Corp	0.85	$116.92	7.85	2.96	3.20	3.90	70.20	73.05	79.40	147.00	170.00	6.82%
Northwest Natural Holding Co.	0.80	$43.88	3.15	1.95	1.97	2.00	34.95	34.65	34.40	36.50	40.00	0.51%
ONE Gas Inc.	0.80	$78.99	5.60	2.60	2.72	3.15	52.70	50.90	64.45	55.50	57.00	5.01%
Spire Inc.	0.80	$65.36	5.50	2.88	3.00	3.45	53.40	59.75	67.10	53.00	55.00	4.77%
Median Beta	***0.9000***											
Average	**0.8887**										Median	**4.96%**
Data Source	Exhibit___(KXD-22)										Avg.	**4.83%**

REPLICATED STAFF DCF APPROACH - APPLY XIRR FORMULA

Company	Retention Rate 2027	Return on Equity 2027	B x R	Increase in Shares	MBR 2023	MBR-1 V Factor	(Accretion /dilutive) Growth S x V	Sustainable Growth	XIRR Formula Long-Form ROE	
ALLETE, Inc.	40.00%	9.34%	3.74%	1.27%	1.22	0.22	0.27%	4.01%	***8.62%***	1
Alliant Energy Corp.	39.74%	12.19%	4.84%	0.12%	2.00	1.00	0.12%	4.96%	***8.72%***	2
Ameren Corp.	40.00%	10.41%	4.17%	1.64%	2.10	1.10	1.80%	5.97%	***9.27%***	3
American Electric Power	38.82%	11.10%	4.31%	1.27%	1.63	0.63	0.80%	5.11%	***9.35%***	4
Avista Corp.	28.33%	8.16%	2.31%	2.50%	1.23	0.23	0.57%	2.88%	***7.84%***	5
Black Hills Corp.	33.11%	8.35%	2.76%	1.27%	1.32	0.32	0.40%	3.17%	***7.68%***	6
Centerpoint Energy	51.28%	10.16%	5.21%	0.12%	1.75	0.75	0.09%	5.30%	***8.09%***	7
CMS Energy Corp.	38.67%	14.47%	5.60%	0.68%	2.38	1.38	0.93%	6.53%	***9.79%***	8
Consolidated Edison,Inc.	35.67%	9.13%	3.26%	0.00%	1.60	0.60	0.00%	3.26%	***7.00%***	9
Dominion Energy, Inc	36.40%	12.24%	4.45%	0.82%	1.59	0.59	0.49%	4.94%	***10.32%***	10
Duke Energy Corp.	38.57%	10.09%	3.89%	0.00%	1.43	0.43	0.00%	3.89%	***8.21%***	11
Edison International	43.41%	13.73%	5.96%	0.39%	1.88	0.88	0.34%	6.30%	***10.76%***	12
Entergy Corp.	23.08%	8.97%	2.07%	1.82%	1.59	0.59	1.07%	3.14%	***7.74%***	13
Evergy Inc.	37.11%	10.34%	3.84%	0.00%	1.39	0.39	0.00%	3.84%	***8.48%***	14
Eversource Energy	37.86%	10.31%	3.90%	0.60%	1.55	0.55	0.33%	4.23%	***8.52%***	15
IDACORP, Inc.	34.13%	9.57%	3.27%	0.97%	1.82	0.82	0.79%	4.06%	***7.55%***	16
NextEra Energy, Inc.	37.73%	15.26%	5.76%	0.31%	3.35	2.35	0.72%	6.48%	***9.50%***	17
Northwestern Corporation	33.49%	8.03%	2.69%	0.00%	1.21	0.21	0.00%	2.69%	***7.24%***	18
OGE Energy Corp.	41.27%	12.35%	5.10%	0.00%	1.62	0.62	0.00%	5.10%	***9.72%***	19
Pinnacle West Capital Corp.	34.21%	9.33%	3.19%	1.40%	1.49	0.49	0.68%	3.88%	***8.17%***	20
PPL Corp.	40.00%	9.52%	3.81%	0.03%	1.38	0.38	0.01%	3.82%	***7.99%***	21
Portland General Electric Co.	35.34%	9.60%	3.39%	0.76%	1.46	0.46	0.35%	3.74%	***8.12%***	22
Public Service Enterprise Group, Inc.	37.78%	13.18%	4.98%	0.00%	2.16	1.16	0.00%	4.98%	***8.94%***	23
Sempra Energy	50.61%	12.00%	6.07%	0.00%	1.73	0.73	0.00%	6.07%	***9.56%***	24
Southern Co. (The)	39.81%	16.17%	6.44%	0.00%	2.55	1.55	0.00%	6.44%	***10.18%***	25
WEC Energy Group	35.59%	14.29%	5.09%	0.00%	2.42	1.42	0.00%	5.09%	***8.78%***	26
Xcel Energy, Inc.	37.41%	11.38%	4.26%	0.45%	2.04	1.04	0.47%	4.73%	***8.33%***	27
Atmos Energy Corp	50.32%	10.02%	5.04%	3.70%	1.67	0.67	2.46%	7.51%	***10.22%***	28
Northwest Natural Holding Co.	36.51%	9.15%	3.34%	2.32%	1.26	0.26	0.59%	3.93%	***8.17%***	29
ONE Gas Inc.	43.75%	9.03%	3.95%	0.67%	1.50	0.50	0.33%	4.28%	***7.84%***	30
Spire Inc.	37.27%	8.36%	3.11%	0.93%	1.22	0.22	0.21%	3.32%	***8.17%***	31
Average	***38.30%***	***10.85%***	***4.19%***	***0.78%***	***172.67%***	***72.67%***	***0.45%***	***4.63%***	***8.67%***	
Median	***37.78%***	***10.16%***	***3.95%***	***0.60%***	***160.40%***	***60.40%***	***0.33%***	***4.28%***	***8.48%***	

REPLICATED STAFF DCF APPROACH - APPLY XIRR FORMULA, EPS GROWTH RATES

Company	Beta	Price	EPS 2026-28	DPS 2023	DPS 2024	DPS 2026-28	BPS 2023	BPS 2024	BPS 2026-28	# of Shares 2023	# of Shares 2026-28	*EPS Growth 2023-2027*
ALLETE, Inc.	0.90	$59.70	5.00	2.71	2.79	3.00	49.10	51.25	54.00	58.00	61.00	***7.40%***
Alliant Energy Corp.	0.85	$53.13	3.80	1.81	1.92	2.29	26.55	27.80	31.90	255.80	257.00	***6.67%***
Ameren Corp.	0.85	$84.24	5.50	2.52	2.65	3.30	40.20	42.90	55.00	267.00	285.00	***6.27%***
American Electric Power	0.75	$85.69	6.80	3.35	3.52	4.16	52.60	55.05	62.55	523.00	550.00	***5.60%***
Avista Corp.	0.90	$40.41	3.00	1.84	1.92	2.15	32.95	34.50	37.25	77.00	85.00	***6.37%***
Black Hills Corp.	1.00	$61.50	4.50	2.53	2.65	3.01	46.75	48.70	55.00	67.50	71.00	***3.53%***
Centerpoint Energy	1.10	$29.46	1.95	0.77	0.83	0.95	16.85	17.70	19.50	631.00	634.00	***7.00%***
CMS Energy Corp.	0.80	$59.89	3.75	1.95	2.04	2.30	25.20	25.45	26.00	292.00	300.00	***7.37%***
Consolidated Edison,Inc.	0.75	$94.15	6.00	3.24	3.34	3.86	58.70	60.55	66.75	345.00	345.00	***4.54%***
Dominion Energy, Inc	0.85	$52.46	5.00	2.67	2.75	3.18	32.95	34.85	42.15	842.00	870.00	***11.17%***
Duke Energy Corp.	0.85	$92.13	7.00	4.06	4.14	4.30	64.50	66.25	70.00	770.00	770.00	***5.61%***
Edison International	1.00	$69.69	6.45	2.95	3.05	3.65	37.00	40.00	48.50	384.00	390.00	***4.24%***
Entergy Corp.	0.90	$100.55	6.50	4.30	4.45	5.00	63.25	65.70	73.90	214.00	230.00	***4.27%***
Evergy Inc.	0.90	$59.37	4.85	2.53	2.61	3.05	42.70	44.10	47.50	230.00	230.00	***5.12%***
Eversource Energy	0.90	$71.89	5.60	2.70	2.86	3.48	46.40	48.65	55.50	351.50	360.00	***6.50%***
IDACORP, Inc.	0.80	$105.08	6.30	3.20	3.40	4.15	57.85	60.20	67.00	51.00	53.00	***4.13%***
NextEra Energy, Inc.	0.95	$74.47	4.40	1.87	2.06	2.74	22.20	23.50	30.00	2025.00	2050.00	***8.90%***
Northwestern Corporation	0.95	$57.45	4.15	2.56	2.60	2.76	47.50	48.50	52.30	62.00	62.00	***4.40%***
OGE Energy Corp.	1.00	$36.08	3.15	1.66	1.78	1.85	22.25	23.10	26.00	200.20	200.20	***5.10%***
Pinnacle West Capital Corp.	0.90	$80.53	5.70	3.48	3.54	3.75	54.10	56.75	62.00	113.50	120.00	***4.97%***
PPL Corp.	1.05	$27.00	2.10	0.96	1.03	1.26	19.50	20.15	22.45	737.00	738.00	***10.87%***
Portland General Electric Co.	0.90	$48.54	3.65	1.88	1.98	2.36	33.25	34.80	38.70	97.00	100.00	***5.63%***
Public Service Enterprise Group, Inc.	0.90	$62.09	4.50	2.28	2.40	2.80	28.75	30.10	35.00	500.00	500.00	***4.57%***
Sempra Energy	1.00	$148.09	12.35	4.76	5.00	6.10	85.55	90.20	105.65	305.00	300.00	***5.35%***
Southern Co. (The)	0.90	$71.38	5.15	2.78	2.86	3.10	28.00	29.90	32.25	1070.00	1070.00	***5.93%***
WEC Energy Group	0.80	$90.35	5.90	3.12	3.33	3.80	37.35	37.90	42.00	315.43	315.43	***5.77%***
Xcel Energy, Inc.	0.85	$64.61	4.25	2.08	2.22	2.66	31.60	33.15	38.25	550.00	560.00	***6.15%***
Atmos Energy Corp	0.85	$116.92	7.85	2.96	3.20	3.90	70.20	73.05	79.40	147.00	170.00	***7.43%***
Northwest Natural Holding Co.	0.80	$43.88	3.15	1.95	1.97	2.00	34.95	34.65	34.40	36.50	40.00	***4.33%***
ONE Gas Inc.	0.80	$78.99	5.60	2.60	2.72	3.15	52.70	50.90	64.45	55.50	57.00	***5.50%***
Spire Inc.	0.80	$65.36	5.50	2.88	3.00	3.45	53.40	59.75	67.10	53.00	55.00	***5.35%***
Median Beta	***0.9000***											
Average	**0.8887**										Median	**5.61%**
Data Source	Exhibit___(KXD-22), Exhibit___(JCN-8R)										Avg.	**6.00%**

REPLICATED STAFF DCF APPROACH - APPLY XIRR FORMULA, EPS GROWTH RATES

Company	Retention Rate 2027	Return on Equity 2027	B x R	Increase in Shares	MBR 2023	MBR-1 V Factor	(Accretion /dilutive) Growth S x V	Sustainable Growth	XIRR Formula Long-Form ROE	
ALLETE, Inc.	40.00%	9.34%	3.74%	1.27%	1.22	0.22	0.27%	4.01%	***9.37%***	1
Alliant Energy Corp.	39.74%	12.19%	4.84%	0.12%	2.00	1.00	0.12%	4.96%	***8.79%***	2
Ameren Corp.	40.00%	10.41%	4.17%	1.64%	2.10	1.10	1.80%	5.97%	***9.20%***	3
American Electric Power	38.82%	11.10%	4.31%	1.27%	1.63	0.63	0.80%	5.11%	***9.35%***	4
Avista Corp.	28.33%	8.16%	2.31%	2.50%	1.23	0.23	0.57%	2.88%	***8.22%***	5
Black Hills Corp.	33.11%	8.35%	2.76%	1.27%	1.32	0.32	0.40%	3.17%	***7.54%***	6
Centerpoint Energy	51.28%	10.16%	5.21%	0.12%	1.75	0.75	0.09%	5.30%	***8.23%***	7
CMS Energy Corp.	38.67%	14.47%	5.60%	0.68%	2.38	1.38	0.93%	6.53%	***10.14%***	8
Consolidated Edison,Inc.	35.67%	9.13%	3.26%	0.00%	1.60	0.60	0.00%	3.26%	***7.02%***	9
Dominion Energy, Inc	36.40%	12.24%	4.45%	0.82%	1.59	0.59	0.49%	4.94%	***11.55%***	10
Duke Energy Corp.	38.57%	10.09%	3.89%	0.00%	1.43	0.43	0.00%	3.89%	***8.83%***	11
Edison International	43.41%	13.73%	5.96%	0.39%	1.88	0.88	0.34%	6.30%	***10.60%***	12
Entergy Corp.	23.08%	8.97%	2.07%	1.82%	1.59	0.59	1.07%	3.14%	***7.81%***	13
Evergy Inc.	37.11%	10.34%	3.84%	0.00%	1.39	0.39	0.00%	3.84%	***8.54%***	14
Eversource Energy	37.86%	10.31%	3.90%	0.60%	1.55	0.55	0.33%	4.23%	***8.51%***	15
IDACORP, Inc.	34.13%	9.57%	3.27%	0.97%	1.82	0.82	0.79%	4.06%	***7.27%***	16
NextEra Energy, Inc.	37.73%	15.26%	5.76%	0.31%	3.35	2.35	0.72%	6.48%	***9.38%***	17
Northwestern Corporation	33.49%	8.03%	2.69%	0.00%	1.21	0.21	0.00%	2.69%	***7.63%***	18
OGE Energy Corp.	41.27%	12.35%	5.10%	0.00%	1.62	0.62	0.00%	5.10%	***10.04%***	19
Pinnacle West Capital Corp.	34.21%	9.33%	3.19%	1.40%	1.49	0.49	0.68%	3.88%	***8.62%***	20
PPL Corp.	40.00%	9.52%	3.81%	0.03%	1.38	0.38	0.01%	3.82%	***8.50%***	21
Portland General Electric Co.	35.34%	9.60%	3.39%	0.76%	1.46	0.46	0.35%	3.74%	***8.09%***	22
Public Service Enterprise Group, Inc.	37.78%	13.18%	4.98%	0.00%	2.16	1.16	0.00%	4.98%	***8.84%***	23
Sempra Energy	50.61%	12.00%	6.07%	0.00%	1.73	0.73	0.00%	6.07%	***9.44%***	24
Southern Co. (The)	39.81%	16.17%	6.44%	0.00%	2.55	1.55	0.00%	6.44%	***10.59%***	25
WEC Energy Group	35.59%	14.29%	5.09%	0.00%	2.42	1.42	0.00%	5.09%	***8.85%***	26
Xcel Energy, Inc.	37.41%	11.38%	4.26%	0.45%	2.04	1.04	0.47%	4.73%	***8.30%***	27
Atmos Energy Corp	50.32%	10.02%	5.04%	3.70%	1.67	0.67	2.46%	7.51%	***10.23%***	28
Northwest Natural Holding Co.	36.51%	9.15%	3.34%	2.32%	1.26	0.26	0.59%	3.93%	***8.71%***	29
ONE Gas Inc.	43.75%	9.03%	3.95%	0.67%	1.50	0.50	0.33%	4.28%	***7.90%***	30
Spire Inc.	37.27%	8.36%	3.11%	0.93%	1.22	0.22	0.21%	3.32%	***8.28%***	31
Average	***38.30%***	***10.85%***	***4.19%***	***0.78%***	***172.67%***	***72.67%***	***0.45%***	***4.63%***	***8.85%***	
Median	***37.78%***	***10.16%***	***3.95%***	***0.60%***	***160.40%***	***60.40%***	***0.33%***	***4.28%***	***8.71%***	

REPLICATED STAFF DCF APPROACH - APPLY XIRR FORMULA, EPS GROWTH RATES, GDP GROWTH RATE

Company	Beta	Price	EPS 2026-28	DPS 2023	DPS 2024	DPS 2026-28	BPS 2023	BPS 2024	BPS 2026-28	# of Shares 2023	# of Shares 2026-28	*EPS Growth 2023-2027*
ALLETE, Inc.	0.90	$59.70	5.00	2.71	2.79	3.00	49.10	51.25	54.00	58.00	61.00	***7.40%***
Alliant Energy Corp.	0.85	$53.13	3.80	1.81	1.92	2.29	26.55	27.80	31.90	255.80	257.00	***6.67%***
Ameren Corp.	0.85	$84.24	5.50	2.52	2.65	3.30	40.20	42.90	55.00	267.00	285.00	***6.27%***
American Electric Power	0.75	$85.69	6.80	3.35	3.52	4.16	52.60	55.05	62.55	523.00	550.00	***5.60%***
Avista Corp.	0.90	$40.41	3.00	1.84	1.92	2.15	32.95	34.50	37.25	77.00	85.00	***6.37%***
Black Hills Corp.	1.00	$61.50	4.50	2.53	2.65	3.01	46.75	48.70	55.00	67.50	71.00	***3.53%***
Centerpoint Energy	1.10	$29.46	1.95	0.77	0.83	0.95	16.85	17.70	19.50	631.00	634.00	***7.00%***
CMS Energy Corp.	0.80	$59.89	3.75	1.95	2.04	2.30	25.20	25.45	26.00	292.00	300.00	***7.37%***
Consolidated Edison,Inc.	0.75	$94.15	6.00	3.24	3.34	3.86	58.70	60.55	66.75	345.00	345.00	***4.54%***
Dominion Energy, Inc	0.85	$52.46	5.00	2.67	2.75	3.18	32.95	34.85	42.15	842.00	870.00	***11.17%***
Duke Energy Corp.	0.85	$92.13	7.00	4.06	4.14	4.30	64.50	66.25	70.00	770.00	770.00	***5.61%***
Edison International	1.00	$69.69	6.45	2.95	3.05	3.65	37.00	40.00	48.50	384.00	390.00	***4.24%***
Entergy Corp.	0.90	$100.55	6.50	4.30	4.45	5.00	63.25	65.70	73.90	214.00	230.00	***4.27%***
Evergy Inc.	0.90	$59.37	4.85	2.53	2.61	3.05	42.70	44.10	47.50	230.00	230.00	***5.12%***
Eversource Energy	0.90	$71.89	5.60	2.70	2.86	3.48	46.40	48.65	55.50	351.50	360.00	***6.50%***
IDACORP, Inc.	0.80	$105.08	6.30	3.20	3.40	4.15	57.85	60.20	67.00	51.00	53.00	***4.13%***
NextEra Energy, Inc.	0.95	$74.47	4.40	1.87	2.06	2.74	22.20	23.50	30.00	2025.00	2050.00	***8.90%***
Northwestern Corporation	0.95	$57.45	4.15	2.56	2.60	2.76	47.50	48.50	52.30	62.00	62.00	***4.40%***
OGE Energy Corp.	1.00	$36.08	3.15	1.66	1.78	1.85	22.25	23.10	26.00	200.20	200.20	***5.10%***
Pinnacle West Capital Corp.	0.90	$80.53	5.70	3.48	3.54	3.75	54.10	56.75	62.00	113.50	120.00	***4.97%***
PPL Corp.	1.05	$27.00	2.10	0.96	1.03	1.26	19.50	20.15	22.45	737.00	738.00	***10.87%***
Portland General Electric Co.	0.90	$48.54	3.65	1.88	1.98	2.36	33.25	34.80	38.70	97.00	100.00	***5.63%***
Public Service Enterprise Group, Inc.	0.90	$62.09	4.50	2.28	2.40	2.80	28.75	30.10	35.00	500.00	500.00	***4.57%***
Sempra Energy	1.00	$148.09	12.35	4.76	5.00	6.10	85.55	90.20	105.65	305.00	300.00	***5.35%***
Southern Co. (The)	0.90	$71.38	5.15	2.78	2.86	3.10	28.00	29.90	32.25	1070.00	1070.00	***5.93%***
WEC Energy Group	0.80	$90.35	5.90	3.12	3.33	3.80	37.35	37.90	42.00	315.43	315.43	***5.77%***
Xcel Energy, Inc.	0.85	$64.61	4.25	2.08	2.22	2.66	31.60	33.15	38.25	550.00	560.00	***6.15%***
Atmos Energy Corp	0.85	$116.92	7.85	2.96	3.20	3.90	70.20	73.05	79.40	147.00	170.00	***7.43%***
Northwest Natural Holding Co.	0.80	$43.88	3.15	1.95	1.97	2.00	34.95	34.65	34.40	36.50	40.00	***4.33%***
ONE Gas Inc.	0.80	$78.99	5.60	2.60	2.72	3.15	52.70	50.90	64.45	55.50	57.00	***5.50%***
Spire Inc.	0.80	$65.36	5.50	2.88	3.00	3.45	53.40	59.75	67.10	53.00	55.00	***5.35%***
Median Beta	***0.9000***											
Average	**0.8887**										Median	**5.61%**
Data Source	Exhibit___(KXD-22), Exhibit___(JCN-8R), Exhibit___(JCN-2R)										Avg.	**6.00%**

REPLICATED STAFF DCF APPROACH - APPLY XIRR FORMULA, EPS GR

Company	Retention Rate 2027	Return on Equity 2027	B x R	Increase in Shares	MBR 2023	MBR-1 V Factor	(Accretion /dilutive) Growth S x V	GDP Growth	XIRR Formula Long-Form ROE	
ALLETE, Inc.	40.00%	9.34%	3.74%	1.27%	1.22	0.22	0.27%	***5.42%***	***10.62%***	1
Alliant Energy Corp.	39.74%	12.19%	4.84%	0.12%	2.00	1.00	0.12%	***5.42%***	***9.21%***	2
Ameren Corp.	40.00%	10.41%	4.17%	1.64%	2.10	1.10	1.80%	***5.42%***	***8.70%***	3
American Electric Power	38.82%	11.10%	4.31%	1.27%	1.63	0.63	0.80%	***5.42%***	***9.64%***	4
Avista Corp.	28.33%	8.16%	2.31%	2.50%	1.23	0.23	0.57%	***5.42%***	***10.47%***	5
Black Hills Corp.	33.11%	8.35%	2.76%	1.27%	1.32	0.32	0.40%	***5.42%***	***9.58%***	6
Centerpoint Energy	51.28%	10.16%	5.21%	0.12%	1.75	0.75	0.09%	***5.42%***	***8.34%***	7
CMS Energy Corp.	38.67%	14.47%	5.60%	0.68%	2.38	1.38	0.93%	***5.42%***	***9.12%***	8
Consolidated Edison,Inc.	35.67%	9.13%	3.26%	0.00%	1.60	0.60	0.00%	***5.42%***	***9.00%***	9
Dominion Energy, Inc	36.40%	12.24%	4.45%	0.82%	1.59	0.59	0.49%	***5.42%***	***11.96%***	10
Duke Energy Corp.	38.57%	10.09%	3.89%	0.00%	1.43	0.43	0.00%	***5.42%***	***10.19%***	11
Edison International	43.41%	13.73%	5.96%	0.39%	1.88	0.88	0.34%	***5.42%***	***9.80%***	12
Entergy Corp.	23.08%	8.97%	2.07%	1.82%	1.59	0.59	1.07%	***5.42%***	***9.85%***	13
Evergy Inc.	37.11%	10.34%	3.84%	0.00%	1.39	0.39	0.00%	***5.42%***	***9.96%***	14
Eversource Energy	37.86%	10.31%	3.90%	0.60%	1.55	0.55	0.33%	***5.42%***	***9.59%***	15
IDACORP, Inc.	34.13%	9.57%	3.27%	0.97%	1.82	0.82	0.79%	***5.42%***	***8.53%***	16
NextEra Energy, Inc.	37.73%	15.26%	5.76%	0.31%	3.35	2.35	0.72%	***5.42%***	***8.40%***	17
Northwestern Corporation	33.49%	8.03%	2.69%	0.00%	1.21	0.21	0.00%	***5.42%***	***10.06%***	18
OGE Energy Corp.	41.27%	12.35%	5.10%	0.00%	1.62	0.62	0.00%	***5.42%***	***10.32%***	19
Pinnacle West Capital Corp.	34.21%	9.33%	3.19%	1.40%	1.49	0.49	0.68%	***5.42%***	***10.00%***	20
PPL Corp.	40.00%	9.52%	3.81%	0.03%	1.38	0.38	0.01%	***5.42%***	***9.93%***	21
Portland General Electric Co.	35.34%	9.60%	3.39%	0.76%	1.46	0.46	0.35%	***5.42%***	***9.60%***	22
Public Service Enterprise Group, Inc.	37.78%	13.18%	4.98%	0.00%	2.16	1.16	0.00%	***5.42%***	***9.25%***	23
Sempra Energy	50.61%	12.00%	6.07%	0.00%	1.73	0.73	0.00%	***5.42%***	***8.84%***	24
Southern Co. (The)	39.81%	16.17%	6.44%	0.00%	2.55	1.55	0.00%	***5.42%***	***9.67%***	25
WEC Energy Group	35.59%	14.29%	5.09%	0.00%	2.42	1.42	0.00%	***5.42%***	***9.15%***	26
Xcel Energy, Inc.	37.41%	11.38%	4.26%	0.45%	2.04	1.04	0.47%	***5.42%***	***8.94%***	27
Atmos Energy Corp	50.32%	10.02%	5.04%	3.70%	1.67	0.67	2.46%	***5.42%***	***8.28%***	28
Northwest Natural Holding Co.	36.51%	9.15%	3.34%	2.32%	1.26	0.26	0.59%	***5.42%***	***10.04%***	29
ONE Gas Inc.	43.75%	9.03%	3.95%	0.67%	1.50	0.50	0.33%	***5.42%***	***8.94%***	30
Spire Inc.	37.27%	8.36%	3.11%	0.93%	1.22	0.22	0.21%	***5.42%***	***10.15%***	31
Average	***38.30%***	***10.85%***	***4.19%***	***0.78%***	***172.67%***	***72.67%***	***0.45%***	***5.42%***	***9.55%***	
Median	***37.78%***	***10.16%***	***3.95%***	***0.60%***	***160.40%***	***60.40%***	***0.33%***	***5.42%***	***9.60%***	

Exhibit__ (JCN-6R)

REPLICATED STAFF CAPM APPROACH - AS FILED

Merrill Lynch Cost of Market[1]

	Implied	Required
May-23	10.70%	11.10%
Jun-23	10.70%	11.10%
Jul-23	10.50%	11.00%

Merrill Lynch Cost of Market 10.85%

Treasury Rates[2]

	10 year	30 year
May-23	3.57%	3.86%
Jun-23	3.75%	3.87%
Jul-23	3.90%	3.96%

Risk-Free Rate (5/23-7/23) 3.82%

Market Risk Premium (5/23-7/23) 7.03%

Proxy Group Beta 0.89

Traditional CAPM Calculation
Risk Free Rate + (Beta * (Market Return - Risk Free Rate)

Traditional CAPM ROE 10.08%

Zero Beta CAPM Calculation
Risk Free Rate + (0.75*Beta * (Market Return - Risk Free Rate))+(0.25*(Market Return - Risk Free Rate))

Zero Beta CAPM ROE 10.27%

Source: Exhibit___(KXD-25)

REPLICATED STAFF CAPM APPROACH - INCLUDE MRP BASED ON S&P 500 MARKET RETURN

Merrill Lynch Cost of Market[1]

	Implied	Required
May-23	10.70%	11.10%
Jun-23	10.70%	11.10%
Jul-23	10.50%	11.00%

Merrill Lynch Cost of Market	10.85%
S&P 500 Market Return	15.14%
Average Market Return	13.00%

Treasury Rates[2]

	10 year	30 year
May-23	3.57%	3.86%
Jun-23	3.75%	3.87%
Jul-23	3.90%	3.96%

Risk-Free Rate (5/23-7/23)	3.82%
Market Risk Premium (5/23-7/23)	9.18%
Proxy Group Beta	0.89

Traditional CAPM Calculation
Risk Free Rate + (Beta * (Market Return - Risk Free Rate)

Traditional CAPM ROE	11.99%

Zero Beta CAPM Calculation
Risk Free Rate + (0.75*Beta * (Market Return - Risk Free Rate))+(0.25*(Market Return - Risk Free Rate))

Zero Beta CAPM ROE	12.24%

Source: Exhibit___(KXD-25), Exhibit___(JCN-4R)

FRB: Federal Reserve Statistical Release H.15 - Historical Data
Website : 'http://federalreserve.gov/releases/h15/

Exhibit__ (JCN-7R)

REPLICATED STAFF APPROACH - EFFECT OF INCREMENTAL ADJUSTMENTS
SUMMARY OF COST OF EQUITY

	As Filed	Apply XIRR Function in DCF	Apply EPS Growth Rates for Near-Term Growth in DCF	Apply GDP Growth Rate for Long-Term Growth in DCF	Apply MRP Based on S&P 500 Expected Return in CAPM
Proxy Group DCF ROE	8.52%	8.67%	8.85%	9.55%	9.55%
Traditional CAPM ROE	10.08%	10.08%	10.08%	10.08%	11.99%
Zero Beta CAPM ROE	10.27%	10.27%	10.27%	10.27%	12.24%
Generic (Average) CAPM ROE	10.18%	10.18%	10.18%	10.18%	12.12%
2/3 DCF & 1/3 CAPM Weighting	9.07%	9.18%	9.29%	9.76%	10.41%
ROUNDED	**9.10%**	**9.20%**	**9.30%**	**9.80%**	**10.40%**
1/2 DCF & 1/2 CAPM Weighting	9.35%	9.43%	9.52%	9.87%	10.84%
ROUNDED	**9.30%**	**9.40%**	**9.50%**	**9.90%**	**10.80%**

Source: Exhibit___(JCN-5R), Exhibit___(JCN-6R)

Exhibit__ (JCN-8R)

STAFF PROXY GROUP - EPS GROWTH RATES

Company	Ticker	Value Line Earnings Growth	Yahoo! Finance Earnings Growth	Zacks Earnings Growth	Average Growth Rate
ALLETE, Inc.	ALE	6.00%	8.10%	8.10%	7.40%
Alliant Energy Corp.	LNT	6.50%	7.00%	6.50%	6.67%
Ameren Corp.	AEE	6.50%	5.90%	6.40%	6.27%
American Electric Power	AEP	6.00%	5.20%	5.60%	5.60%
Avista Corp.	AVA	6.50%	6.30%	6.30%	6.37%
Black Hills Corp.	BKH	3.00%	5.40%	2.20%	3.53%
Centerpoint Energy	CNP	6.50%	NA	7.50%	7.00%
CMS Energy Corp.	CMS	6.50%	7.80%	7.80%	7.37%
Consolidated Edison,Inc.	ED	5.50%	6.12%	2.00%	4.54%
Dominion Energy, Inc	D	4.50%	9.00%	20.00%	11.17%
Duke Energy Corp.	DUK	5.00%	5.74%	6.10%	5.61%
Edison International	EIX	4.50%	4.53%	3.70%	4.24%
Entergy Corp.	ETR	0.50%	6.60%	5.70%	4.27%
Evergy Inc.	EVRG	7.50%	2.67%	5.20%	5.12%
Eversource Energy	ES	6.50%	6.70%	6.30%	6.50%
IDACORP, Inc.	IDA	5.00%	3.70%	3.70%	4.13%
NextEra Energy, Inc.	NEE	9.50%	8.80%	8.40%	8.90%
Northwestern Corporation	NWE	3.50%	4.50%	5.20%	4.40%
OGE Energy Corp.	OGE	6.50%	NA	3.70%	5.10%
Pinnacle West Capital Corp.	PNW	2.50%	6.10%	6.30%	4.97%
PPL Corp.	PPL	8.00%	17.21%	7.40%	10.87%
Portland General Electric Co.	POR	5.00%	5.90%	6.00%	5.63%
Public Service Enterprise Group, Inc.	PEG	4.00%	4.90%	4.80%	4.57%
Sempra Energy	SRE	7.00%	4.14%	4.90%	5.35%
Southern Co. (The)	SO	6.50%	7.30%	4.00%	5.93%
WEC Energy Group	WEC	6.00%	5.50%	5.80%	5.77%
Xcel Energy, Inc.	XEL	6.00%	6.15%	6.30%	6.15%
Atmos Energy Corp	ATO	7.00%	7.80%	7.50%	7.43%
Northwest Natural Holding Co.	NWN	6.50%	2.80%	3.70%	4.33%
ONE Gas Inc.	OGS	6.50%	5.00%	5.00%	5.50%
Spire Inc.	SRE	7.00%	4.14%	4.90%	5.35%

Sources: Value Line, Yahoo! Finance, Zacks, as of July 31, 2023

Exhibit__ (JCN-9R)

STAFF OF THE DEPARTMENT OF PUBLIC SERVICE
RESPONSE TO INTERROGATORY/DOCUMENT REQUEST

The Brooklyn Union Gas Company d/b/a National Grid NY
KeySpan Gas East Corporation d/b/a National Grid
Cases 23-G-0225 & 23-G-0226
Gas Rates

Request No.:	NG-27
Requested By	KEDNY/KEDLI Cost of Equity Witness
Information Requested of:	Staff Witness Duah
Date of Request:	September 6, 2023
Response Due Date:	September 18, 2023
Date of Response:	September 15, 2023
Name & Position of Respondent:	Kwaku Duah
Subject:	Return on Equity

Request:

1. Please refer to page 73 of Staff Witness Duah's testimony and the discussion related to NiSource Inc.

    a. Is it Staff's view that NiSource Inc.'s sale of a minority interest in one if its operating subsidiaries will be a "transformational financial transaction" that will distort the common stock price of NiSource Inc.?

    b. If so, has Staff performed an analysis of whether NiSource Inc.'s stock price was distorted by the transaction? Please provide any analysis relied upon by Staff.

2. Please refer to the following statement in Staff Witness Duah's testimony: "I find Companies' witness Nowak's requirement that a company must have 70 percent of its net operating income from regulated utility operations to be too restrictive. Operating income is the amount of revenue left over after accounting for all the expenses necessary to keep the business running" (page 104).

    a. Is it true that "common equity investors only earn a return after the payment of all other expenses," as stated in page 14 of Staff Witness Duah's testimony?

    b. If so, from a common equity investors' perspective, is income or revenue a more relevant metric for evaluating a company's expected return?

3. Please refer to Exhibit __ (KXD-22), page 2 and confirm the following:

    a. The column "Long-Form ROE" relies on the Excel formula "IRR".

STAFF OF THE DEPARTMENT OF PUBLIC SERVICE
RESPONSE TO INTERROGATORY/DOCUMENT REQUEST

The Brooklyn Union Gas Company d/b/a National Grid NY
KeySpan Gas East Corporation d/b/a National Grid
Cases 23-G-0225 & 23-G-0226
Gas Rates

Response to NG-27 continued.

b. The column "Long-Form ROE" is calculated based on a projected dividend stream that uses 200 years of annual dividend payments.

c. The application of the Excel IRR formula in the calculation of the "Long-Form ROE" implicitly assumes that the first dividend is received at the end of the first projected year.

4. Please refer to the following statements in Staff Witness Duah's testimony: "If companies are not paying dividends, then the whole DCF calculation is flawed because a DCF cannot be performed" (page 125); and "The DCF approach can be applied to any investment instrument that has an intrinsic value" (page 75). Is it Staff Witness Duah's contention that a company not paying dividends has no "intrinsic value"?

5. Does Staff Witness Duah agree that Treasury rates are set by the capital markets and thus outside the Companies' control? If not, please explain.

6. Does Staff Witness Duah agree that in the event a multi-year settlement was agreed upon – as opposed to a single year, litigated case - there would be an increased risk of interest rates deviating from their current levels over the term of the rate plan, and as such there would be increased risk that the Companies may either over- or under-collect their actual debt costs absent a true-up mechanism for interest rate expense associated with new debt issuances? If not, please explain.

Response:

1.

a. The transaction between NiSource and Blackstone Infrastructure Partners is significant and transformative, given the substantial capital involved, the strategic implications for NiSource's future, and stock price movement as noted in the Standard and Poor's report entitled, "*NiSource touts premium valuation in sale of Ind. utility stake to Blackstone*" published on June 20, 2023. This report is attached as DPS Response - NG-27, Attachment 1. Therefore, it is reasonable to eliminate NiSource, Inc. from the proxy group.

b. See response to 1a above.

2.

a. Yes, as stated on p. 14, lines 19 to 23 of Staff Witness Duah's testimony, common equity investors only earn a return after the payment of all other expenses.

STAFF OF THE DEPARTMENT OF PUBLIC SERVICE
RESPONSE TO INTERROGATORY/DOCUMENT REQUEST

The Brooklyn Union Gas Company d/b/a National Grid NY
KeySpan Gas East Corporation d/b/a National Grid
Cases 23-G-0225 & 23-G-0226
Gas Rates

Response to NG-27 continued.

b. Operating income gives a snapshot of a company's profitability from its core operations, while revenues represent the total sales or turnover. For equity investors, both are important but serve different purposes. Revenue is key for gauging a company's size, market share, and growth potential. Companies with revenue growth can attract investors seeking growth, even if current profitability isn't strong. Therefore, revenues are a more accurate measure of exposure to the risks of competitive operations, especially since competitive ventures can sometimes be unprofitable. As stated on p. 104 of Staff Witness Duah's testimony, using operating income to establish a proxy group might include companies with much higher risk than a typical regulated utility. Additionally, as stated on pp. 105 and 106 of Staff Witness Duah's testimony, different accounting policy choices, such as inventory valuation methods and depreciation choices, can significantly impact reported operating income across companies. These variations in accounting practices tend to have a more pronounced effect on operating income than on revenues, further emphasizing the relevance of using revenues as a superior metric when constructing a proxy group.

3.

a. Yes, the column labeled "Long-Form ROE" on p. 2 of Exhibit___(KXD-22) relies on the Excel formula for internal rate of return (IRR).
b. Yes, the "Long-Form ROE" on p. 2 of Exhibit____(KXD-22) is calculated based on a projected dividend stream that uses 200 years of annual dividend payments.
c. Yes, the first dividend is received at the end of the first projected year and subsequent dividend payments are received on a regular periodic basis for the next 200 years.

4. No. While Staff Witness Duah has emphasized the importance of dividends in the DCF for purposes of determining the return on common equity for regulated utilities, the intent was not to imply that a company not paying dividends lacks intrinsic value.
5. Yes, Staff Witness Duah agrees that Treasury rates are determined by the capital markets and are outside of the Companies' control.
6. Staff Witness Duah cannot speculate on the risk of interest rates in the event of a multi-year rate plan, as his testimony specifically addresses a one-year rate plan.

Name of Respondent:
Kwaku Duah

Date of Reply:
September 15, 2023